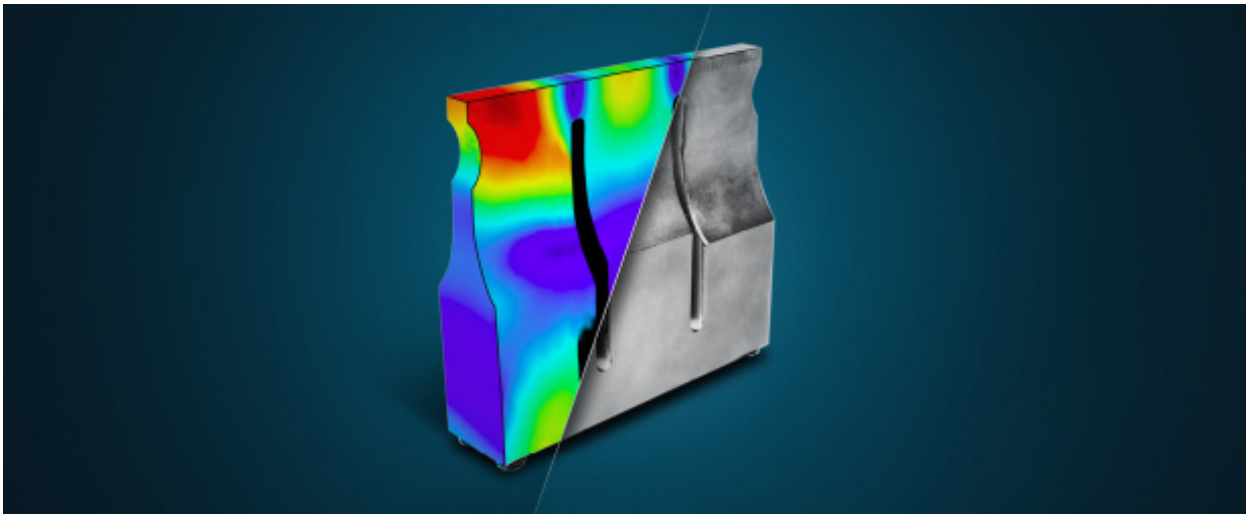
Polytec

Software Manual

Polytec Scanning Vibrometer

Presentation



MSA

MSA-3D

PSV/PSV-3D

STAGE

UHF

Version 10.1

Trademarks

Brand or product names mentioned in this documentation could be trademarks or registered trademarks of their respective companies or organizations.

41597-Man-Vib-PSV-Soft10.1-PRES-0422-01en

Contents

1 First Steps 1-1

1.1 System Requirements 1-1
1.2 Starting Data Management System and Installing the Software 1-2
1.3 Starting the Software 1-4
1.4 Modes in the Software: Acquisition and Presentation 1-5
1.5 Windows in the Software 1-6
1.5.1 Application Window 1-8
1.5.2 Presentation Window 1-10
1.5.3 Analyzers 1-13
1.5.4 Project Browser 1-15
1.5.5 Signal Processor 1-19
1.5.6 Arranging Windows 1-20
1.5.7 Linking Windows 1-21
1.5.8 Zooming in Windows 1-22
1.5.9 Printing Windows 1-22
1.6 Help Texts and Software Information 1-23

2 Setting Up Presentation 2-1

2.1 Setup: dB Reference 2-2
2.2 Setup: Display 2-3
2.3 Setup: Messages 2-5

3 Displaying Data 3-1

3.1 Setting Up Display Properties 3-2
3.1.1 Setting Up the Object 3-3
3.1.2 Setting up View Style of Measurement Data 3-4
3.1.3 Setting Up 3D View 3-6
3.2 Working in Analyzers 3-10
3.2.1 Setting Cursors and Reading Data 3-11
3.2.2 Linking Windows 3-16
3.2.3 Zooming in Analyzers 3-17
3.2.4 Scaling Axes Ranges 3-18
3.2.5 Displaying Vibrational Direction (only PSV-3D) 3-19
3.2.6 Setting up Presentation of Graphs 3-20
3.2.7 Setting up Diagrams 3-21
3.3 Working in Presentation Windows 3-22
3.3.1 Setting Up the View of the Presentation Window 3-23
3.3.2 Selecting the View Style of the Data 3-25
3.3.3 Setting Cursors and Reading Data in Presentation Windows 3-26
3.3.4 Displaying Data as Readout 3-27
3.3.5 Selecting the Color Palette 3-28

3.4 Selecting the Data Set Displayed 3-28
3.5 Defining Clipping Planes 3-33
3.6 Generating Combined 3D Files (Stitching) 3-35
3.6.1 Merging Individual Measurements 3-37
3.6.2 Working with Combined Files 3-41
3.7 Measurement Data in the Signal Processor 3-44
3.7.1 Copying and Calculating Measurement Data 3-45
3.7.2 Available Functions 3-49
3.7.3 Calculation Examples for Functions 3-58
3.7.4 Setting up Signal Description 3-60
3.7.5 Saving Calculated Data (User Defined Data Sets or Bands) 3-61
3.7.6 Displaying Measurement Results 3-63
3.7.7 Saving Formulae 3-64
3.7.8 Finding and Replacing in Formulae 3-65
3.7.9 Selecting Files 3-65
3.7.10 Application Examples for Processing Measurement Data 3-67
3.8 Creating a New Scan File 3-79

4 Evaluating Measurement Data 4-1

4.1 Processing Data 4-1
4.1.1 Status of Scan Points 4-2
4.1.2 Filtering Data 4-6
4.1.3 Interpolating Points 4-6
4.1.4 Changing Vibrational Direction 4-11
4.2 Evaluating Sections of a Spectrum 4-12
4.2.1 Evaluating Band Data (not in Time Measurement Mode) 4-12
4.2.2 Evaluating Point Data (Cursor Mode) 4-19
4.3 Displaying Profiles 4-21
4.4 Animating Vibration 4-23
4.4.1 Animation in FFT, FastScan or MultiFrame Measurement Mode 4-23
4.4.2 Animation in Time Measurement Mode 4-27
4.5 Editing Scan Points 4-30
4.5.1 Selecting Scan Points 4-30
4.5.2 Selecting Outliers 4-32
4.5.3 Changing Indices 4-35
4.5.4 Defining Connections between Scan Points 4-37
4.5.5 Defining Geometry Components 4-38

5 Managing Data 5-1

5.1 Opening, Loading, Importing 5-1
5.1.1 Opening or Loading Files 5-1
5.1.2 Opening Scans, Single Point Measurements and Signal Processor Documents 5-3
5.1.3 Loading Frequency Band Definition 5-5
5.1.4 Importing Data in Universal File, ME'Scope or ASAM ODS ATFX Format 5-6
5.2 Saving and Exporting 5-11
5.2.1 Saving a Scan 5-11
5.2.2 Saving Single Point Measurement and Average Spectrum 5-13
5.2.3 Saving Frequency Band Definition 5-14
5.2.4 Saving Graphics 5-16
5.2.5 Saving Animation 5-17
5.2.6 Copying Data onto Clipboard 5-18
5.2.7 Exporting Measurement Data in ASCII Format 5-20
5.2.8 Exporting Universal File 5-21
5.2.9 Exporting Measurement Data in Wave Sound Format 5-25
5.2.10 Exporting ME'Scope 5-26
5.2.11 Exporting ASAM ODS ATFX 5-29
5.2.12 Exporting Geometries in STL Format 5-32

Appendix A: Key Combinations in the Software

Index

1 First Steps

With the PSV software you can perform the following tasks in the measurement system:

- Controlling the hardware
- Defining scan points
- Setting the parameters for data acquisition
- Generating excitation signals (as an option)
- Acquiring digital data
- Saving data
- Displaying data and evaluating it

First read the following sections to familiarize yourself with the basic concepts of the PSV software.

Check whether your system meets the required conditions. See SECTION 1.1 on this.

Start up the data management system and configure it. See SECTION 1.2 on this.

Start the software. See SECTION 1.3 on this.

In the software you work in two different modes: Acquisition and Presentation. You can start an instance of the software in each of the Presentation and Acquisition modes and use them together. See SECTION 1.4 on this.

In the software, you work with different windows in Presentation. See SECTION 1.5 on this.

The software offers you help on operation and information about the software. See SECTION 1.6 on this.

1.1 System Requirements

In order to work optimally with the software, adhere to the following requirements for data acquisition systems:

- Operating system: Windows 10 (64 bit)
- Microsoft DirectX 11
- The respective drivers for your measurement system.
  You will find detailed information on this in the Release Notes.

In order to work optimally with the software, adhere to the following requirements for desktop systems:

- Operating system: Windows 10 (64 bit) or Windows 11 (64 bit)
- Microsoft DirectX 11
- Monitor with 32 bit color resolution, 1280 x 1024 pixel

  You will find detailed information on this in the Release Notes.

INFORMATION

First of all, carry out the installations as described in the Release Notes. See SECTION 1.2 on this as well.

1.2 Starting Data Management System and Installing the Software

Start Polytec DMS

To start the Polytec data management system for the first time, proceed as follows:

1. Switch on the data management system.

   Windows setup is started.
2. Select the display language.

   The default language for the operating system is set.
3. Accept the license conditions displayed.

ⓘ The Windows operating system has already been activated.

4. Set up a new user.

   On the Sign in with Microsoft page, click Domain join instead to create a local account or to use a domain account.
5. Choose a password.
6. Allocate a computer name.
7. Follow any other instructions.

   You have now logged into the system and can start using the measurement system.

Adapt Windows 10 settings

The settings for the behavior of the Windows operating system are preset by Polytec. If you reinstall the operating system, you need to adjust the settings as follows:

1. Press the Windows and X key combination. Click System. Click Advanced System Settings. Display the Hardware page in the System Properties dialog. Click Device Installation Settings and select No.

   Automatic installation of updates is suppressed.

2. Press the Windows and X key combination. Click System. Click Advanced System Settings. Display the Advanced page in the System Properties dialog. Click Settings in the Startup and Recovery field. Deactivate the Automatically restart check box in the System failure field.

   Automatic restart after a Blue Screen is suppressed.

3. Press the Windows button. Click Settings and select Power & sleep. Under Related settings, click Additional power settings. In the Power Options dialog, select High performance.

   The Data Management System is not automatically switched off by power saving mode.

Install desktop software

To install a desktop software, perform the following steps. If any problems occur during installation, contact your local Polytec representative.

1. Switch on your PC and log in with administrator rights. See your operating system manual on this.
2. Run Polytec License Update that you have received from Polytec on a USB stick or as a link in an e-mail.

   Your license is activated and you can use Polytec Update to download Polytec software.

3. If Polytec Update is already installed on your PC, proceed to step 6.
4. To download Polytec Update, copy the following link to your web browser:

   http://swdownload.polytec.com/polyupdate/Setup32.exe

5. Install Polytec Update. Follow the installation instructions.
6. Start Polytec Update.
7. Download the Polytec software that is available for your PC and install it. You will find further information in the Polytec Update software manual.

ⓘ You will find information on updating PCs that are not connected to the Internet in the Polytec Update software manual.

Alternatively, install the required software as described in the Release Notes on your Polytec USB stick.

Storage of setting files

The files with settings for the PSV software are saved in different directories. The following table shows where the various setting files are saved.

ⓘ If the directories are not displayed, open the Folder and search options and activate the Show hidden files, folders, and drives check box on the View page.

Setting/File	Directory path
Example files	C:\ProgramData\Polytec\PSV\'Version'\Examples
Window settings - AnalysisSettings.set - BrowserSettings.set - WindowSettings.set	C:\Users\'Account name'\AppData\Roaming\Polytec\PSV\'Version'
Acquisition settings - <FrontEndName>_Ch<NumberOfChannels>.set (e.g. PSV-F-500_Ch8.set)	C:\Users\'Account name'\AppData\Local\Polytec\PSV\'Version'
Settings for camera and alignment - AcquisitionSettings.set	C:\Users\'Account name'\AppData\Local\Polytec\PSV\'Version'
Objective calibration (only MSA, MSA-3D and UHF)[1] - LensCalibrations.set	C:\ProgramData\Polytec
Settings for frequency response (only UHF)[2] - frequencyresponse.cal	C:\ProgramData\Polytec
License files	C:\ProgramData\Polytec\Licenses

[1] This file must never be deleted.

[2] This file must never be deleted.

1.3 Starting the Software

Prepare

Before you start the software, check whether one of the options applies:

- You have correctly installed the hardlock (dongle) on the PC.
- You have correctly installed a license file using Polytec License Update.

Start

To start the software, proceed as follows:

1. Switch on the Data Management System.
2. After a short while you will be asked to log in. See the operating system documentation on this or contact your administrator. You do not need administrator rights to start the software.
3. Start the software in Presentation mode by double-clicking the 

icon on the desktop.

   The application window appears.

You can now work with the software.

1.4 Modes in the Software: Acquisition and Presentation

In the software you work in two different modes: Acquisition and Presentation. You can start an instance of the software in each of the Presentation and Acquisition modes and use them together. You can switch between the windows of the software instances. Some functions are available for both instances.

ⓘ In the desktop version of the PSV software, the acquisition is not available.

Acquisition

When starting acquisition mode, the software searches the interfaces of the data management system for external devices and initializes them.

ⓘ You may receive an error message and information on it. You can switch off error messages as described in SECTION 2.3.

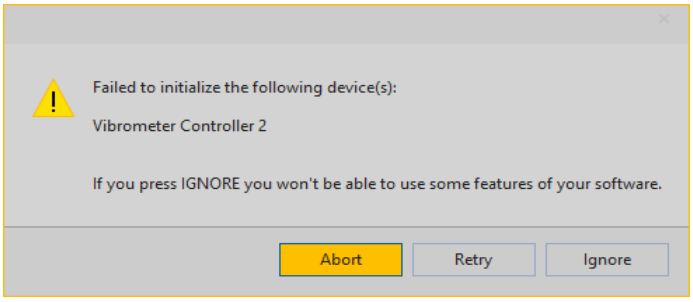


Figure 1.1: Error message when changing into acquisition mode

To check the connection between the devices and to correct it if necessary, select Setup > Preferences in the menu bar and display the Devices page. The connections to external devices are displayed.

ⓘ Some Polytec instruments are automatically initialized in acquisition mode and can not be changed. When ending Acquisition, all the hardware used is enabled. For this reason, you can use PSV software in parallel with other Polytec software products.

To change from Presentation to Acquisition, you have the following options:

- Select View > Acquisition in the menu bar.
- In the toolbar, click  .
- In the Windows task bar, click on the software instance for acquisition.
- Press the Alt + Tab key combination and select the software instance for Acquisition.

Presentation

To change from Acquisition to Presentation, you have the following options:

- Select View > Presentation in the menu bar.
- In the toolbar, click  .
- In the Windows task bar, click on the software instance for presentation.
- Press the Alt + Tab key combination and select the software instance for Presentation.

1.5 Windows in the Software

Read the following sections to familiarize yourself with the various windows in the software.

Application window

In the application window of the software, you work with windows to display and evaluate data. In order to do this, you have an arbitrary number of presentation windows available. See SECTION 1.5.1 on this.

Analysis window

You analyze measurements in 2D and 3D diagrams in presentation windows. See SECTION 1.5.2 on this.

Analyzer

The analyzer is part of the presentation window and displays measurement data and evaluated data in xy-diagrams. In analyzers you can display data from a file and analyze them. See SECTION 1.5.3 on this.

Project browser

In the project browser, you can display folders with the subdirectories and files they contain in a tree structure. See SECTION 1.5.4 on this.

Signal processor

You can use the signal processor to recalculate measurement data for further analysis and then compare the results with the original data. See SECTION 1.5.5 on this.

Arrange windows

You can arrange all windows with the commands in the Window menu. See SECTION 1.5.6 on this.

Link windows

You can link area views and analyzers to display data with the same settings. See SECTION 1.5.7 on this.

Zoom in windows

You can zoom images sections in different windows. Zooming in helps you to display the data more precisely. See SECTION 1.5.8 on this.

Print windows

You can print out analyzers and signal processors. See SECTION 1.5.9 on this.

General icons Several icons from the toolbars of the windows are similar to those in other software applications.

Table 1.1: General icons in the software

Icon	Key combination	Menu navigation	Function
	-	File > New > Signal Processor	Create new signal processor document
	-	File > New > Combined File or Scan File	Create new document
	Ctrl + O	File > Open	Open existing document
	Ctrl + S	File > Save	Save existing document
	Ctrl + I	File > Info	Display file information
	Ctrl + P	File > Print	Print active document
	Ctrl + X	Edit > Cut	Cut selection and copy onto the clipboard
	Ctrl + C	Edit > Copy	Copy selection onto the clipboard
	Ctrl + V	Edit > Paste	Insert selection from the clipboard
	Delete	Edit > Delete	Delete selection (refers to adding user defined data)
	Ctrl + Z	Edit > Undo	Undo last action
	Ctrl + Y	Edit > Redo	Redo previously undone action

You will find a description of the other icons in the following sections and chapters.

1.5.1 Application Window

After starting the software, the application window appears. At the top edge you will see the title bar, menu bar and toolbar, at the bottom the status bar. In the View menu, you can show/hide the toolbar and the status bar.

If you start PSV Presentation for the first time, the application window will show the project browser.

The windows are described in the following sections. Click  to generate new signal processor documents, new combined files or new scan files.

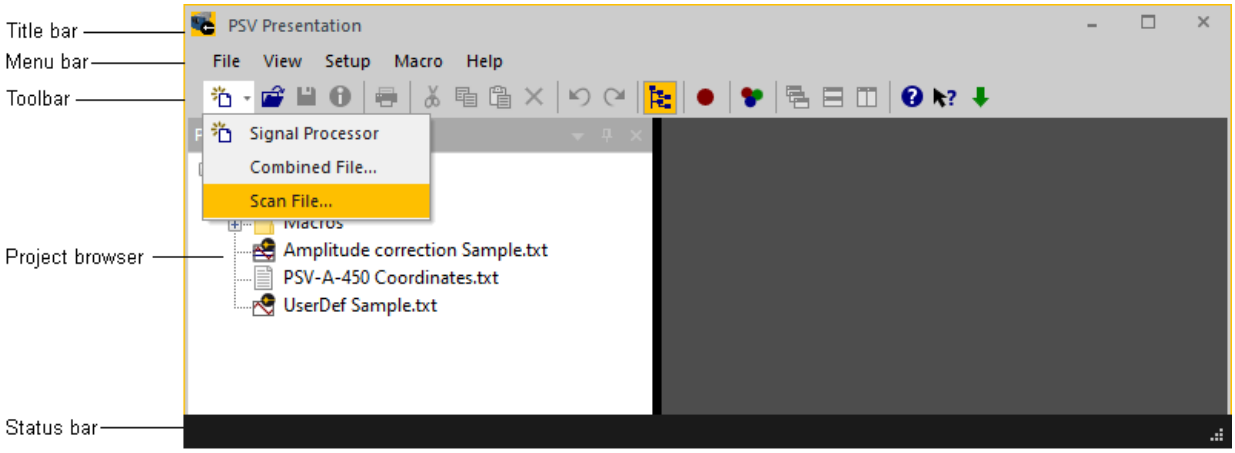


Figure 1.2: Example of the application window in presentation mode

If you open a measurement file, you will see the presentation window. The windows are described in the following sections.

In the application window you can display and evaluate data. In order to do this, you have an arbitrary number of presentation windows available:

When you restart the software, then the layout and signal selection of the windows opened most recently will be restored. Read SECTION 1.5.6 on how to arrange windows.

Title bar In the title bar, you will see the name of the software. If you have already saved the measurement or have opened a measurement, you will see the file name of the measurement in the title bar as well.

Menu bar In the menu bar, every menu contains a group of commands for working with the software. Some menus are only visible if an analyzer or a presentation window is open.

Toolbar By clicking the icons, you can execute some commands quickly which you would otherwise have to select in the menus. The appropriate icons are available to you for every window that you have opened.

Software-specific icons

Apart from the general icons, you will find more software-specific icons in the application window.

Table 1.2: Software-specific icons in the application window

Icon	Key combination	Menu navigation	Function
	-	View > Project Browser	Show/hide project browser window
	Ctrl + Q	View > Acquisition	Start Acquisition (software instance)
	-	Setup > Color Palette	Select color palette
	-	Window > Cascade	Arrange windows so they overlap
	-	Window > Tile Horizontally	Arrange windows horizontally
	-	Window > Tile Vertically	Arrange windows vertically
	F1	Help > Contents	Display the table of contents for the online help
	-	Help > Context Help	Display help for icons, menus and windows
	-	Help > Check for Updates	Search for Polytec Software updates
	-	Help > Check for Updates	Install Polytec Software updates

Status bar

On the left you will see a short help text in certain situations. For example, point the mouse at one of the icons and leave the mouse pointer there for a moment. Then on the left you will see which action it is for. You will find more information on how to get help in the software in SECTION 1.6.

1.5.2 Presentation Window

You analyze scans in 2D and 3D diagrams in presentation windows. Read SECTION 3.3 on how to work in presentation windows.

Open

A presentation window appears when you open a scan. The menu bar of the application window is then extended by menus for working with presentation windows. You can then open further presentation windows by selecting Window > New Window.

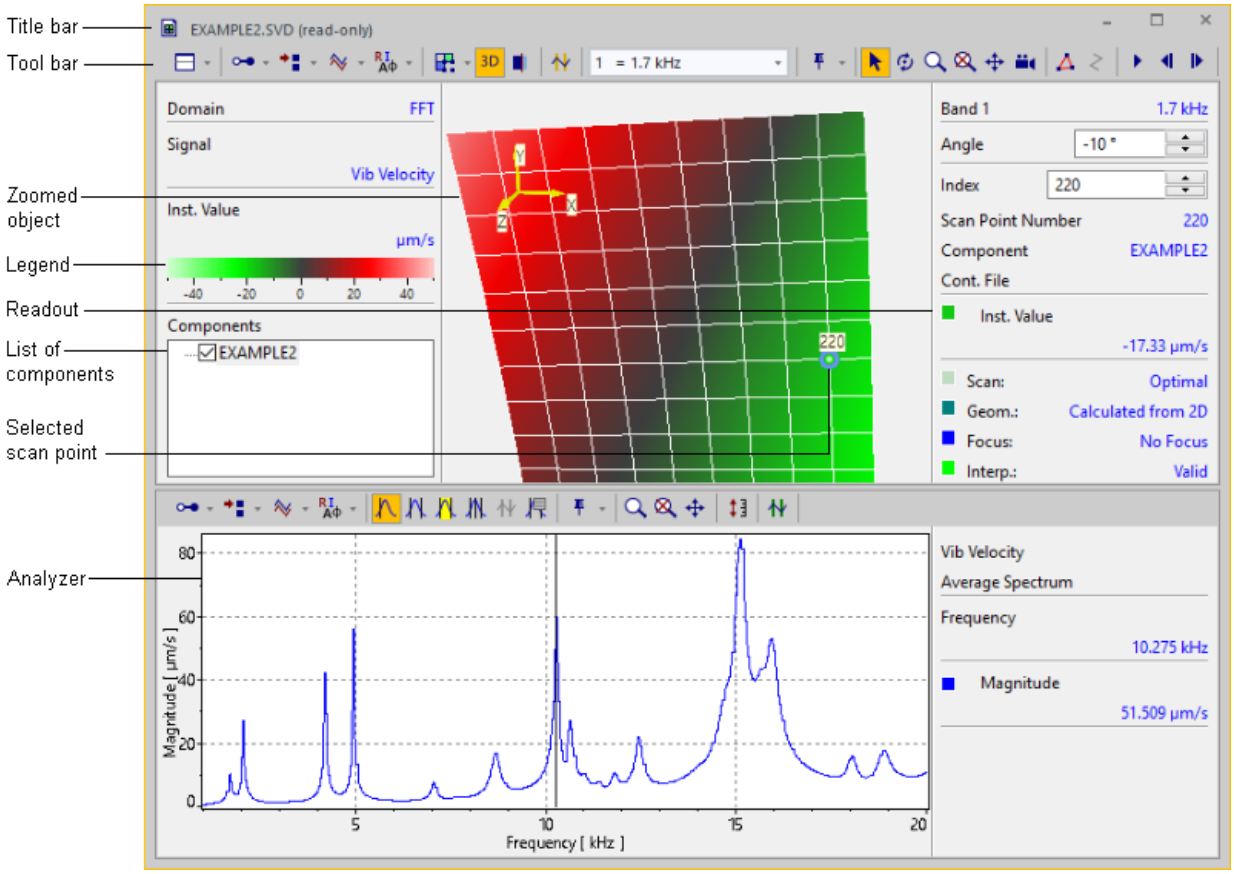


Figure 1.3: Example of the presentation window in Average Spectrum view

Toolbar

You can show/hide the toolbar by selecting Presentation > Toolbar. By clicking the icons you can quickly execute some commands from the Setup, Presentation, and Animation menus.

Icons in the presentation window

In the following table, you will find the icons of the surface view in the presentation window summarized.

Table 1.3: Icons in the presentation window

Icon	Menu navigation (key combination)	Function
	Presentation > View	Select a view
	Presentation > Domain	Select a domain
	Presentation > Channel	Select a channel or combination of channels
	Presentation > Signal	Select signal
	Presentation > Display Type	Select display type
	Presentation > View Style	Select a view style for surface data
	Presentation > 2D/3D View	Toggle between 2D and 3D view
	Presentation > Define Clipping Planes	Define clipping planes for 3D view
	Setup > Frequency Bands (Ctrl + F)	Define frequency bands
	Pin	Select surface view settings to link the windows
	Presentation > Select	Select scan points with the mouse
	Presentation > Rotate	Rotate the object in 3D view
	Presentation > Zoom In	Zoom the image section
	Presentation > Zoom Out	Undo zooming the image section
	Presentation > Pan	Pan the image section
	Presentation > Camera Sight (Shift + Y)	Sight of the camera in 3D view
	Presentation > Connect Points	Connect points
	-	Add a profile line in profile view
	Animation > Start or Animation > Stop (space bar)	Start or stop animation
	Animation > Step Backward	Display the previous animation image
	Animation > Step Forward	Display the next animation image
	Presentation > Scale Up	Scale up data for the current range
	Presentation > Auto Scale	Automatically scale data for the current range

Icon	Menu navigation (key combination)	Function
	Presentation > Scale Down	Scale down data for the current range
	-	Show the X-component of data (also for Y- or Z-component)

View You can display the presentation window in different views. See SECTION 3.3.1 on this as well.

Object You see the object in all views. You can display the object data in different ways. See SECTION 3.3.2 on this as well.

Legend On the left you can show (hide) a legend for color-coding and the geometry components. To do so, select Presentation > Legend in the menu bar. Additionally you can show/hide the status of the scan points as a legend. To do so, select Presentation > Status.

Readout On the right you can show/hide a readout for the data displayed. To do so, select Presentation > Readout in the menu bar.

List of components In acquisition and presentation mode you can define geometry components and assign scan points to the individual components. See SECTION 4.5.5 on this as well.

Analyzer In the Single Scan Point, Average Spectrum and Profile views you will see an analyzer at the bottom of the presentation window. See SECTION 1.5.3 on this.

Zoom functions In the presentation window you can zoom on the object in all views and in all view styles and also pan the zoomed section. You will find the appropriate tools in the toolbar of the presentation window. See SECTION 1.5.8 on this as well.

1.5.3 Analyzers

Analyzers show measured raw data and evaluated data in xy-diagrams. In analyzers, you can display measurement data from a file. In presentation mode, an analyzer can also be part of a presentation window. See SECTION 1.5.2 on this. Read SECTION 3.2 on how to work in analyzers.

Open

An analyzer appears if you:

- open a scan (as part of the presentation window) or
- open a single point measurement.

If an analyzer is open, the menu bar of the application window is extended by menus for working with analyzers. To open more presentation windows or analyzers, select Window > New Window.

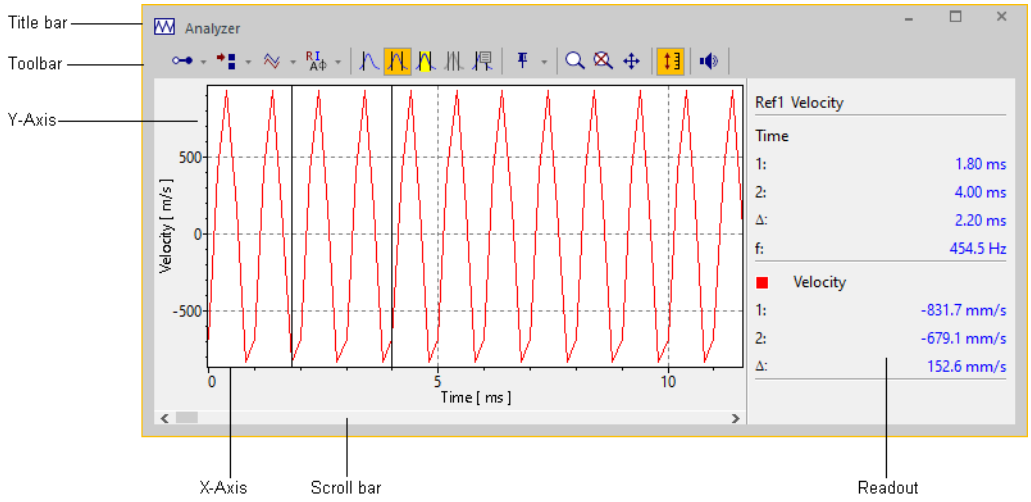


Figure 1.4: Example of an analyzer

Title bar

As soon as you open a single point measurement, you will see the file name in the title bar.

Toolbar

You can show/hide the toolbar by selecting Analyzer > Toolbar. By clicking the icons you can quickly execute some commands from the Analyzer menu.

Icons in the analyzer

In the following table, you will find the default icons of the analyzer summarized.

Table 1.4: Icons in the analyzer

Icon	Menu navigation (key combination)	Function
	Analyzer > Domain	Select a domain
	Analyzer > Channel	Select a channel or combination of channels
	Analyzer > Signal	Select a signal
	Analyzer > Display Type	Select a display type
	Analyzer > Cursor	Place a cursor in the graph
	Analyzer > Differential Cursor	Place two cursors in the graph
	Analyzer > Band Cursor	Display a band cursor in the graph
	Analyzer > Harmonic Cursor	Display a harmonic cursor in the graph
	Analyzer > Cursor Properties	Edit cursor properties
	Pin	Select analyzer settings to link the windows
	Analyzer > Zoom In	Zoom graph's x- and y-ranges simultaneously
	Analyzer > Zoom Out	Undo last zoom action; press Shift key to undo all actions
	Analyzer > Pan	Pan the zoomed section
	Analyzer > Auto Scale (Ctrl + K)	Scale the ordinate to min/max values of current data

Zoom functions

In the analyzer you can zoom in on diagram sections and can pan the zoomed section. You will find the appropriate tools in the toolbar of the analyzer.

Diagram

The data are shown as a graph in xy diagrams. You can set up the look of the diagrams and the graphs individually.

Readout

On the right you can show/hide the readout for the diagram by selecting Analyzer > Readout in the menu bar.

Scroll bar

You can use the scroll bar to scroll if you have zoomed the abscissa.

1.5.4 Project Browser

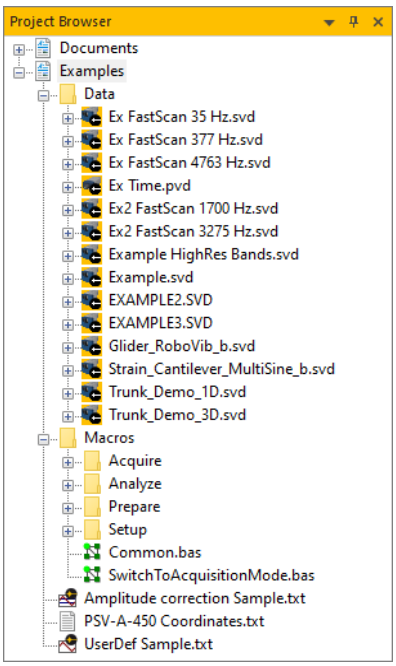


Figure 1.5: Example of the project browser

Description

In the project browser, you can display folders with the subdirectories and files they contain in a tree structure. To do so, open a program on file management, e.g. Windows Explorer and use the mouse to drag the required folder or folders into the project browser. You can now manage these folders from here.

ⓘ The folders in the project browser are not copies but direct links to the folders on your hard disk or network drives. All actions, such as deleting or renaming subdirectories and files, are directly executed in the respective folder.

Open/position

The contents of the project browser can be configured separately for every Windows user. The contents of the project browser and selected entries are saved when the program ends and are displayed again the next time the software is started. You can arrange the docking window of the project browser in different ways in the application window. The following overview shows the available functions.

Table 1.5: Icons for the project browser

Icon	Menu navigation or name	Function
	View > Project Browser	Show/hide project browser window
	Window Position	Select the window position for the docking window from the list: - Floating - Docking - Auto Hide - Hide
	Auto Hide	Hide the docking window automatically, that means the docking window is shown minimized at the edge. As soon as you point the mouse pointer there, the docking window is displayed completely.
	Docking	Fix the docking window at the edge
	Close	Close the docking window

You can adapt the display of the project browser, i.e. reduce its size and position it in a different place on the screen. If you drag the project browser with the mouse, then positioning aids appear at the edges and in the middle of the application window. Drag the project browser to a suitable position and release the mouse button. The project browser is automatically placed at the corresponding window edge.

Add folder

To add another project folder to the project browser, you have the following options:

- Click the project browser with the right mouse button, however not a file or a folder, and select Add Project Folder in the context menu.
- In the menu bar, select File > Add Project Folder.

The Select Folder dialog appears. Navigate to the required folder and click Select.

ⓘ You cannot transfer individual files to the project browser.

Remove folder

You can remove folders of the highest level. To do so, click the folder with the right mouse button and select Remove in the context menu.

ⓘ You can also delete individual files.

Copy and move

You can copy files or folders in the project browser into other directories. To do so, mark them and holding the left mouse button pressed, drag the marked objects onto the required directory.

To move files or folders in the project browser, proceed as described above. In addition, hold the Shift key pressed while dragging with the mouse.

Commands in the project browser

Click a file or a folder with the right mouse button to edit the contents in the project browser. The commands shown in the context menu differ depending on the level in the tree structure, the type and number of objects selected and the mode you are currently working in (acquisition or evaluation). The individual commands are explained in the following:

Explorer: Click here if you want to have the selected folder displayed in the Windows Explorer. The command is only available for folders.

New Folder: Click here if you want to create a new subfolder in the selected folder. The subfolder appears in the project browser and can immediately be renamed respectively.

Update: Click here to update the display of folders and files in the project browser.

Remove: Click here if you want to remove the selected objects from the project browser, but do not want to delete them. The command is only available for folders on the top level.

Rename: Click here if you want to give the selected object a different name.

Delete: Click here if you want to irrevocably delete the selected objects. Before permanently deleting it, you have to confirm asecurity popup. The command is only available for files.

Load: Click here to load the setting file in Settings Files (*.set) format you selected in the project browser.

Open: In principle, you can open files from all programs installed on your PC in the project browser. Click Open and the selected file will be opened using the corresponding program. If you open a file in Signal Processor Data (*.spd) format, then the signal processor opens (SigPro option).

ⓘ The PSV software is run in 64 bit Windows systems as a 64 bit process. This means that all applications started from the PSV software which are available as 64 bit and 32 bit versions will automatically be started as a 64 bit process, too. This can cause some applications to behave differently depending on whether they are started in the PSV software or in Windows Explorer respectively.

New Window: Click here if you want to display the file you have already opened again in a second window.

Info: Click here if you want to display the file information of the file selected. Refer also to SECTION 5.1.1.

Close: Click here if you want to close the file you have already opened.

Save As: Click here if you want to save the selected file under a different name or in another directory.

Paste: This command is only available if you are using the optional signal processor. See SECTION 3.7 on this as well. If you have copied data in the signal processor, you can insert this data into the selected file by using the Paste command.

Edit: This command is available for macros in the WinWrapBasic (*.bas, *.obm, *.cls) format and for project files in the (*.wbp) format. Click Edit and a macro window appears with the macro you have selected in the project browser.

ⓘ Macros which are edited or run via the project browser appear as a list of files opened most recently in the Macro menu (max. 4). The lists are managed separately for the acquisition and presentation mode.

Run: Click here to run the macro you selected in the project browser.

Additional commands

Create Scan File: Click here if you want to generate a new scan file from the selected file. A prerequisite for this is that there is a 3D geometry in the file. Only the 3D geometry is then taken over into the new scan file. You can now add user defined data sets and save the file.

You can generate a new scan file from different file formats. See SECTION 3.8 on this.

Attributes: This command is only available to you for combined files in presentation mode. You will see the Create Combined File dialog with the individual files in the set order. See SECTION 3.6 on this as well.

Import: In presentation mode, you can import measurement data in Universal File, ME'Scope or ASAM ODS ATFX format. See SECTION 5.1.4 on this as well.

1.5.5 Signal Processor

You can use the signal processor to recalculate measurement data for further analysis and then compare the results with the original data. Read SECTION 3.7 on how to work with the signal processor.

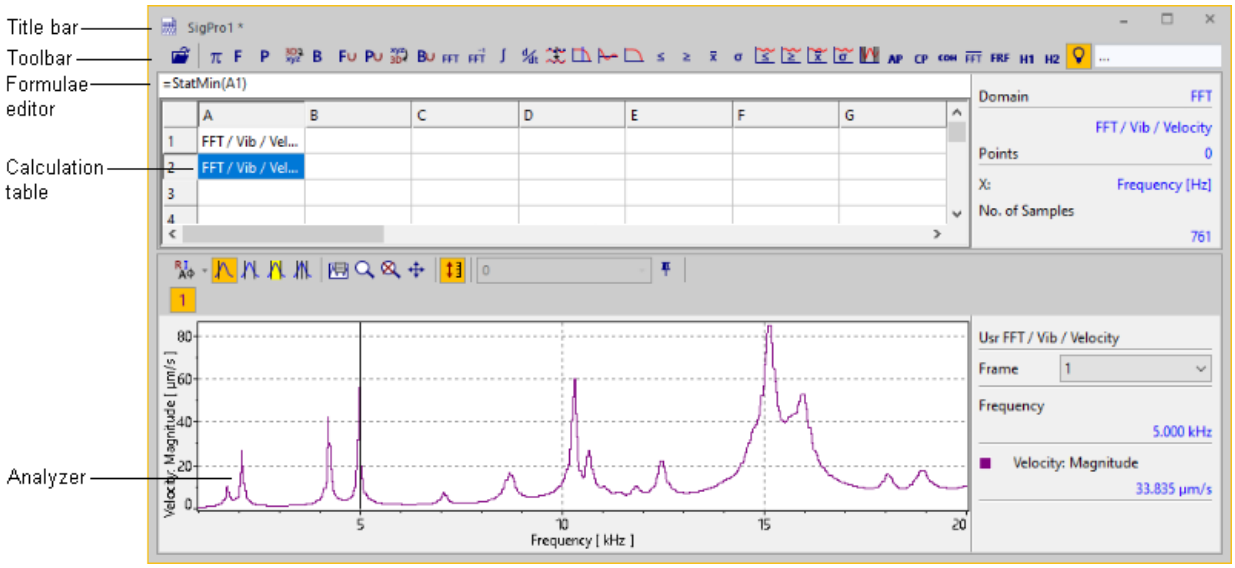


Figure 1.6: Example of recalculating measurement data in the signal processor

Toolbar

In the toolbar you will find the most important arithmetic operations you can execute with the signal processor. The icons are active as soon as the cursor is in the formulae editor.

You have the following options to enter the formulae in the formulae editor:

- Enter the formula using the keyboard.
- Click one of the icons.
- Select the formula in the list on the right of the toolbar.
- Click  and the desired icon to call up the function wizard. See SECTION 3.7.1 on this as well.

Formulae editor

The contents of the selected table cell is shown in the formulae editor. Here, you enter text or formulae to be contained in the selected cell.

Calculation table

You enter the measurement data and the formulae in the cells of the calculation table. By linking the individual cells via the formulae, the software can make various arithmetic operations using the measurement data and then display the results at any point in the computing chain. Apart from that, you can also enter comments into the cells.

ⓘ In the table cells, you can display either the formula or the result. To toggle, select View > Formulae in the menu bar (key combination: Ctrl + #).

For linking individual cells you establish references. A reference designates a cell or a cell range in the table and informs the signal processor where the values or data used in the formula are located. Using references, you can use data or values of a cell in different formulae.

Depending on the operation that you want to execute in the signal processor, you can either use relative or absolute cell links. Relative cell links are references to cells that are given in relative to the position of the formula, absolute references are cell links that always refer to cells at a certain position. If a dollar symbol is prefixed to the letter and/or the number, e.g. A1, then the column and/or cell link is absolute. Relative references are automatically adapted in case of copying, absolute references are not adapted.

Analyzer

The signal processing results are shown as a graph in the analyzer.

1.5.6 Arranging Windows

You can arrange all open windows in the application window using the commands in the Window menu.

New window

To open more presentation windows, select Window > New Window in the menu bar.

Cascade

To arrange all windows/analyzers so that only the title bar and the ordinate name are visible respectively, select Window > Cascade in the menu bar (toolbar: ).

Tile horizontally and vertically

To display all windows/analyzers so that they are completely and maximum size, select Window > Tile Horizontally or Window > Tile Vertically in the menu bar (toolbar:  or ).

Close all

To close all windows/analyzers, select Window > Close All in the menu bar.

Window list

At the bottom of the Window menu, you will find a list of opened windows/ analyzers. To activate a window/analyzer – i.e. move it to the foreground – select it in the list.

Window settings

Window settings which you have made are automatically restored by the software when you newly open the file. For this purpose, the window settings are saved in a database under the complete path name of the file. This database is located on the local hard disk of your PC. Each user who is logged in Windows has their own database.

Change window settings

You can deactivate recovering the window layout and also use the settings for the active window as default for opening other files which have not yet been opened. See SECTION 2.2 on this.

1.5.7 Linking Windows

You can link several analyzers or presentation windows (surface view) to each other so that you display data with the same settings.

While doing so, you can select the following properties:

- General
- Signal
- View (only for surface view)
- Lines (only for analyzers)
- Abscissa (only for analyzers)
- Ordinate (only for analyzers)
- Cursor

Pin

To link several windows together, proceed as follows:

1. Open several windows.
2. Click  in the window in which you want to link properties together.
   The icon is permanently highlighted.
3. Click the arrow of the  icon and select the properties in the pull-down menu which are to be linked.

1.5.8 Zooming in Windows

You can zoom images sections in the presentation window. You will find the appropriate tools in the toolbar of the presentation window.

Zooming in on the object helps you to display the data more precisely. This is particularly advantageous if your object has complicated geometry or your scan points are very close together.

Zoom in

If you would like to take a closer look at a certain image section, you can zoom in on it. To do so, proceed as follows:

1. Click  .

   The mouse pointer becomes a magnifying glass icon  .
2. Holding the left mouse button pressed, draw a rectangle around the area of the image which you want to zoom in on. You can repeat this step several times.

ⓘ You can also zoom by scrolling with the mouse wheel. If you point with the mouse pointer at the area you want to zoom, and then scroll with the mouse wheel, exactly this area is zoomed.

Pan image section

You can pan the zoomed image section. To do so, you have the following options:

- Click in the zoomed view using the middle mouse button (mouse wheel). Hold the mouse button pressed and pan the image section. If you have given the middle button a special function (e.g. double-click), then you will have to proceed as described below.
- Click  . The mouse pointer becomes an arrow cross icon  . Hold the left mouse button pressed and pan the image section.

Zoom out

You can zoom out again. To do so, scroll up with the mouse wheel. To undo all previous zoom actions at once, click  .

End zooming

To leave the zoom function again, click  . The last image section selected remains.

1.5.9 Printing Windows

You can print out presentation windows and signal processors.

ⓘ The title of the window as well as the date and time are automatically printed as a header.

Set up printer

To set the paper size, feed and orientation, select File > Print Setup in the menu bar. The Print Setup dialog appears. Here you can select the printer and set up the page.

Preview

To see a print preview, select File > Print Preview in the menu bar.

Start printing

To print out the active window, proceed as follows:

1. In the menu bar, select File > Print (toolbar: ).

   If you can see two views in the active window (e.g. in the signal processor), then the Select View dialog will appear. Otherwise the Print dialog appears. Then proceed with step 4.
2. Select whether you want to print out the top, the bottom, or both graphics.
3. Click OK.

   The Print dialog appears.
4. Select the printer and set additional print properties.
5. To start printing, click OK in the Print dialog.

   The corresponding window is printed.

1.6 Help Texts and Software Information

You can get help in the software as online help, as context-sensitive help and through brief explanations of commands. You can also display the theory manual and the Basic Engine manual as well as the version of the software and the software options which are installed.

Online help

To open the online help, you have the following options:

- Press the F1 key on the keyboard.
- In the toolbar, click  .
- In the menu bar, select Help > Contents.
- In the menu bar, select Help > Search.
- In the menu bar, select Help > Index.

Context-sensitive help

For dialogs and windows you can get context-sensitive online help. To do so, proceed as follows:

1. Activate the window or dialog you need help with. To do so, click him.
2. Press the F1 key or click in the dialog Help.

   The corresponding page of the online help appears.

ⓘ You can also get context-sensitive help if you click  and then the corresponding window or element.

Command description

You can also get explanations for icons and commands without opening the online help. To do so, point the mouse at the command or the icon and leave the mouse pointer there for a moment. The explanation appears on the left in the status bar. For icons, next to the mouse pointer you will see a brief description.

Theory and Basic Engine

For background information about various features in the software, you can display the theory manual of the PSV software. To do so, select Help > Theory Manual in the menu bar.

For interesting facts from Basic Engine programming, you can display the Basic Engine manual of the PSV software. To do so, select Help > Basic Engine Manual in the menu bar.

Software version and options

To display the version of the software, select Help > About PSV in the menu bar. The About Polytec Scanning Vibrometer dialog appears.



Figure 1.7: About Polytec Scanning Vibrometer dialog

To display a system overview, click System Info. The System Information dialog appears. Here you will find all information on the operating system (version, serial number, etc.) as well as the resources (memory, available hard disk space, etc.) of the Data Management System.

To display the software options installed, click Options.

2 Setting Up Presentation

For the Presentation you must set up the software appropriately.

ⓘ In almost all views and modes, you can use the right mouse button to call up via the context menu several commands and functions that are often needed at this point.

ⓘ If you are in Acquisition mode, select View > Presentation in the menu bar to change to Presentation mode (toolbar: ) or start the software in presentation mode.

Options

To set up the software, proceed as follows:

1. In the menu bar, select Setup > Preferences.

   The Preferences dialog appears.
2. The dialog has several pages, each with a group of parameters. To display a certain page, click the name of the group.
3. Set the parameters on all pages.
4. When you have set all parameters, click OK.

You will find information on this in the following sections:

dB Reference

On the dB Reference page you can define dB = 0, either generally or for the displayed data set. See SECTION 2.1 on this.

Display

On the Display page, select the behavior of the window settings, the appearance of an animated spectrum and the samples in the Time domain. See SECTION 2.2 on this.

Messages

On the Messages page, you select which system messages you would like to have displayed. See SECTION 2.3 on this.

2.1 Setup: dB Reference

On the dB Reference page you can define dB = 0, either generally or for the displayed data set.

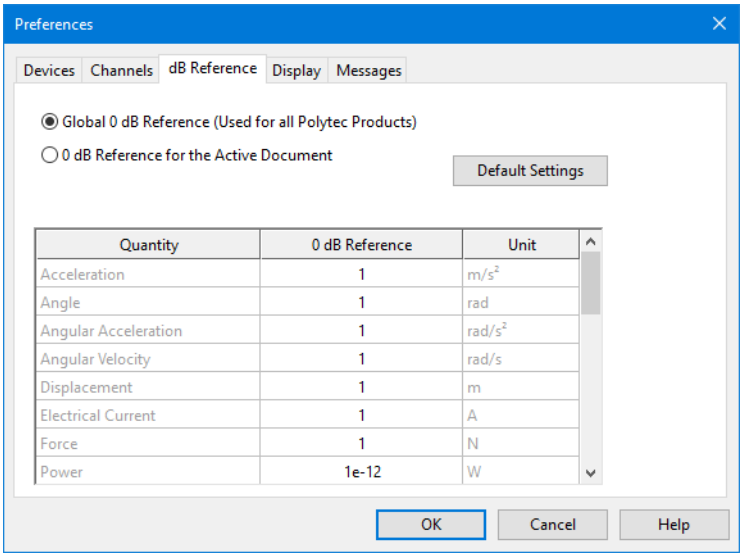


Figure 2.1: dB Reference page

Global 0dB Reference (Used for all Polytec Products): Click here if the settings should be valid for every Polytec software installed on the PC.

0dB Reference for the Active Document: Click here if the settings should only be valid for the file that is actually open.

Settings

Default settings: If you want to recover the original settings for the 0 dB Reference, click here.

Quantity: All available physical quantities are shown here.

dB Reference: Enter the factor for the respective unit here.

Unit: The unit for the respective physical quantity is shown here.

Example of 0 dB Reference

The 0 dB reference for displacement is 2 mm. Enter 0.002 or 2e-3 in the 0 dB Reference column.

Quantity	0 dB Reference	Unit
Displacement	0.002	m

Figure 2.2: Example of 0 dB Reference

2.2 Setup: Display

On the Display page, select the behavior of the window settings, the appearance of an animated spectrum and the samples in the Time domain. You will find how to work with windows in principle in SECTION 1.5.6. You will find how to animate spectrums in SECTION 4.4.1.

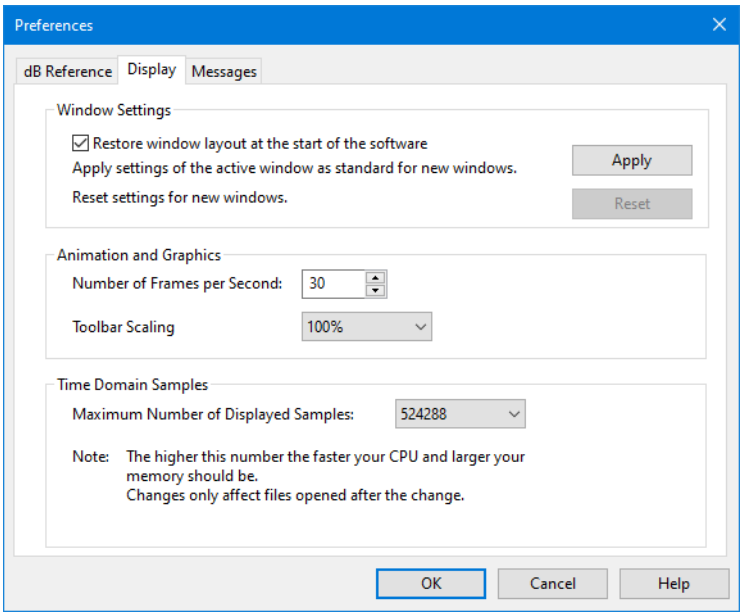


Figure 2.3: Display page

Window settings

Restore Window Layout at the Start of the Software: Tick the check box so that next time the software starts, the window layout is restored as it was when the software was last closed. If you deactivate the box, the software will open with an empty application window.

ⓘ The software needs a few minutes to open files in which the window settings are to be restored. You cannot abort the process.

Apply: Click here if you want to apply the window settings of the active window (only presentation window) to other new measured files of the same type. To activate the window, click it. The window settings will be applied to every file which is opened for the first time.

Reset: Click here if you want to reset the window settings for newly opened files to default values.

Animation and graphics

Number of Frames per Second: Enter the frame rate for animations, i.e. the number of frames per second that will be shown during animation. The minimum setting is 10, the default setting is 30. The frame rate is also used for stored animations.

Toolbar Scaling: Select the scaling for the display of the toolbars in the software. The selection of the scaling affects the display in Presentation mode, and at the next start also in Acquisition mode.

Time domain samples

Maximum Number of Displayed Samples: Enter an upper limit for the number of samples to be shown simultaneously in the Time domain.

2.3 Setup: Messages

On the Messages page, you select which system messages you would like to have displayed.

ⓘ Message windows which contain queries or which advise you of faulty input will appear in any case.

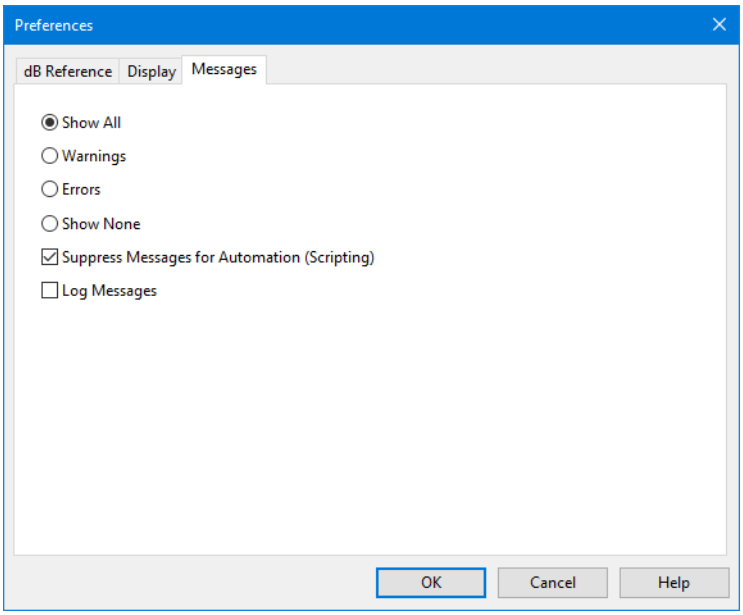


Figure 2.4: Messages page

Settings

Show All: All messages are displayed (notes, errors and warnings).

Warnings: Only warnings and errors are displayed.

Errors: Only error messages are displayed.

Show None: No system message is displayed.

Suppress Messages for Automation (Scripting): Tick the check box if you want to suppress messages during an automatic activity (Scripting). Otherwise it is possible that the activity of macros is disrupted by these messages.

Log Messages: Tick the check box if you want to log all messages. You will find the log file in the Event Viewer (refer to the following section).

You can save the log file in the following file formats:

- Event Files (*.evtx)
- Xml (Xml File) (*.xml)
- Text (Tab delimited) (*.txt)
- CSV (Comma Separated) (*.csv)

Log file (Windows)

To open the log file under Windows, proceed as follows:

1. Press the Windows key on your keyboard.
2. Enter Event Viewer in the search field.
3. Click Event Viewer in the files shown.

   The Event Viewer window appears.
4. Open the Application and Services Logs folder.
5. Click the Polytec directory.

   The logged messages will be displayed.

To save the log file, proceed as follows:

6. Click the Polytec directory with the right mouse button and select Save All Events As in the context menu.

   The Save As dialog appears.
7. Navigate to the saving location, select Save as type and enter the file name.
8. Click Save.

3 Displaying Data

You can set up the properties of the data displayed. See SECTION 3.1 on this.

You work with the following windows in the software to display data:

- In acquisition mode: a video window, any number of analyzers and signal processors
- In presentation mode: any number of presentation windows, analyzers and signal processors

You will find a description of the analyzer in SECTION 1.5.3.

Analyzers show measured raw data and evaluated data in xy-diagrams. You can work in analyzers as described in SECTION 3.2.

To evaluate a scan, open the presentation window and set up the window and the view of the object. Read SECTION 3.3 on this.

You select the data set to be displayed in the analyzer as described in SECTION 3.4.

You can define clipping planes in 3D view to control complex geometries. Read SECTION 3.5 on this.

If you are working with 3D geometries, you can merge several individual scans to a combined file (stitching). Read SECTION 3.6 on this.

You can use your measurement data as the basis for further calculations. Read SECTION 3.7 on this.

If you want to add user defined data sets to geometries, you can generate a new scan file for this purpose. Read SECTION 3.8 on this.

3.1 Setting Up Display Properties

In the Display Properties dialog, you can make the following settings:

- Set up title and color for background and readout for the object - See SECTION 3.1.1 on this.
- Set up video image, grid and indices for the view style of the measurement data - See SECTION 3.1.2 on this.
- Set up sight, illumination model, projection, and clipping planes as well as the scan range for 3D view - See SECTION 3.1.3 on this.

In presentation mode

If you are in presentation mode, open the dialog as follows:

- Double-click the object in the presentation window and display the desired page.
- Click the presentation window with the right mouse button and select General Properties, Data Properties or 3D View Properties in the context menu.
- Activate the presentation window (to do so, click it) and select Presentation > Properties in the menu bar. Display the desired page.

To set up the 3D view in presentation mode, you have to change to the 3D view. To do so, click 

in the toolbar of the presentation window.

ⓘ In presentation mode you can only switch to 3D view if you have selected the Object, Single Scan Point or Average Spectrum view.

3.1.1 Setting Up the Object

ⓘ The content of these pages differs depending on whether you are in acquisition or presentation mode. Depending on the type of measurement and the geometry, not all settings options are shown in the dialog.

If you are in presentation mode, open the dialog as follows:

- Double-click the object in the presentation window and display the General page.
- Click the presentation window with the right mouse button and select General Properties in the context menu.
- Activate the presentation window (to do so, click it) and select Presentation > Properties. Display the General page.

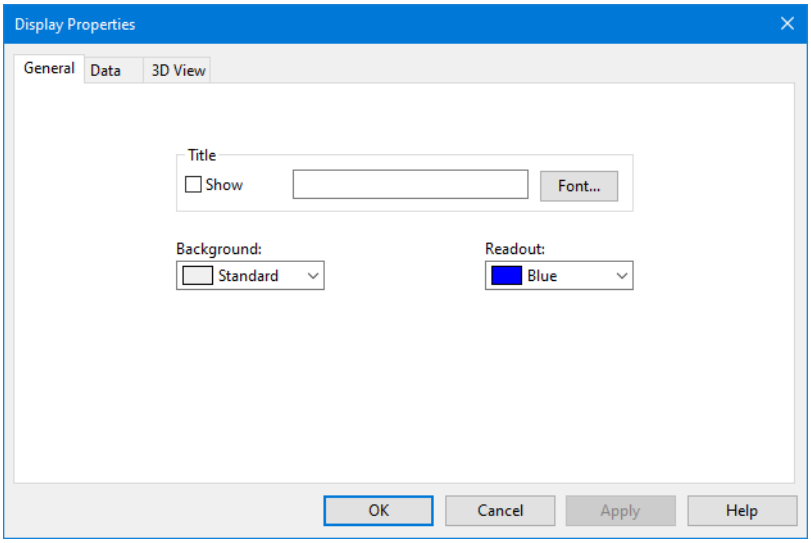


Figure 3.1: General page in presentation mode

ⓘ Changing the settings on this page is effective immediately.

Title

Set up the object title.

Show: If you would like to show a title over the object, tick the check box. You enter the title on the right.

Font: Click here to format the title. The Font dialog appears. Set up the font, font style, size, and if applicable effects, and click OK.

Colors

Select the background color of the object and the color of the readout.

Background: Select the background color of the presentation window.

Readout: Select which color is to be used to display the readout beside the object.

3.1.2 Setting up View Style of Measurement Data

ⓘ The content of these pages differs depending on whether you are in acquisition or presentation mode. Depending on the type of measurement and the geometry, not all settings options are shown in the dialog.

If you are in presentation mode, open the dialog as follows:

- Double-click the object in the presentation window and display the Data page.
- Click the presentation window with the right mouse button and select Data Properties in the context menu.
- Activate the presentation window (to do so, click it) and select Presentation > Properties. Display the Data page.

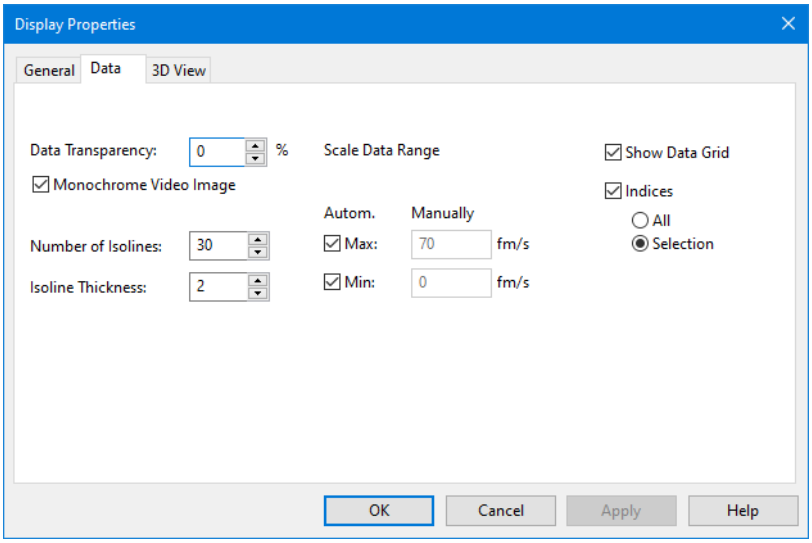


Figure 3.2: Data page in presentation mode

ⓘ Changing the settings of the data view and the isolines on this page is effective immediately. To scale the z-range, you have to click Apply.

View style of measurement data

Data Transparency: Enter how transparently the superimposed data is to be shown in the presentation window.

ⓘ In the Wireframe view style this selection is not active.

Monochrome Video Image: If you want to change the video image from color display to black&white, tick the check box.

Show Data Grid: If you want to show the grid on which the scan points are located, tick the chek box.

ⓘ In the Wireframe view style this selection is not active.

Indices: If you want to show the index of scan points, tick the check box.

You have the following options:

- All: The indices of all scan points are shown.
- Selection: The indices of selected scan points are shown.

Set up isolines

Number of Isolines: Select how close the isolines are to each other.

Isoline Thickness: Select the width of the isolines.

Scale data range

Left, Right: If you have selected the Real & Imag. display type for example, select which part you want to scale. For setting the display type, see SECTION 3.4.

Autom.: The software can individually autoscale the maximum and minimum of the data range. To do so, tick the appropriate Max and Min box.

Manually: You can manually scale the data by making entries in the Max and Min fields. To do so, you have to deactivate the corresponding Autom. box. Enter the range which the data is to be displayed on.

3.1.3 Setting Up 3D View

ⓘ The content of these pages differs depending on whether you are in acquisition or presentation mode. Depending on the type of measurement and the geometry, not all settings options are shown in the dialog.

If you are in presentation mode (3D view), open the dialog as follows:

- Double-click the object in the presentation window and display the 3D View page.
- Click the presentation window with the right mouse button and select 3D View Properties in the context menu.
- Activate the presentation window (to do so, click it) and select Presentation > Properties in the menu bar. Display the 3D View page.

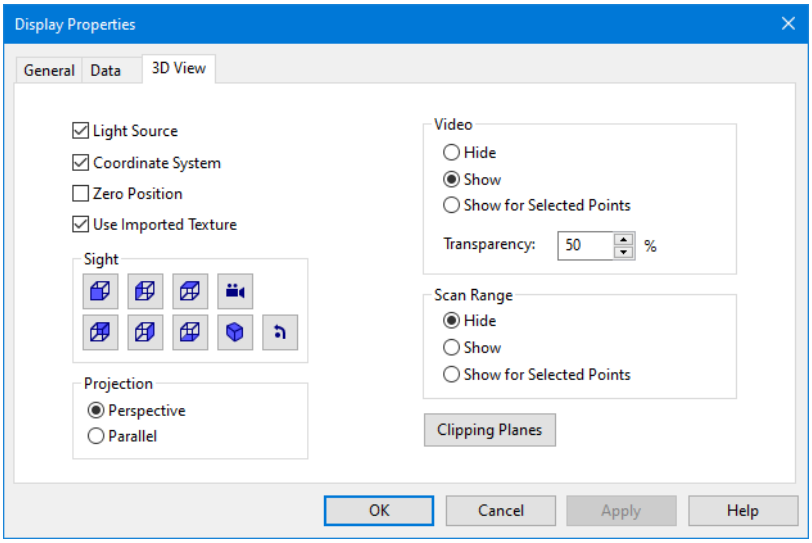


Figure 3.3: 3D View page, e.g. for the PSV-3D in presentation mode

ⓘ Changing the settings on this page is effective immediately.

Light Source: Tick the check box to switch on the illumination model. It simulates illuminating the object from the direction of the observer.

ⓘ Depending on the shape of the object, the display of light and shadow may have a detrimental effect on the presentation quality. In this case, deactivate the check box to switch off the illumination model.

Coordinate System: Tick the check box to display a coordinate system at the top left next to the object.

Zero Position: Tick the check box to show the object in its original position in addition to viewing the measurement results.

Use Imported Video Image as Texture: Tick the check box to show the imported video image of the object in its original view. The function is only available with the GeoPro option and for geometry import from suitable files (e.g. OBJ files).

Sight

The icons in the Sight field show the line of sight of the object. By clicking the icons, you can move the object more precisely than by dragging it with the mouse.

 View of the object from the front

 View of the object from behind

 View of the object from the left

 View of the object from the right

 View of the object from above

 View of the object from below

 View of the object at an angle from front, left and above

 Sight of the video camera (only for 3D geometries; for this purpose the same limitations apply as with the display of the video image, see above under Video)

ⓘ The sight of the video camera is not available for combined files.

 Rotating the object around the line of sight

You can show/hide a coordinate system which shows you the resulting orientation of the object (see above, Axes).

ⓘ You can also use the mouse to set the line of sight directly. To select the Rotate mode, click  . Click on the object and drag it in the required direction with the mouse. Hold the Control key pressed and the object only turns around the line of sight.

Projection Select the type of projection of the object.

Perspective: The object is shown in perspective, i.e. parts at the front of the picture are shown larger, parts towards the back of the picture are shown smaller.

Parallel: All parts of the object are shown the same size, regardless of their distance from the observer.

ⓘ Parallel projection makes it easier to distinguish between in-plane and out-of-plane vibrations. Particularly with out-of-plane vibrations, the view in perspective in top view can give the impression that the object is expanding and contracting again.

Video For 3D geometries, the video image is only shown if the following is applicable:

- The 2D alignment is good quality, i.e. the laser beam hits the precise position on the video image.
- 3D coordinates are available for the scan points.
- You have defined at least four scan points which cover a larger area of the video image.

Select one of the following options to display the video image.

Hide: Hide the video image outside the defined scan points.

Display: To video image will be fully displayed.

Show for Selected Points: The video images will only be fully displayed for the video planes in which scan points have been selected (recommended for combined files).

ⓘ If you have not selected any scan points the video image will be hidden outside the defined scan points.

Transparency: Enter in % how transparent the video image is to be displayed.

Scan Range

You can display reference lines which show you the position of the scanning head(s) in front of the object. In this way you can see whether the 3D alignment has provided a useful result.

Hide: The reference lines for the scanning head(s) are hidden.

Show: The reference lines for the scanning head(s) are shown for all scan points.

Show for Selected Points: The reference lines for the scanning head(s) are shown for selected scan points. You can select scan points as described in SECTION 4.5.1.

ⓘ You can only show the scan range for measurements with 3D geometries.

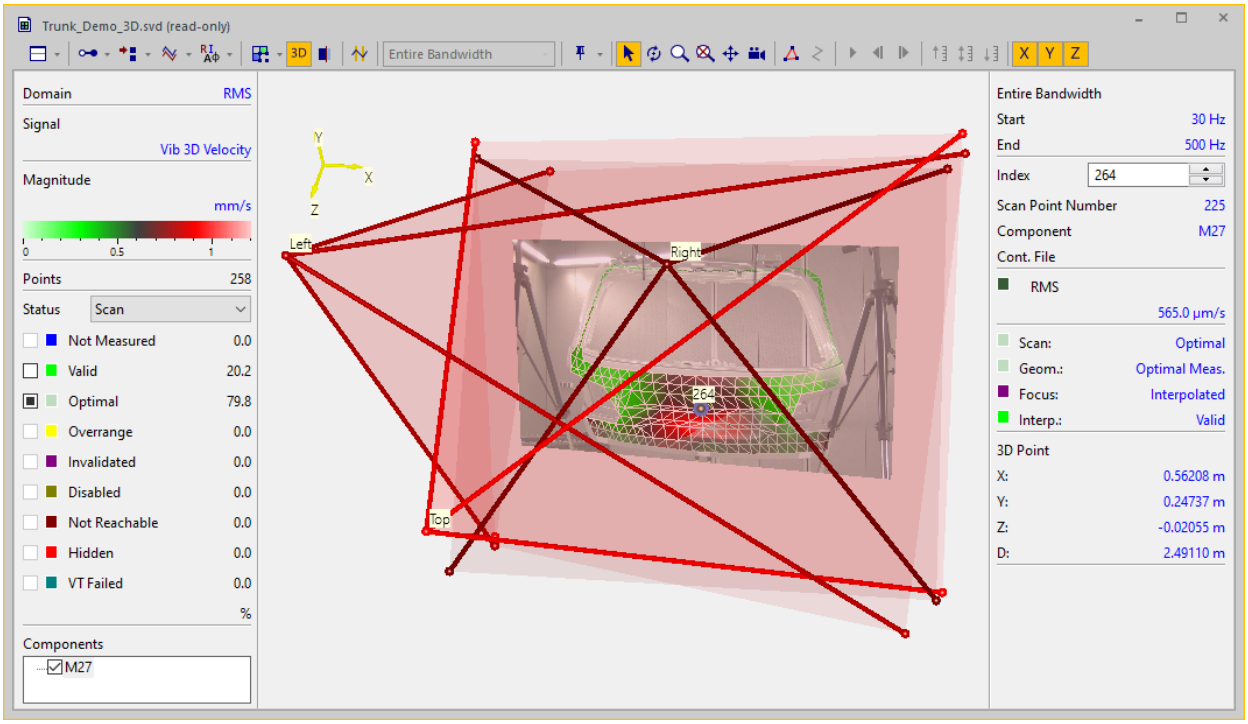


Figure 3.4: View of the scan range of the three scanning heads (PSV-3D)

Clipping planes

In 3D view, you can define clipping planes to check your geometry. Thereby, it does not matter whether you are in the APS mode or not. The 3D view is only available if you have defined scan points. To define clipping planes, click Clipping Planes. You will find a description of the dialog in SECTION 3.5.

3.2 Working in Analyzers

Analyzers show measured raw data and evaluated data in xy-diagrams. You will find a description of the analyzer in SECTION 1.5.3.

You select the data set to be displayed in the analyzer as described in SECTION 3.4.

Analyzers offer other user-friendly functions to present data quickly and sensibly:

- You can set cursors in the analyzer to read individual data points. See SECTION 3.2.1 on this.
- You can link several analyzers with each other to ensure you are showing data with the same settings. See SECTION 3.2.2 on this.
- You can zoom in on sections of the diagrams (zooming). See SECTION 3.2.3 on this.
- You can scale manually or autoscale, i.e. the displayed ordinate is selected according to the y-value range of the data. See SECTION 3.2.4 on this.
- You can display defined frequency bands in the analyzer. To do so, click  . Read SECTION 4.2.1 on how to define frequency bands.
- You can listen to the measured signal in the analyzer. To do so, you need the Audio option. To listen to the signal via the PC loudspeaker, click  in the analyzer. This function is also available to you in the signal processor. See SECTION 3.7 on this.
- If you have made a measurement with the PSV-3D, then you can individually display the vibrations of the object in x-, y- and z-direction in the analyzer. See SECTION 3.2.5 on this.

You will find more functions in the Analyzer Properties dialog.

- You can scale graphs, and set line style and color of the graphs. See SECTION 3.2.6 on this.
- You set up the presentation of graphs in the Analyzer Properties dialog. See SECTION 3.2.7 on this.

The analyzer simultaneously shows up to 0.5 Megasamples for time series as standard. If the measurement contains more data, then this is reloaded during scrolling. If you want to see more time values at once, you can increase the maximum number of time values shown simultaneously as described in SECTION 4.4.2.

ⓘ You can increase the maximum number of time values shown at the same time only in presentation mode. In acquisition mode, the analyzer always shows only a maximum of 0.5 Megasamples.

Features

In the Nyquist display type you can not set any cursors.

You will find details on how to animate time data in SECTION 4.4.2.

3.2.1 Setting Cursors and Reading Data

You can set cursors in the analyzer to read individual data points. Cursors are represented by thin vertical lines.

Single cursor

To set a single cursor, proceed as follows:

1. In the menu bar, select Analyzer > Cursor (toolbar of the analyzer: ).
2. Click in the diagram at the desired position.

   The cursor appears as a thin vertical line.
3. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursor subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout is simultaneously updated.

Read data

4. You can see the cursor's coordinates in the readout. If the readout is not visible on the right in the analyzer, select Analyzer > Readout in the menu bar to display it.

ⓘ If you have selected two data sets to be displayed, then the cursor is visible in both diagrams and the data from both diagrams is shown in the readout.

Cursor properties

In the cursor properties, you can set what is to be shown in the readout or in the diagram. To display the cursor properties, continue as follows:

5. Select Analyzer > Cursor in the menu bar (toolbar of the analyzer: ).

   The Cursor Properties dialog appears.
6. Select the required additional properties and click OK.

   The data you have selected in the cursor properties will appear in the readout or in the diagram.

Cursor Properties

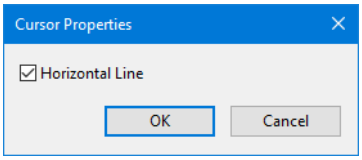


Figure 3.5: Cursor Properties dialog

Horizontal Line: Tick the check box if a horizontal line is to be shown respectively at the cursor position in the diagram.

Differential cursor

You can set two cursors and read the difference between the coordinates. To do so, proceed as follows:

1. Select Analyzer > Differential Cursor in the menu bar (toolbar of the analyzer: ).
2. In the diagram, point at the first desired point.
3. Press the mouse button.

   The mouse pointer becomes a double arrow ↔.
4. Drag the mouse to the left or right to the second point.

   The difference of the coordinates is also displayed in the readout.
5. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursors subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout is simultaneously updated.
6. To display the cursor properties, select Analyzer > Cursor Properties in the menu bar (toolbar of the analyzer: ).

   The Differential Cursor Properties dialog appears.
7. Select the required additional properties and click OK.

   The data you have selected in the cursor properties will appear in the readout or in the diagram.

Differential Cursor Properties

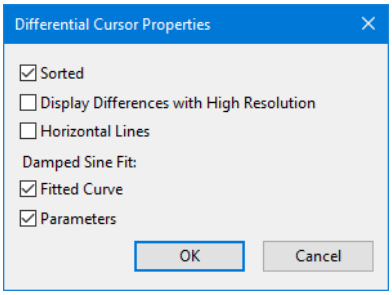


Figure 3.6: Differential Cursor Properties dialog

Sorted: Tick the check box if the cursor on the left is always to be named 1. The cursor on the right is then named 2. If you move a cursor so that the order changes, the numbers also change. If the check box is not marked, the cursors keep their numbers from the readout, independently of their position in the diagram.

Display Differences with High Resolution: Tick the check box to show the difference more clearly for values that are close to each other.

Horizontal Lines: Tick the check box if a horizontal line is to be shown respectively at the cursor positions in the diagram.

Damped Sine Fit: This property is only available for time signals. Tick the Fitted Curve check box to allow the adaptation of a damped sine fit to the signal being displayed between the cursor lines. Tick the Parameters check box to allow the adapted frequency f and the damping coefficient δ being displayed in the readout.

Band cursor

You can set a band cursor and can read various data about the marked area, e.g. the RMS value, the minimum and maximum, the mean value as well as the standard deviation. Apart from that, the software can fit the results to the frequency response of a one-dimensional oscillator and calculate the position of the peak as well as the attenuation. To do so, proceed as follows:

1. Select Analyzer > Band Cursor in the menu bar (toolbar of the analyzer: ).
2. In the diagram, point at the first desired point.
3. Press the mouse button.

   The mouse pointer becomes a double arrow ↔.
4. Drag the mouse to the left or right to the second point.

   The data you have selected in the cursor properties will appear in the readout or in the diagram.
5. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursors subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout or in the diagram is simultaneously updated.
6. To display the cursor properties, select Analyzer > Cursor Properties in the menu bar (toolbar of the analyzer: ).

   The Band Cursor Properties dialog appears.
7. Select the required additional properties and click OK.

   The data you have selected in the cursor properties will appear in the readout or in the diagram.

Band Cursor Properties

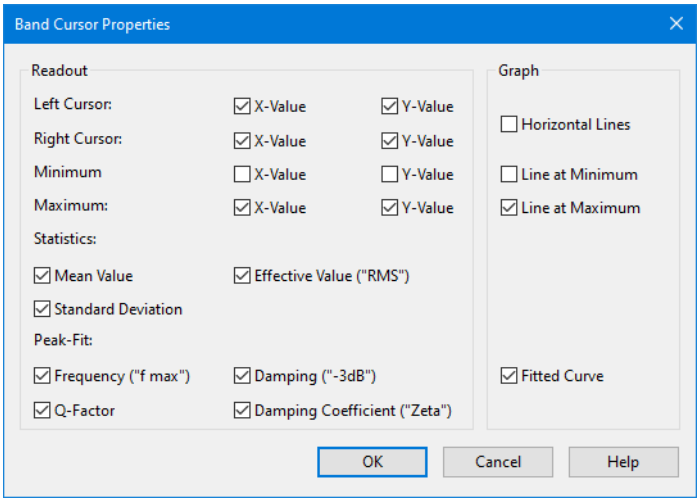


Figure 3.7: Band Cursor Properties dialog

Left Cursor, Right Cursor, Minimum, Maximum: Tick the required check boxes, so that the corresponding ordinate values are shown. In the Graph field on the right, you can select the corresponding line display in the diagram.

Statistics: Tick the desired check boxes for Mean Value, Effective Value ("RMS"), and Standard Deviation to see the corresponding ordinate values.

Peak-Fit: The Peak-Fit allows the frequency response function of a one-dimensional, harmonic oscillator being adapted to a real or complex spectrum. The function uses the least squares method for the adaptation. For a Peak-Fit, you can display the following values.

ⓘ Peak-Fit is only available for spectra.

Frequency ("fmax"): Tick the check box to display the frequency of the maximum of the fitted curve.

Damping ("–3 dB"): Tick the check box to show the damping. The damping corresponds to the full width of the fitted curve at a height of 3 dB below the maximum amplitude.

Q-Factor: Tick the check box to show the Q-Factor. The Q-factor is defined as follows:

$$\text{Q-factor} = f_{max}/\text{Damping} \qquad \text{Equation 3.1}$$

Damping Coefficient ("Zeta"): Tick the check box to show the damping coefficient. The damping coefficient is defined as follows:

$$\text{Zeta} = \text{Damping}/(2 \cdot f_{max}) \qquad \text{Equation 3.2}$$

ⓘ Calculating the Damping Coefficient ("Zeta") is only valid for small dampings.

Fitted Curve: Tick the check box to display the fitted curve for the selected band cursor in the diagram.

Harmonic cursor

You can set a harmonic cursor which allows you to display up to 12 cursor lines of the 2nd, 3rd, 4th etc. order of the fundamental frequency. To do so, proceed as follows:

1. Select Analyzer > Harmonic Cursor in the menu bar (toolbar of the analyzer: ).
2. Click the required frequency in the diagram.

   The displays you selected in the cursor properties will appear in the diagram. The data at the cursor position you selected in the analyzer will be shown in the readout. Additionally for simple spectra, the amplitude ratio (THD = Total Harmonic Distortion) will be displayed in [%] or [dB] respectively. THD is calculated as follows:

$$THD_{\%audio} = \frac{\sqrt{A_2^2 + A_3^2 + A_4^2 + \ldots + A_n^2}}{A_1}$$

ⓘ No THD is displayed for power spectra, PSD, and spectra from multiple channels.

3. Once you move the mouse pointer onto the cursor, it becomes a double arrow ↔. In this way, you can move the cursors subsequently. To do so, click the cursor and drag it to the right or left.

   The data in the readout or in the diagram is simultaneously updated.

4. To display the cursor properties, select Analyzer > Cursor Properties in the menu bar (toolbar of the analyzer: ).

   The Harmonic Cursor Properties dialog appears.

5. Select the required additional properties and click OK.

   The data you have selected in the cursor properties will appear in the readout or in the diagram.

Harmonic Cursor Properties

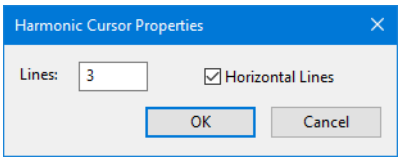


Figure 3.8: Harmonic Cursor Properties dialog

Lines: Enter the required number of lines that should be displayed.

Horizontal Lines: Tick the check box if a horizontal line is to be shown respectively at the cursor positions in the diagram.

Move cursor

You can move the cursor to the right and left. To do so, proceed as follows:

1. Point at the cursor.

   The mouse pointer becomes a double arrow ↔.

2. Press the mouse button and drag to the left or right to the required point.

   The data in the readout or in the diagram is simultaneously updated.

ⓘ You can also move a single cursor with the → and ← keys.

Delete cursor

To delete cursors which has been set, click another cursor icon ( ,  ,  or ).

Copy cursor values

You can use the clipboard to copy the currently set cursor values into a text file. To do so, proceed as follows:

1. Set the required single cursor, differential cursor, band cursor or harmonic cursor.
2. Click the diagram with the right mouse button and select Copy Cursor Values in the context menu.
3. You can now open a text editor, for example, and paste the cursor values from the clipboard there.

3.2.2 Linking Windows

You can link several analyzers with each other to ensure you are showing data with the same settings.

While doing so, you can select the following properties:

- General
- Lines
- Abscissa
- Ordinate
- Cursor

Pin

To link several windows together, proceed as follows:

1. Open several windows.
2. Click  in the window in which you want to link properties together.

   The icon is permanently highlighted.

3. Click the arrow of the  icon and select the properties in the pull-down menu which are to be linked.

3.2.3 Zooming in Analyzers

You can zoom in on sections of the diagrams (zooming). You will find the appropriate tools in the toolbar of the analyzer.

Zoom abscissa or ordinate

To zoom the abscissa or ordinate individually, proceed as follows:

1. Point at an edge point of the required section on the abscissa or ordinate.

   The mouse pointer becomes a magnifying glass icon  .
2. Press the mouse button and drag horizontally or vertically to the other edge point.

Scroll the abscissa

The scroll bar below the diagram shows the position of the zoomed section. You can scroll along the abscissa with the set zoom factor. To do so, move the bar in the scroll bar to the right or to the left. You can also click the arrows to the left and right of the scroll bar.

Zoom abscissa and ordinate

To zoom both abscissa and ordinate simultaneously, proceed as follows:

1. In the menu bar, select Analyzer > Zoom In (toolbar of the analyzer: ).

   The mouse pointer becomes a magnifying glass icon  .
2. In the diagram, point at an edge point of the required section.
3. Press the mouse button and drag diagonally to the other corner point.

Pan the section

You can pan the zoomed diagram section. To do so, you have the following options:

- Click in the zoomed diagram with the middle mouse button (mouse wheel), hold the mouse button pressed and pan the diagram section. If you have assigned a special function to the middle button (e.g. double-click), then you need to proceed as described below.
- In the menu bar, select Analyzer > Pan (toolbar of the analyzer: ). You can now pan the diagram section while pressing the left mouse button.

Zoom out

To undo the last zoom or autoscale, select Analyzer > Zoom Out in the menu bar (toolbar of the analyzer: ).

To undo all zoom actions, keep the Shift key pressed and select Analyzer > Zoom Out (toolbar of the analyzer: with pressed Shift key: ).

3.2.4 Scaling Axes Ranges

Auto scale ordinate

If you have started a measurement or are displaying a different data set, then the ordinate range is often unsuitable. You can then autoscale, i.e. the software selects the ordinate displayed according to the y-range of data. To switch on autoscale, select Analyzer > Autoscale in the menu bar (toolbar of the analyzer: ). To switch off autoscale again, repeat this action.

Scale manually

In the Analyzer Properties dialog, you can scale manually and define additional scaling properties.

You have the following options to open the dialog:

- Point with the mouse pointer at an axis (the mouse pointer becomes a magnifying glass ) and double-click the axis.
- Click the diagram with the right mouse button and select Range Properties in the context menu.
- Activate the analyzer (to do so, click it), select Analyzer > Properties in the menu bar and display the Ranges page.

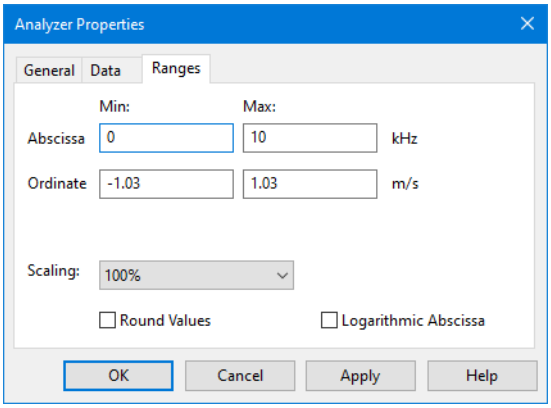


Figure 3.9: Ranges page

Abscissa Min, Max: Enter the range for the abscissa that you want to display.

Ordinate Min, Max: Enter the range for the ordinate that you want to display. If you want to display two diagrams (e.g. Mag.&Phase), you can respectively define the ordinate.

ⓘ The settings for abscissa and ordinate are only displayed temporarily. Any zooming or auto scaling will overwrite the values entered.

Scale: Select the scaling of the ordinate and abscissa from the list.

Round Values: Tick the check box if the values for abscissa and ordinate are to be displayed as round numbers. The software will then round the values in Min and Max appropriately. Round Values is also active when zooming. The zoomed section is then usually slightly larger than selected.

Logarithmic Abscissa: This check box only appears if the active analyzer is showing a spectrum. Tick the check box if you want to scale the abscissa logarithmically. Deactivate the check box for linear scaling of the abscissa.

3.2.5 Displaying Vibrational Direction (only PSV-3D)

If you have made a measurement with the PSV-3D, then you can individually display the vibrations of the object in x-, y- and z-direction in the analyzer. You can also combine displaying several vibrational directions together. To do so, select the Vib 3D channel.

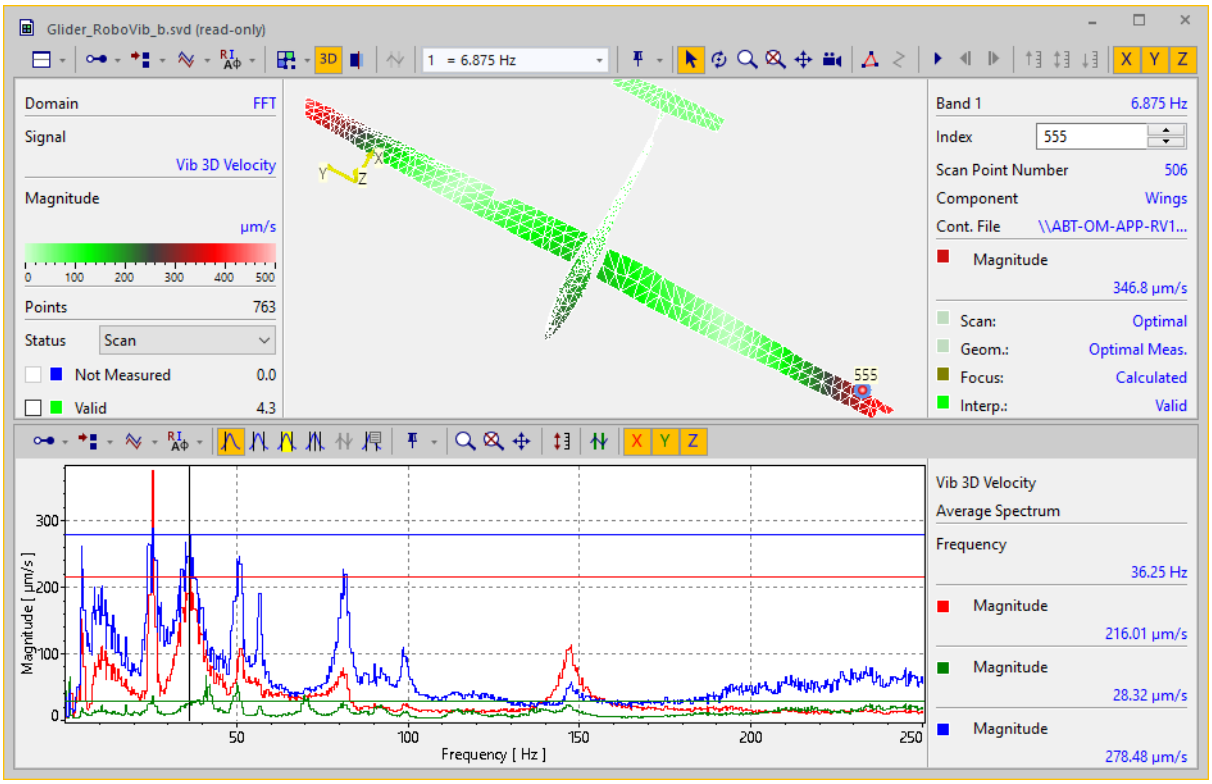


Figure 3.10: Displaying the vibrational direction in the analyzer

To show/hide the vibrations in the various directions, click the X, Y and Z icons. If the respective icon is activated, then the vibration is being shown in the respective direction. If the icon is deactivated, then the vibration is not being shown in the respective direction.

If you want to animate vibrations and display the directions in surface view, see SECTION 4.4.1.

3.2.6 Setting up Presentation of Graphs

You set up the presentation of graphs in the Analyzer Properties dialog.

You have the following options to open the dialog:

- Double-click the diagram area.
- Click the diagram with the right mouse button and select Data Properties in the context menu.
- Activate the analyzer (to do so, click it), select Analyzer > Properties in the menu bar and display the Data page.

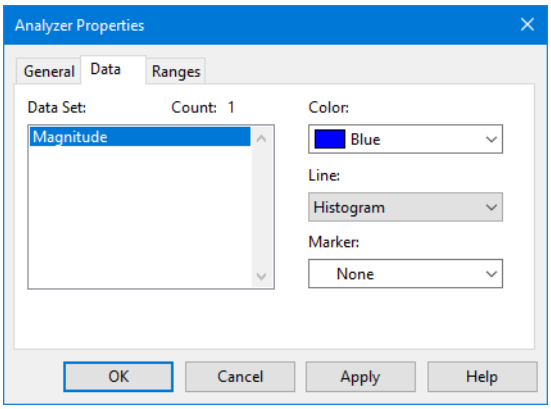


Figure 3.11: Data page

Data Set: Mark the data set which you want to set up. If the analyzer is showing several diagrams – e.g. magnitude and phase – select the data set which you want to set the layout for. To change the properties of multiple data sets at the same time, select the data sets before using the Control key.

ⓘ If you want to display the spectrum of different frames, then you can select the respective frame here as well (e.g. 1, 2, 3 etc.) and can set up the way it is displayed.

Color: Select the color of the graph.

ⓘ Changing the color affects the data set displayed in all display types.

Line, Marker: Select the view style of the graph and, if applicable, the marker. In the Marker, Histogram, Thin Bars and Thick Bars view styles, only measured y-values are shown. In the other view styles, the software interpolates between the measured values.

3.2.7 Setting up Diagrams

Set up the look of the diagram in the Analyzer Properties dialog. The current settings in the dialog also apply for all analyzers which you open afterwards.

You have the following options to open the dialog:

- Double-click an axis name.
- Click the diagram with the right mouse button and select General Properties in the context menu.
- Activate the analyzer (to do so, click it), select Analyzer > Properties in the menu bar and display the General page.

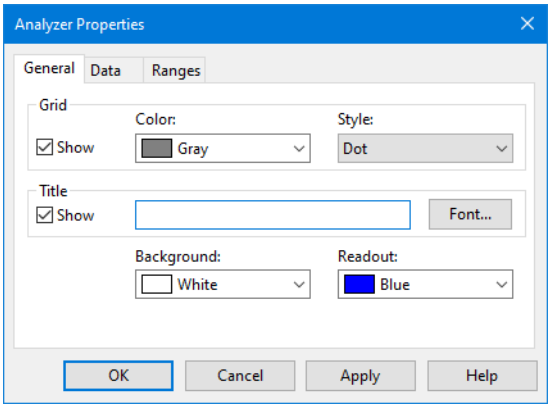


Figure 3.12: General page

Grid

Display: Tick the check box to display a grid in the diagram.

Color, Style: Select the color and the line style for the grid.

Title

Display: Mark the check box if you want to display a title above the diagram. You enter the title on the right.

Font: Click here to format the title. The Font dialog appears. Set up the Font, Font style, Size, and if applicable Effects and click OK.

Colors

Background: Select the background color of the diagram.

Readout: Select which color is to be used to display the readout beside the diagram.

3.3 Working in Presentation Windows

You analyze scans in presentation windows. You will find a description of the presentation window in SECTION 1.5.2.

Depending on which functions you are working with in the presentation window, you display it in different views. The following functions are available to you:

- Depending on which data and functions you are working with in the presentation window, you can display them in different views. See SECTION 3.3.1 on this.
- In the presentation window you see the data superimposed on the video image. You can present the data in different ways. See SECTION 3.3.2 on this.
- You will find more presentation options in the Display Properties dialog. You can scale there and set up the appearance of different views. See SECTION 3.1 on this.

The presentation window also offers user-friendly functions to present data quickly and usefully:

- You can zoom images sections in the presentation window. Zooming in helps you to display the data more precisely. See SECTION 1.5.8 on this.
- If you would like to read data at single scan points, you can set a cursor on the object. You see the cursor as a blue marker. See SECTION 3.3.3 on this.
- In the presentation window, various data is displayed in the readout next to the object. See SECTION 3.3.4 on this.
- You can choose between different color palettes for color-coding in the presentation window. See SECTION 3.3.5 on this.

You can select the data set that is displayed in the presentation window. Read SECTION 3.4 on this.

Apart from the display functions, in the presentation window you also have possibilities to evaluate the measurement data. Read CHAPTER 4 on this.

3.3.1 Setting Up the View of the Presentation Window

Depending on which data and functions you are working with in the presentation window, you display them in different views:

First of all, select which data you want to display:

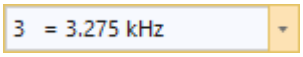
- Band data
- Point data

Then select the view:

- Object
- Single Scan Point
- Average Spectrum
- Profile

ⓘ For animations, you can display single vibrational directions (x, y, z) and scale the degree of deflection. See SECTION 4.4 on this.

For selecting the view, proceed as follows:

1. Select Presentation > View in the menu bar and select Band Data or Point Data in the pull-down menu (toolbar of the presentation window: ).
2. If you select Band Data, you need to select the required frequency band in the 

list.
3. Select Presentation > View and select the required view in the pull-down menu (toolbar of the presentation window: ).

In the Object view, you can see the data superimposed on the video image. See SECTION 3.3.2 on this as well. In the other three views you see the presentation window divided horizontally: at the top the Object view and at the bottom an analyzer. Work in the analyzer as described in SECTION 3.2.

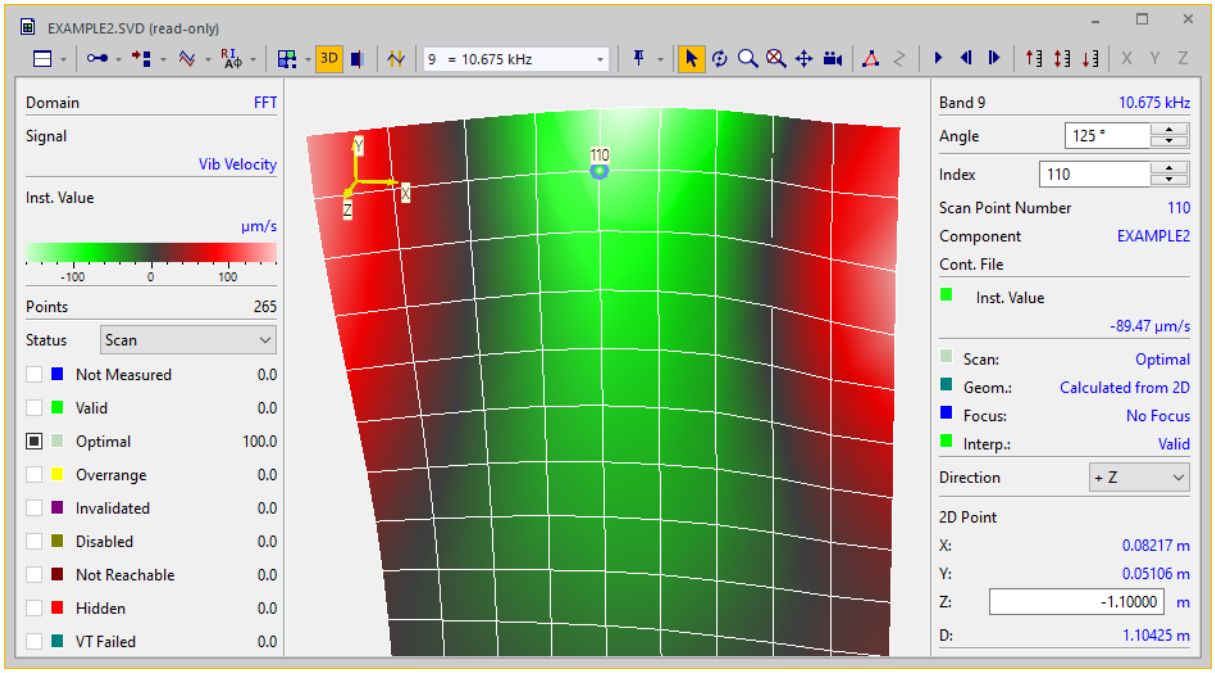


Figure 3.13: Presentation window in the Object view (3D view)

Object

You will see the data superimposed on the video image. You can present the data itself in different ways, see SECTION 3.3.2 on this. You can set up the look of the object and the video image (background color, title, transparency etc.) as described in SECTION 3.1.1.

Single Scan Point

In the upper part of the presentation window you can see the object. At the bottom you see an analyzer. The analyzer displays the spectrum at the single scan points. At the top, you select the scan point for which the spectrum is shown. To do so, set a cursor at this scan point as described in SECTION 3.3.3.

Average Spectrum

At the top you see the object. At the bottom the analyzer displays the average spectrum of all scan points. The average spectrum is calculated from the amplitude spectra of the valid scan points. See SECTION 4.1 on this as well.

Profile

At the top you see the object. You can draw profile lines on the object as described in SECTION 4.3. You will see the profiles at the bottom in the analyzer.

ⓘ You can not draw any profile lines in the 3D view.

3.3.2 Selecting the View Style of the Data

In the presentation window you see the data superimposed on the video image. You can present the data in different ways:

- Surface
- Wireframe
- Isolines
- Scan Points
- Arrows
- Status Points

You can switch between 2D and 3D view in all view styles by selecting Presentation > 2D/3D View in the menu bar (toolbar of the presentation window: ).

You will see the data color-coded in all view styles apart from Status Points. You can select the color palette for this as described in SECTION 3.3.5. You can show or hide a legend for color-coding to the left of the object by selecting Presentation > Legend.

ⓘ Regardless of the view style of the data, you can show/hide a legend for the status of the scan points on the left of the object by selecting Presentation > Status in the menu bar. If the legend on the left of the presentation window is not visible, previously select Presentation > Legend to display it.

To adapt the display of the data, select Presentation > View Style in the menu bar and then the respective view style (toolbar of the presentation window: ).

You can display the live video image outside the scan points.

Surface The software interpolates between the scan points.

Wireframe You will see the connecting lines between the scan points as a color-coded grid.

Isolines You will see equal data values connected with color-coded isolines. You can set up the view style of the isolines (number, thickness) as described in SECTION 3.1.2.

Scan Points You will see each scan point as a color-coded marker which corresponds to its data value. In addition to that, you can display the index of each scan point. See SECTION 3.1.2 on this.

Arrows You will see each scan point as a color-coded arrow which corresponds to its data value. In addition, you can see the deflection by the length of the arrow.

Status points

You see the scan points as markers in different colors. The color shows the status of the scan point. To show/hide a legend for the status to the left of the object, select Presentation > Status in the menu bar. If the legend is not visible on the left of the presentation window, previously select Presentation > Legend to display it. All status conditions for a scan point are displayed in the readout on the right of the object.

For an explanation of the Interpolate status, see SECTION 4.1.3. For an explanation of the other statuses such as Scan, Geometry etc., see SECTION 4.1.1.

3D view

You can set up the 3D view style (projection, colors, size) as described in SECTION 3.1.3. You can to set the line of sight in the presentation window with the mouse by proceeding as follows:

1. Select Presentation > Rotate in the menu bar (toolbar of the presentation window: ).

   The mouse pointer becomes a rotation symbol  .
2. Drag the mouse in the direction you require. Hold the Control key pressed while dragging and the object only rotates around the line of sight.

3D geometry

You will find detailed information on the presentation of 3D geometries in your PSV theory manual.

3.3.3 Setting Cursors and Reading Data in Presentation Windows

If you would like to read data at single scan points, you can set a cursor on the object. You see the cursor as a blue marker. You cannot set a cursor in Profile view.

Set a cursor

To set a cursor, proceed as follows:

1. Select a suitable view (Object, Single Scan Point or Average Spectrum). To do so, select Presentation > View in the menu bar and select the respective view in the pull-down menu (toolbar of the presentation window: ).
2. Display any view style. To do so, select Presentation > View Style in the menu bar and select the view style in the pull-down menu (toolbar of the presentation window: ).

Select scan point

3. You can select scan points in surface view as well as in 3D view. To do so, point with the mouse pointer at the scan point and click it.

   The scan point is shown as a blue marker.

Read data

4. In the readout you can see the data of the scan point where the cursor is. If the readout is not visible on the right of the object, select Presentation > Readout in the menu bar to display it.

ⓘ You will see the spectrum at this scan point at the bottom in the analyzer in Single Scan Point view.

Move cursor

5. You can move the cursor to other scan points using the mouse.

   The readout is updated simultaneously.

ⓘ In all views (apart from Profile) you can also move the cursor within a cohesive area using the arrow keys.

You can zoom in on the view to be able to see details more clearly. Read SECTION 1.5.8 on this.

Delete cursor

6. You can delete the cursor or undo the selection by clicking  in the toolbar of the presentation window.

3.3.4 Displaying Data as Readout

The following data is displayed as readout in the presentation window:

- Index and scan point number - See SECTION 4.5.3 on this.
- Name of component - See SECTION 4.5.5 on this.
- For combined files the name of the file contained (Cont. File) which provides the data for the selected scan point - See SECTION 3.6 on this.
- Display type, how the vibration is displayed (e.g. Magnitude, Inst. Value, etc.) - See SECTION 3.4 on this.
- Status of the selected scan point - For Scan Status, Geometry Status and Laser Focus Status, see SECTION 4.1.1. For the Interpolate status, see SECTION 4.1.3.
- Coordinates of the selected scan point (2D Point or 3D Point) as well as the distance of the scan point to the scanning head.

3.3.5 Selecting the Color Palette

You can choose between different color palettes for color-coding in the presentation window. To do so, proceed as follows:

1. Select Setup > Color Palette in the menu bar (toolbar: ).

   The Color Palette Selection dialog appears.

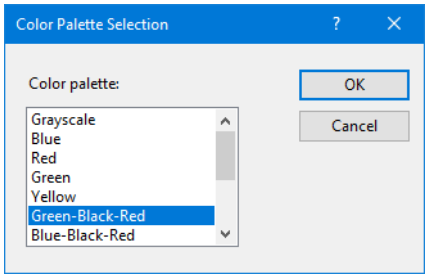


Figure 3.14: Color Palette Selection dialog

2. Mark the required color palette.

ⓘ The Green-Black-Red color palette is set by default.

3. Click OK.

ⓘ You can also use the right mouse button to click on the legend on the left in the presentation window and select the required color palette in the context menu.

3.4 Selecting the Data Set Displayed

In the presentation window or in the analyzer of the presentation window, you select the data set displayed. See SECTION 3.3.1 on this as well. The analyzer shows the same data set. However, changes to the data set in the analyzer do not have any impact on the data set in the presentation window.

Selection

To display a certain data set, select in the following order:

1. Time signal or type of spectrum (domain)
2. Channel or combination of channels
3. Signal or function
4. Display type used to display the signal or the function
5. 0 dB Reference
6. Frequency band - only in presentation windows of data you have already captured in FFT mode

ⓘ First of all you have to define frequency bands. Read SECTION 4.2.1 on this.

ⓘ With the PSV-3D and MSA-3D you can also select the channels Vib 3D or Vib X, Vib Y and Vib Z as well as combinations of channels such as Vib X & Ref 1.

ⓘ You can also select the corresponding command in the Presentation menu instead of the Analyzer menu.

Requirement

The selection depends on the following settings:

- Measurement mode
- Averaging, trigger and reference signal - See PSV theory manual on this as well.
- Which displays have already been selected - e.g. you can display a frequency response only for frequency spectra and a combination of channels.

To display a certain data set, proceed as follows:

Domain

1. In the menu bar, select Analyzer > Domain (toolbar: ).
2. Select the required setting (e.g. time signal, spectrum) in the pull-down menu or in the context menu.

Channel

3. In the menu bar, select Analyzer > Channel (toolbar: ).
4. Select a channel or a combination of channels in the pull-down menu or in the context menu. If you want to use user defined data sets, select the Usr, Usr X, Usr Y, Usr Z or Usr 3D channel. You will find information on this in the Polytec File Access documentation.

Signal

5. In the menu bar, select Analyzer > Signal (toolbar: ).
6. Select a signal or a function in the pull-down menu or in the context menu. The abbreviations used are explained in the following table.

Table 3.1: Abbreviations for functions

Abbreviation	Function
AP	Autopower
CP	Crosspower
FRF	Frequency Response Function
H1	Frequency Response
H2	Frequency Response
PSD[1]	Power Spectral Density
ESD[1]	Energy Spectral Density

[1] The PSD and ESD signals are only available in the analyzer.

ⓘ If you have more than 24 signals to choose from (Usr channel), select Analyzer > Signal > More Signals in the menu bar. The Signals dialog appears. Select the desired signal and click OK.

Display type

7. Select Analyzer > Display Type in the menu bar (toolbar: ).
8. Select the way that the vibration is to be presented, in the pull-down menu or in the context menu.

ⓘ In the Magnitude [dB (A)] display type, the measured amplitude is frequency-weighed according to the standard EN 61672-1 (IEC 61672-1) (A-weighting). The A-weighting approximately describes the human hearing range at low amplitudes. The following figure shows the frequency-weighting curve. Outside the range shown, the software evaluates as follows: under 10.0 Hz with –70.4 dB and over 20 000 Hz with –9,3 dB.

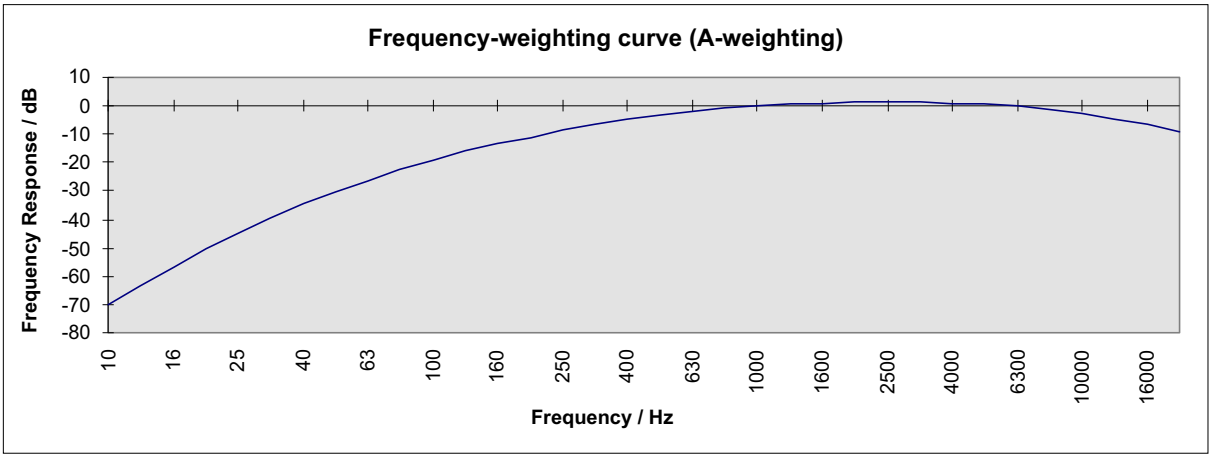


Figure 3.15: Frequency-weighting curve for the Magnitude [dB (A)] display type

ⓘ In the Nyquist display type, the real part of the amplitude is plotted along the x-axis and the imaginary part along the y-axis. You can limit the frequency range of the Nyquist display type by initially zooming the x-axis in a different display type so that the required frequency range is shown and then change to the Nyquist display type.

Features

Domain — FFT and MultiFrame measurement mode: To calculate a frequency spectrum and to display it, the software needs to have acquired the entire measurement time. Therefore, during a continuous measurement, the spectrum of the last complete measurement is displayed at the same time as the time signal of the current measurement.

Signal — Abbreviation Principal Inputs = Principal Component Analysis (MIMO)

In TABLE 3.2 you will find a summary of how to select Domain, Channel, Signal and Display Type.

Table 3.2: Selection of the data set displayed

Click	Or select Analyzer > or Presentation >	Display of	Examples
	Domain	Time signal, spectrum or RMS signal	FFT, 1/3 Octave RMS Time
	Channel	Channel or combination of channels	Vib, Vib 3D Ref 1, Ref 2, Ref 3 References (MIMO) Vib & Ref 1, Vib & Ref 2, Vib & Ref 3 Vib 3D & Ref 1 Ref 1 & Ref 2, Ref 1 & Ref 3, etc.
	Signal	Signal or function	Displacement, Velocity, Voltage, Acceleration FRF Displacement / Voltage AP Displacement^2, AP Velocity^2, AP Voltage^2, AP Acceleration^2 CP Voltage * Velocity H1 Displacement / Voltage, H1 Velocity / Voltage, H1 Acceleration / Voltage H2 Displacement / Voltage, H2 Velocity / Voltage, H2 Acceleration / Voltage Coherence Multiple Coherence (MIMO) Virtual Coherence (MIMO) Principal Inputs Voltage^2 (MIMO) PSD Displacement^2/Frequency PSD Velocity^2/Frequency PSD Acceleration^2/Frequency ESD Displacement^2 * Time / Frequency ESD Velocity^2 * Time / Frequency ESD Acceleration^2 * Time / Frequency
	Display Type	Way in which the signal or function is shown	Magnitude Magnitude [dB], Magnitude [dB(A)] Mag. & Phase, Mag. [dB] & Phase Phase Real Imaginary Real & Imag. Inst. Value Nyquist

0 dB Reference

Set the 0 dB reference in the Preferences dialog.

1. In the menu bar, select Setup > Preferences and display the dB Reference page.

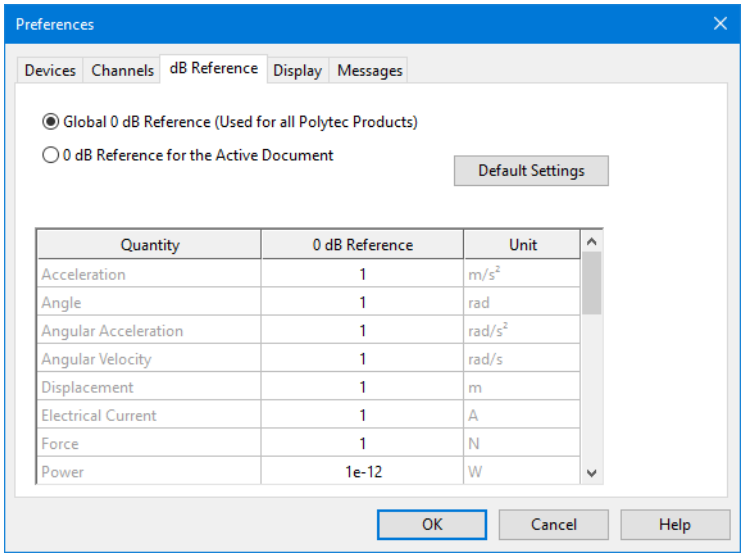


Figure 3.16: dB Reference page in presentation mode

2. Select Global 0 dB Reference (Used for all Polytec Products) if you want the settings to apply to any Polytec software installed on the data management system.

   Select 0 dB Reference for the Active Document if you want the settings to apply to the currently open file only.

   The Quantity column shows the available physical quantities, the Unit column the respective applicable unit (ISO).

3. Define the factor for the respective unit in the 0 dB Reference column.

ⓘ Depending on the size of the opened file, it may take some time to transfer the new value.

4. If you want to evaluate combined signals (e.g. Vib & Ref 1), then define the 0 dB Reference for both quantities involved.

   The software will then automatically determine the 0 dB Reference of the combined signals.

ⓘ If you want to recover the original settings for the 0 dB Reference, click Default Settings.

Example of 0 dB Reference

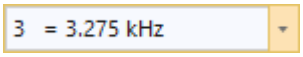
The 0 dB reference for displacement is 2 mm. Enter 0.002 or 2e-3 in the 0 dB Reference column.

Quantity	0 dB Reference	Unit
Displacement	0.002	m

Figure 3.17: Example of 0 dB Reference

Display frequency band

You have different options to select the frequency band displayed:

- Select the frequency band in the toolbar of surface view in the 

list. If no frequency bands are defined, only the Entire Bandwidth entry is displayed in the list.
- Press the Page Up ↑ or Page Down ↓ key. You will see the next or previous frequency band.
- You select the frequency band in the analyzer in the Single Scan Point or Average Spectrum view. To do so, select Analyzer > Show Bands in the menu bar (toolbar: ). You see the frequency bands in the analyzer in color. Click the respective frequency band there and the software will display it in surface view.

3.5 Defining Clipping Planes

You can define clipping planes in 3D view to control complex geometries.

Depending on your measurement system, you can define a front and/or rear clipping plane respectively perpendicular to the coordinate axes and the sight of the camera. All parts of the geometry which lie outside the cutting area are removed from displaying. This makes it easier to analyze complex geometries and deflection shapes.

To open the dialog for defining clipping planes, you have the following options:

- In the menu bar, select Presentation > Define Clipping Planes.
- In the toolbar of the presentation window, click  .
- Click the presentation window (video image) with the right mouse button and select Define Clipping Planes in the context menu.
- Open the Display Properties dialog, display the 3D View page, and click Clipping Planes.

Clipping planes

To define clipping planes, proceed as follows:

1. Switch to 3D view (toolbar: 

).
2. Open the Clipping Planes dialog (see above).

   The Clipping Planes dialog appears.

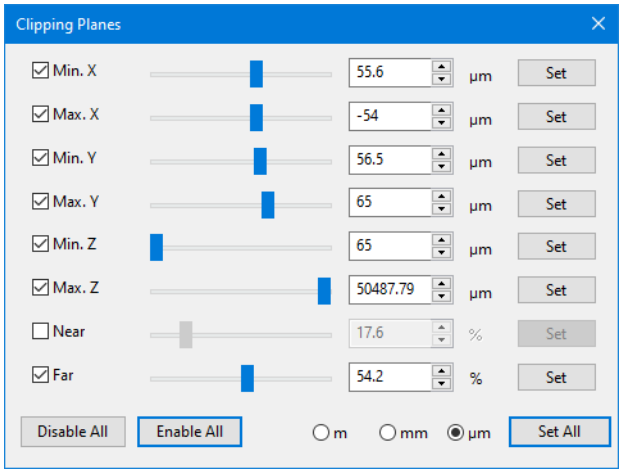


Figure 3.18: Clipping Planes dialog

3. Define the clipping planes as described below.

4. To finish the definition and to exit the dialog, click  .

   The current configuration of clipping planes is saved and restored when you reopen the dialog.

Description

Tick the check boxes of the clipping planes that you want to define. If you want to set up all available clipping planes, click Enable All. Only activated clipping planes can be configured.

Min. X/Y/Z and Max. X/Y/Z: Define the position of a clipping plane either using the associated slider or entering the directional coordinate value in the associated input field.

m, mm, µm: Select the appropriate length unit for the X/Y/Z-values of the positions of the clipping planes. Each length unit has its own step size for adapting the position values using the sliders and the arrow keys. The step sizes are defined as follows:

Length unit	Step size
m	10 mm
mm	10 µm
µm	100 nm

Near, Far: Define the position of these clipping planes either using the sliders or entering percentage values into the input fields. The whole percentage range covers the expansion of the amount of all selected scan points from the camera's perspective.

Set, Set All: To position a clipping plane at the appropriate edge of the smallest cuboid that contains all selected scan points, click Set. To configure this for all clipping planes, click Set All.

Disable All: To deactivate all clipping planes, click Disable All.

3.6 Generating Combined 3D Files (Stitching)

If 3D geometries are available, you can generate combined 3D files with the different measurement systems. To do so, proceed as follows:

- You merge individual measurements into a combined file. See SECTION 3.6.1 on this.
- For working with combined files, the same functions are available as for individual measurements. See SECTION 3.6.2 on this.

ⓘ You can generate a new combined filed from several combined files. To do this, proceed in the same way as when merging individual measurements.

Depending on the measurement system, different conditions apply:

PSV-3D/PSV-1D If you are using PSV-3D, you can merge individual measurements with 3D geometries and 3D vibration data to form combined files. This also works with PSV-1D with 3D geometries which you have either imported or determined using a geometry scan. For this, the following conditions have to be met:

- You have performed the individual measurements with PSV-3D.
- You have performed the individual measurements with PSV-1D and 3D geometries which you have either determined through a geometry scan or imported (ImpGro or GeoPro option).
- The coordinate systems of all individual measurements must have the same origin and the same orientation. This means that for 3D alignment, you have used the same coordinate system for the alignment points of all files.
- You have performed all individual measurements with the same basic settings. These include the number of channels, the definition of the reference channel, units, averaging mode and trigger settings as well as the setting for the optional Principal Component Analysis (MIMO).
- However, you can select different vibrometer settings, generator signals, input ranges, vibrational directions of the channels and calibration factors.

ⓘ If you combine measurements with different vibrometer settings, the possibly different time delays will only be corrected under the following conditions:

- As of OFV-5000 (Firmware Version 2.0) and PSV-Software Version 8.3
- As of PSV-F-500 and PSV software version 9.0
- Measurement data is acquired in the spectral range (FFT)
- No correction is made in the time domain.

- If the indices of scan points appear twice, then these are marked red in the Point Indices column. If the 3D coordinates of the scan points are different, then the software shows a corresponding warning message when combining the files. However, you can continue combining.

ⓘ You can not create 3D vibration data by combining 1D vibration data with different vibrational directions that exist for scan points with the same indices.

MSA/UHF

If you are using MSA, MSA-3D or UHF with an XY positioning stage from Polytec, you can merge individual measurements with 3D geometries and 3D vibration data into combined files. For this, the following conditions have to be met:

- MSA-600/MSA-650/MSA-3D: You have performed the individual measurements with 3D geometries which you have determined with the aid of a geometry scan (optional automated z-axis from Polytec). As an alternative, you have performed the individual measurements with the aid of imported 3D geometries (ImpGro or GeoPro option).
- MSA-650: You have measured the individual measurements with their specific path length alignments respectively.
- MSA-500: You have performed the individual measurements with 3D geometries which you have determined with the help of a geometry scan (MSAGeo option). As an alternative, you have performed the individual measurements with the aid of imported 3D geometries (ImpGro or MSAGeo option).
- UHF: You have performed the individual measurements with the aid of imported 3D geometries (ImpGro or GeoPro option).
- The coordinate systems of all individual measurements must have the same origin and the same orientation. This means that for 3D alignment, you have used the same coordinate system for the alignment points of all files.
- You have performed all individual measurements with the same basic settings. This includes the number of channels, definition of the reference channel, units, averaging mode and trigger settings.
- However, you can select different vibrometer settings, generator signals, input ranges, vibrational directions of the channels and calibration factors.
- If the indices of scan points appear twice, then these are marked red in the Point Indices column. If the 3D coordinates of the scan points are different, then the software shows a corresponding warning message when combining the files. However, you can continue combining.

3.6.1 Merging Individual Measurements

Criteria for data selection

If scan points with the same indices occur in the single shots, then the software selects the data and the 3D coordinates to be used for the combined file according to the following criteria in the given order:

- Valid data (Scan status Valid or Optimal) have priority over invalid data (Scan status Not Measured, Overrange, Invalidate, Disabled, Not Reachable or VT Failed).
- Optimal data (Scan status Optimal) has priority over data which is not optimal.
- Available data (Scan status Overrange or Invalidate) has priority over data which is not available (Scan status Not Measured, Disabled, Not Reachable or VT Failed).
- Disabled points have priority over points which have not been disabled (Scan status Not Measured).
- If both scan points have the Optimal scan status, then the data with the lower noise level has priority.

If no decision can be made from these criteria, then the order of the individual measurements in the Create Combined File dialog is decisive. To change the order of the individual measurements in the list, mark the individual measurement and click  or  .

To merge several individual measurements to a combined file, proceed as follows:

1. Select File > New > Combined File in the menu bar (toolbar:  > Combined File).

   The Create Combined File dialog appears.

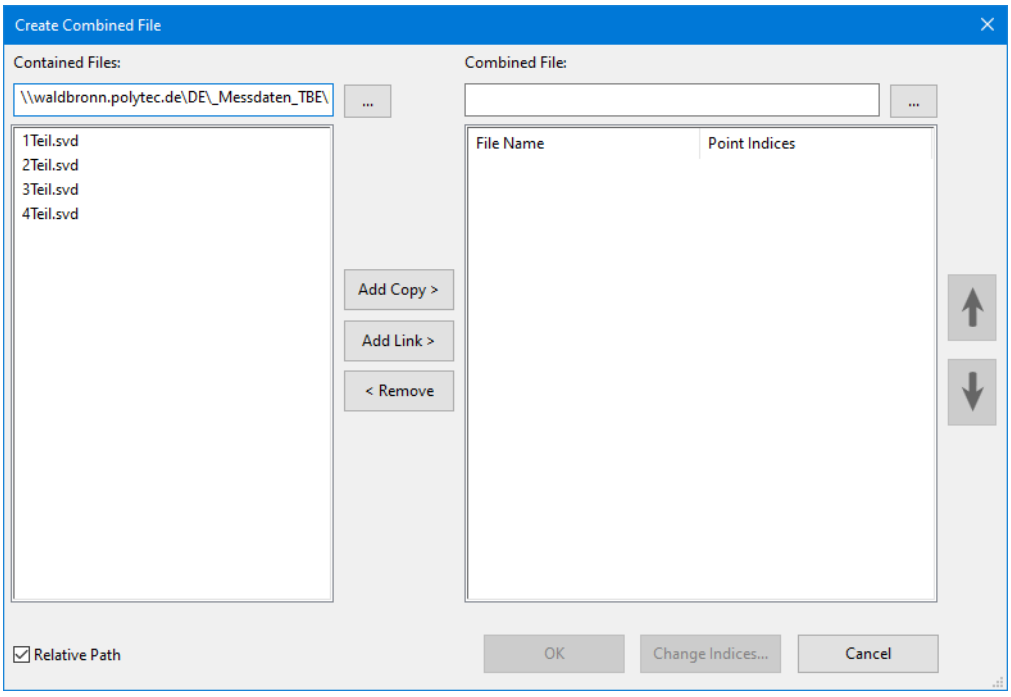


Figure 3.19: Create Combined File dialog

Add individual measurement

2. Click ... next to the Contained Files field.

   The Select Folder dialog appears.

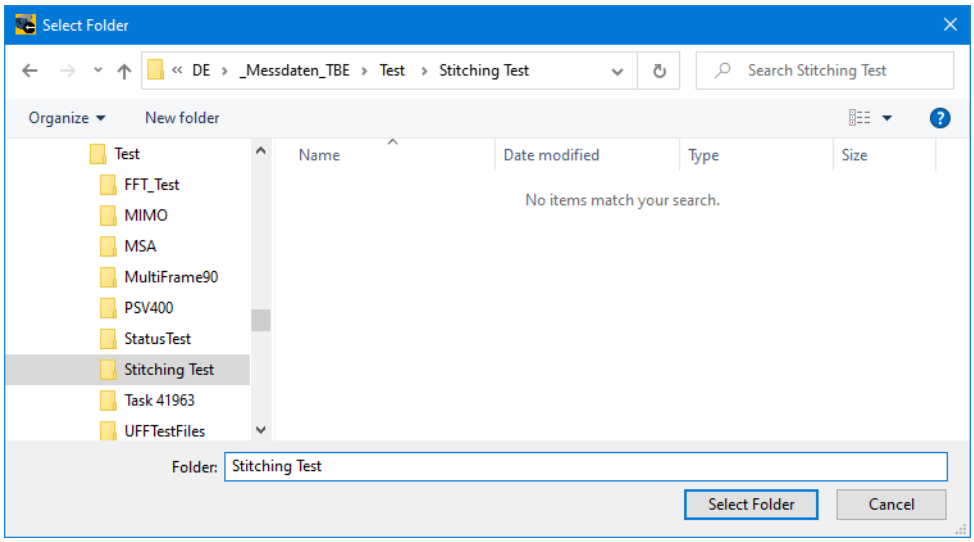


Figure 3.20: Select Folder dialog

3. Navigate to the directory containing the individual measurements you want to merge and click Select Folder.

Select individual measurements

4. Mark the files with the individual measurements you would like to merge.

ⓘ To select several individual files at the same time, hold the Control key pressed and mark the files using the mouse. To select several consecutive files at the same time, hold the Shift key pressed and mark the first and last file using the mouse.

5. If you want to apply the selected files in the combined file, click Add Copy.

   If you want to link the selected files with the combined file, click Add Link.

ⓘ If you link individual measurements, later modifications to these measurements will have an effect on the combined file.

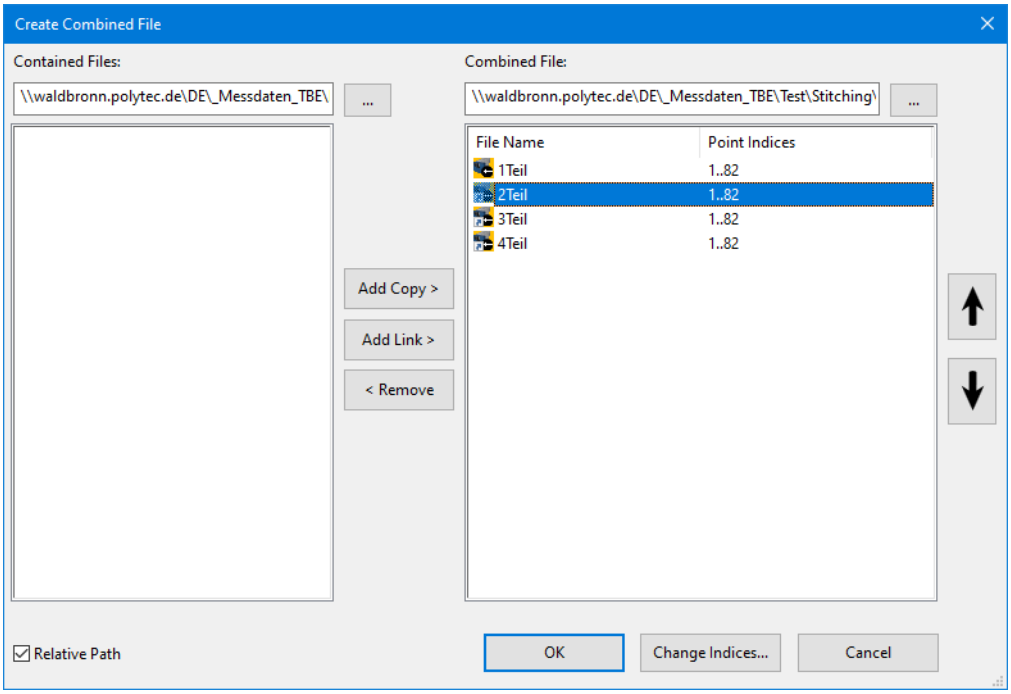


Figure 3.21: Selecting the files in the Create Combined File dialog

ⓘ In the Point Indices column, the indices of the scan points of all individual files are displayed. Possible overlaps are marked in red. You can change the indices directly in this dialog. To do so, click Change Indices. The Change Indices dialog appears. See SECTION 4.5.3 on this as well.

6. If for linked files only the position in the file system relative to the combined file is to be saved, tick the Relative Path check box.

   In this case, the links also remain valid if you copy the combined file with the linked files e.g. into another directory and thereby retain the relative folder structure.

Save combined file

7. Click ... next to the Combined File field.

   The Save As dialog appears.

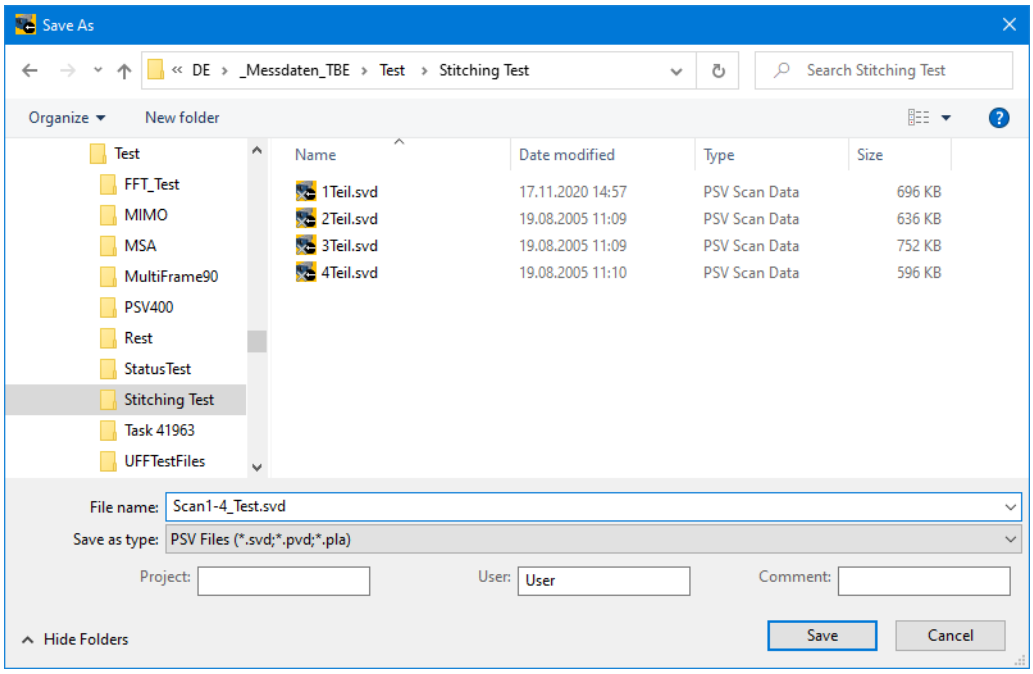


Figure 3.22: Save As dialog for combined files

8. Navigate to the directory which you want to save the combined file in and enter the file name.

   You can define the file more accurately under Project, User and Comment.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

9. Click Save.

ⓘ If you have defined geometry components in the individual measurements, then they will be shown in the combined file with the respective hierarchy. Read SECTION 4.5.5 on how to define geometry components.

Remove individual measurements

10. To remove files from the list of individual measurements for the combined file, mark them as described in step 4 and click Remove.
11. In the Create Combined File dialog, click OK.

    The combined fill will be saved in selected directory and will be opened in the application window.

3.6.2 Working with Combined Files

The same functions are available to you for combined files as they are for individual measurements. However, after merging, all other operations will only be saved in the combined file. The individual measurements remain unaffected by this.

As a general rule, every change to a file is marked with an asterisk (*) in the file name as soon as you make the change.

Display combined file

To open the combined file, double-click the file name in the project browser or click it with the right mouse button and select Open in the context menu.

3D View — In the 3D view you can simultaneously see the geometry and the vibration data from all parts of the object that you scanned in the individual measurements in three dimensions. Between the geometry parts, you will only see connecting lines if the scan points which are involved have identical coordinates. This is the case for example if you make each of your individual measurements on the same complete geometry. You can also add and edit connecting lines retrospectively. See SECTION 4.5.4 (Connect Automatically) on this.

2D View — In the 2D view you can see the geometry and the vibration data of the individual measurements. To show the individual measurements consecutively, select them in the Contained File list.

ⓘ To draw a profile line, you have to select 2D view. For profile lines, see SECTION 4.3.

If you have defined geometry components in the individual measurements, you will see these in the Components list. You can edit the geometry components as described in SECTION 4.5.5.

ⓘ If you select a scan point in 3D view, the file path with file name of the individual measurement is displayed in the readout which contains the selected scan point (Cont. File).

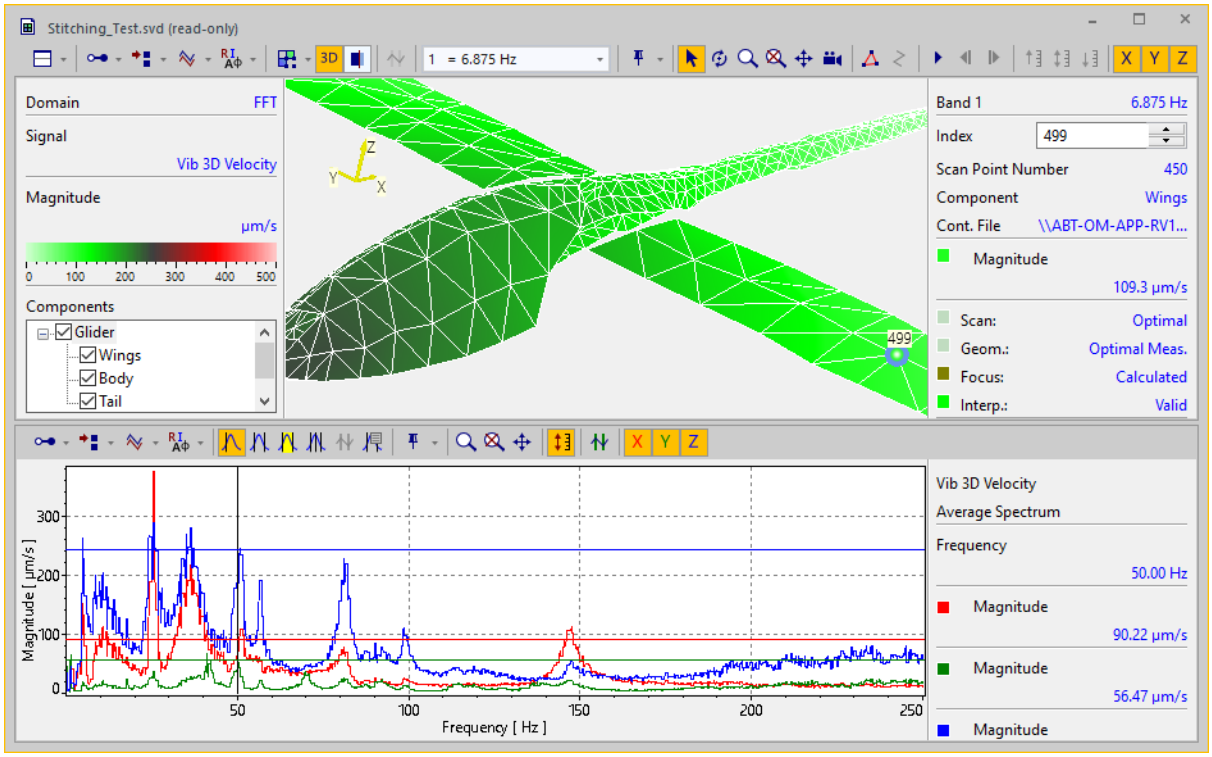


Figure 3.23: 2D view of an individual measurement of the combined file

In the project browser your can see the individual measurements as well as the settings which were used to make these measurements. To do so, click the plus sign (+) in front of the name of the combined file or in front of the name of the individual measurements.

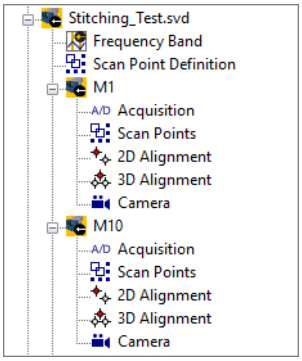


Figure 3.24: Combined file, individual measurements and settings

ⓘ You can load the scan point definition of the combined file directly from the project browser. However, the scan point definition of a combined file does not contain the positions of the scan points on the video image. When loading the scan point definition, these positions will be recalculated based on the 2D alignment which is currently valid.

Overlap of individual measurements

Depending on how you defined the scan ranges of the individual measurements, parts of the combined file in the 3D view can overlap.

If scan points with the same indices occur in single shots, the software initially selects the data and the 3D coordinates to be used for the combined file according to the criteria as described in SECTION 3.6.1. If no decision can be made from these criteria, the software prioritizes the data in the order of the individual measurements in the Create Combined File dialog: The individual measurement which is in first place in the Combined File list is given priority 1. See FIGURE 3.21 on this as well. With the combined file, for scan points which have been measured twice, the video image is displayed with the geometry data, measurement data, etc. from the individual measurement which contains these scan points and has the highest priority.

Sort individual measurements

To adapt the display, you have to change the order of the files. To do so, proceed as follows:

1. Open the Create Combined File dialog. To do so, click the combined file with the right mouse button in the project browser and select Properties in the context menu.

ⓘ You can only select the properties if the combined file is not opened.

2. In the Combined File list, mark the individual measurement which you want to move.
3. To move the selected individual measurement up or down in the list, click  or  .
4. Click OK.

Apply frequency bands

You can apply frequency bands from one of the individual measurements to the combined file. To do so, proceed as follows:

1. Open the combined file.
2. Open the frequency band definition. See SECTION 4.2.1 on this as well.
3. Holding the left mouse button pressed, drag the Frequency Band entry of the required individual measurement from the project browser to frequency band definition.
4. Close the frequency band definition.

   A message window appears asking whether the frequency bands are to be recalculated.
5. Click Yes.

3.7 Measurement Data in the Signal Processor

ⓘ When a signal processor document is active, in the menu bar the Analyzer menu is displayed.

In acquisition mode

You can display data in the signal processor in the acquisition mode. To do so, you can apply data from the analyzer during the measurement or open a scan or multi point file and apply the data from there into the signal processor. This allows you to check whether certain measurement conditions have not yet been met during acquisition.

In the menu bar, select File > New Signal Processor to open the signal processor in acquisition mode (toolbar: ).

You use the signal processor in acquisition mode in principle as described below for presentation.

In presentation mode

If you want to continue to analyze measurement data which has already been acquired, then you can recalculate it and show the results in direct comparison with the original data. For this purpose a signal processor is available which you can transfer the data to and can then process it further. The data calculated this way is shown in the analyzer of the signal processor. You can also save this data with the scan and display it again at any time. Read the following sections on this:

- To edit the measurement data in the signal processor, you need to open the respective measurement, apply the measurement data to the signal processor and carry out the required arithmetic operations there. See SECTION 3.7.1 on this.
- In tables, you will find an overview of all functions you can carry out. See SECTION 3.7.2 on this.
- Here you will find different calculation examples for specific functions. See SECTION 3.7.3 on this.
- You can adapt the presentation of the recalculated measurement data in the analyzer. See SECTION 3.7.4 on this.
- You can add the measurement data which you calculated in the signal processor to the original file as user defined data sets or bands. See SECTION 3.7.5 on this.
- At the same time as recalculating the data, the updated graph is shown in the analyzer. See SECTION 3.7.6 on this.
- You can save the formulae which you have used to recalculate the measurement data. See SECTION 3.7.7 on this.
- You can search for functions or expressions in formulae and replace them respectively. See SECTION 3.7.8 on this.
- You can directly select a file in the signal processor to replace the copied measurement data. See SECTION 3.7.9 on this.
- Application examples for processing measurement data - See SECTION 3.7.10 on this.

3.7.1 Copying and Calculating Measurement Data

To edit the measurement data in the signal processor, you need to open the respective measurement, apply the measurement data to the signal processor and carry out the required arithmetic operations there.

Open signal processor

To open the signal processor, proceed as follows:

1. Select File > New > Signal Processor in the menu bar (toolbar: > Signal Processor).

   The signal processor appears.

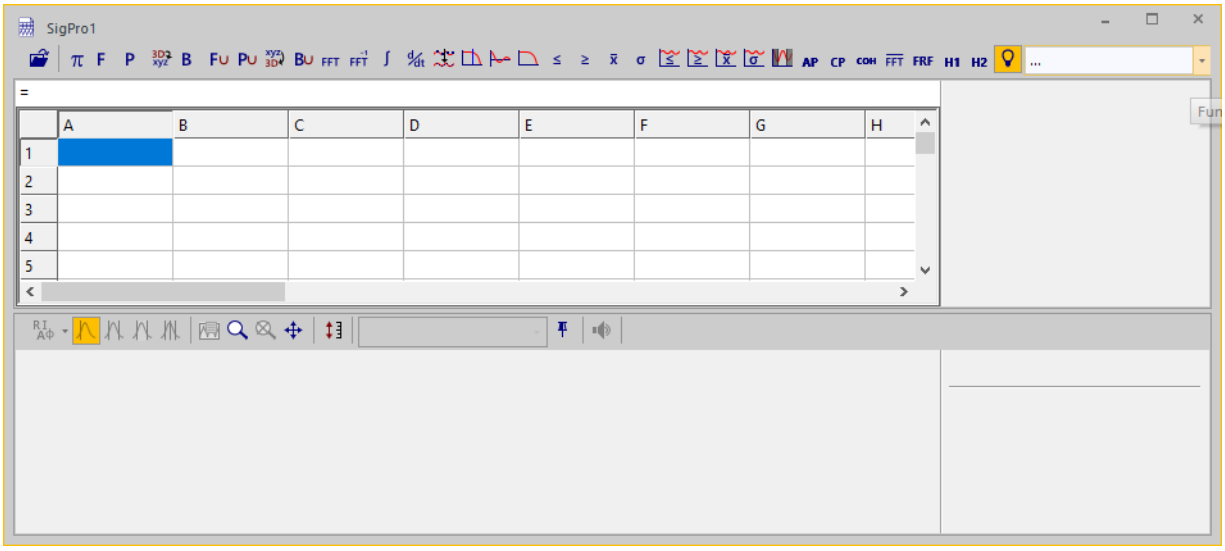


Figure 3.25: New signal processor

ⓘ You will find a description of the signal processor in SECTION 1.5.5.

Open scan

Open the scan in presentation mode. To do so, proceed as follows:

2. Select File > Open in the menu bar (toolbar:).

   The Open dialog appears.

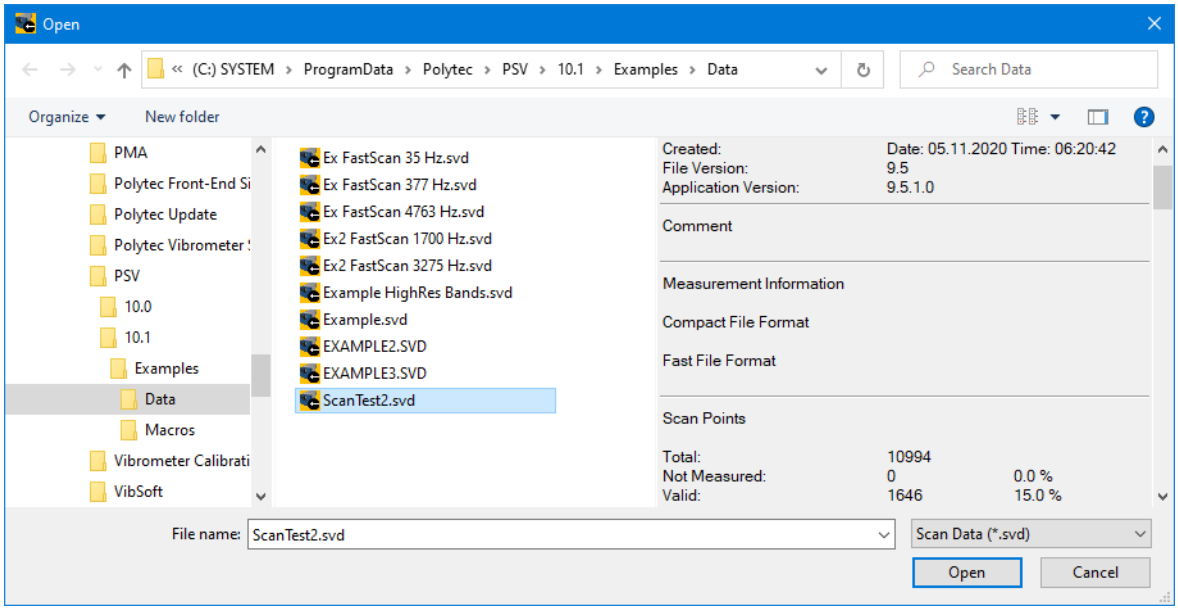


Figure 3.26: Open dialog for a scan

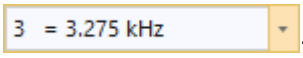
3. Navigate to the saving location and select Files of type and File name.
4. To identify the files more easily, show the preview of the meta information by clicking  .

   You will receive information on the project, on scan points, the hardware and the measurement settings of the data.
5. Click Open.

   If you open a scan, a presentation window appears. If you open a single point measurement, an analyzer appears.

Copy measurement data

You can apply point data or band data in the signal processor and create user defined data sets. To do this, copy the measurement data into the signal processor. To do so, proceed as follows:

6. To copy Point Data or Band Data, click  and select the desired in the pull-down menu. When copying band data, select the required frequency band in the frequency band list  .
7. In the presentation window, use the mouse to select one or more scan points, click the right mouse button and select Copy Selected Points in the context menu for point data. If you have selected a frequency band, select Copy Selected Band.

   The data from the selected scan points or the frequency band is copied into the signal processor.

ⓘ If you select Copy All Points or Copy All Bands in the context menu, then the data of all scan points or frequency bands is copied into the signal processor.

8. If you have selected the Average Spectrum view, click the analyzer with the right mouse button and select Copy in the context menu.

   The Copy dialog appears.

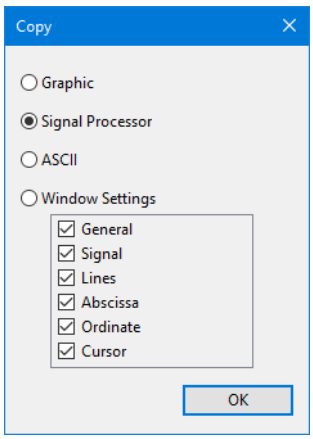


Figure 3.27: Copy dialog for signal processor

9. Select Signal Processor and click OK.

   The data is copied.

ⓘ You can also press the Ctrl + C key combination. The Copy dialog appears as long as you hold the Control key pressed. Select which data you want to copy and release the Control key. The data is copied.

Paste measurement data

10. Change to the signal processor.
11. Click a cell (e.g. A1) with the right mouse button and select Paste or Paste Link in the context menu.

    The copied data is inserted in the signal processor or inserted and linked. When inserting data from surface view, the signals of all channels are merged into one 3D signal for the defined scan points. In addition, a coordinate transformation into the coordinate system of the scan points will be carried out. When entering raw data, no transformation is carried out.

ⓘ Inserted data contains a reference to the original file and remains in the corresponding cell, even if you select other data in the sequence or close the scan or the signal processor. If you are using the RelativePath function, this ensures that the link still works if you copy or move the signal processor file with the original file. But while doing so, you need to retain the relative path names in the file system.

Linked data in contrast is updated in the formula as soon as you select other scan points and is lost as soon as you close the scan or the signal processor.

12. If you have selected and inserted several scan points or frequency bands, and would like to display the data from a single scan point or frequency band, then select the index or the frequency band in the list. See the following figure on this as well.
13. If you have applied the 3D data, you can select one of the vibrational directions for the display (X, Y, Z).

14. If you would like to display specific frames of the selected scan point or frequency band, click the number of the respective frame (1, 2, 3, 4, etc.). You can display an individual graph for each frame. See the following figure and SECTION 3.7.3 on this as well.

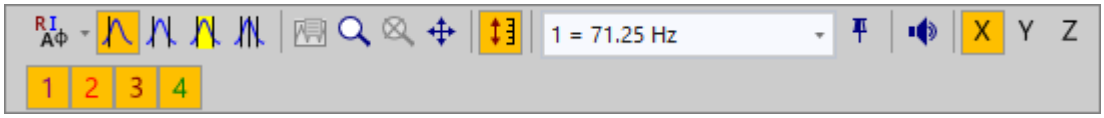


Figure 3.28: Selecting index/frequency band, vibrational direction and frames in the analyzer of the signal processor

Edit measurement data

15. Click another cell (e.g. A2) and activate the input line by clicking it.

ⓘ You can insert a reference to another cell into the formula by holding the Alt key pressed and then clicking the required cell.

16. Enter the required function and confirm with Enter.

ⓘ You can also enter any comments you like as text in the individual cells. In this case you just do not set an equal sign at the start of the respective text. Invalid formulae are shown in red.

Toolbar

You have the following options to enter a function in the formulae editor:

- Click the respective icon the toolbar.



Figure 3.29: Icons for functions in the signal processor

- Select the function in the ... list, right next to the icon for the function wizard.
- Use the function wizard.

  To activate the function wizard, click . As long as the function wizard is active ( ), click one of the icons to open the Function Wizard dialog, which will help you enter the parameters.

The signal processor carries out this calculation and shows the result in the analyzer.

Example

You will find an example for the function wizard in SECTION 3.7.10.4.

ⓘ You will find an overview of the functions stored in the signal processor in SECTION 3.7.2. You will find calculation examples for the functions in SECTION 3.7.3. Read SECTION 3.7.4 on how to set up the signal description.

3.7.2 Available Functions

In the following tables, you will find an overview of all functions you can carry out.

Table 3.3: Overview of the functions of the signal processor

Function	Icon	Description
Add (+)	-	By entering + you add two scalars, two strings, two Vib data sets or one Vib data set and one scalar or join two sets.
Sub (–)	-	By entering – you subtract two scalars, two Vib data sets or one Vib data set and a scalar or two sets.
Mul (*)	-	By entering * you multiply two scalars, two Vib data sets or one Vib data set with a scalar or intersect two sets.
Div (/)	-	By entering / you divide two scalars or two Vib data sets by each other, one Vib data set by a scalar or one scalar by a Vib data set.
Power (^)	-	By entering ^ you raise the value of x to the power of y (written: x^y, i.e. x^y); x and y can be scalars or Vib data sets.
RelativePath (TargetPath, SourcePath, UseRelative)	-	Entering RelativePath ensures that the link to the original file still works if you copy or move the signal processor file with the original file. But while doing so, you need to retain the relative path names in the file system. If UseRelative = 1, then the path is formed from TargetPath and the relative SourcePath given for it. Otherwise the path is only formed from the SourcePath.
CopyVibData (FilePath, Domain, Channel, Signal, DataType, 3D)	-	The CopyVibData (RelativePath) operation is used to copy specified measurement data from a measurement file into the signal processor (Paste command in the context menu).
LinkVibData (FilePath, Domain, Channel, Signal, DataType, 3D)	-	The LinkVibData operation is used to link specified measurement data from a measurement file into the signal processor (Paste Link command in the context menu). After making changes to the display in the measurement file the data will be updated.
LinkAcqData (Domain, Channel, Signal)	-	The LinkAcqData operation is used to display the specified signal from the measurement running in acquisition mode (Paste command in the context menu). The data is automatically updated after each single shot or after each measured scan point.

Several functions apply to one scalar (x) or Vib data set (VibData). Optional parameters are marked [ ].

Function	Icon	Description
Re(x), Re(VibData)	-	By entering Re you calculate the real part.
Im(x), Im(VibData)	-	By entering Im you calculate the imaginary part.
Abs(x), Abs(VibData)	-	By entering Abs you calculate the absolute value.
Arg(x), Arg(VibData)	-	By entering Arg you calculate the phase.
ArgMin(VibData) ArgMax(VibData)	-	By entering ArgMin or ArgMax the abscissa (x-coordinate) of the minimum or maximum in a real Vib data set which contains an individual sample point is given.
Exp(x), Exp(VibData)	-	By entering Exp you calculate the exponential function.
Log(x), Log(VibData)	-	By entering Log you calculate the logarithm (base 10).
Ln(x), Ln(VibData)	-	By entering Ln you calculate the natural logarithm (base e).
Sqrt(x), Sqrt(VibData)	-	By entering Sqrt you calculate the square root.
Sin(x), Sin(VibData)	-	By entering Sin you calculate the sine (radian).
Cos(x), Cos(VibData)	-	By entering Cos you calculate the cosine (radian).
Tan(x), Tan(VibData)	-	By entering Tan you calculate the tangent (radian).
ASin(x), ASin(VibData)	-	By entering ASin you calculate the arc sine (radian).
ACos(x), ACos(VibData)	-	By entering ACos you calculate the arc cosine (radian).
ATan(x), ATan(VibData)	-	By entering ATan you calculate the arc tangent (radian).
Sign(x), Sign(VibData)	-	By entering Sign you calculate the sign of the value x. The result is –1 for x = negative, +1 for x = positive and 0 for x = 0.
Deg(x), Deg(VibData)	-	By entering Deg you convert radians into degrees.
Rad(x), Rad(VibData)	-	By entering Rad you convert degrees into radians.
Round(x)	-	By entering Round you round real numbers to the next integer e.g. Round(1.1) = 1 or Round(1.9) = 2.
Ceil(x)	-	By entering Ceil you round up real numbers to the next integer e.g. Ceil(1.1) = 2 or Ceil(1.9) = 2.
Floor(x)	-	By entering Floor you round down real numbers to the next integer e.g. Floor(1.1) = 1 or Floor(1.9) = 1.
Card(x)	-	By entering Card you calculate the cardinality (number of elements) in a set.
Pi()		By entering Pi you add the constant π to the formula.
Frame(VibData, IndexOrIndexSet)		By entering Frame you extract the specified frame or frames from the Vib data set. If the second parameter is a scalar, its real part is rounded up or down to the next integer and its imaginary part is ignored.
Point(VibData, IndexOrIndexSet)		By entering Point you extract the specified sample point or points from the Vib data set. If the second parameter is a scalar, its real part is rounded up or down to the next integer and its imaginary part is ignored.
Points(VibData)	-	By entering Points the set of sample points in a Vib data set is given.

Function	Icon	Description
Direction(VibData, dir)		By entering Direction you extract the data in vibrational direction of the 3D Vib data set. The direction is set with the variable dir (0 for x-direction, 1 for y-direction, 2 for z-direction)
Band(VibData, IndexOrIndexSet)		By entering Band you extract the specified band or bands from the Vib data set. If the second parameter is a scalar, its real part is rounded up or down to the next integer and its imaginary part is ignored.
FrameUnion (VibData, …)		By entering FrameUnion you combine the frames of up to 10 Vib data sets. The maximum total number of combined frames is 12. You can use this to compare a sample point with an identical index from different measurements.
PointUnion (VibData, …)		By entering PointUnion you combine the sample points of up to 10 Vib data sets. If sample points occur in more than one data set, those found first are used.
Make3D(VibData)		By entering Make3D you combine 1D Vib data sets to a 3D Vib data set. If you combine less than three Vib data sets, the missing ones are replaced with 0.
BandUnion (VibData, …)		By entering BandUnion you combine the frequency bands of up to 10 Vib data sets.
DisplayUnion (VibData, …)	-	By entering DisplayUnion you combine the frames of up to 10 Vib data sets. The maximum total number of combined frames is 12. You can use this to compare sample points with a different index from different measurements if each of the measurements only contains one sample point respectively.
FFT(VibData, Lines, Window, WindowParameter, WindowParameter)		By entering FFT you calculate the Fast Fourier Transformation of the Vib data set with real data. Parameter: - For Lines enter the number of spectral lines which are to be calculated. The lines must be in the range of 25…1048576. If you do not specify the lines, the maximum possible number will be assumed. - The length of the vector for the Vib data set must be at least twice the number of lines, rounded up to the next power of 2. - For Window, enter the window function used to multiply the time data before the transformation. Use one of the following character strings: "Rectangle", "Bartlett", "Blackman Harris", "Exponential", "Flat Top", "Force", "Hamming", "Hanning", "Tapered Hanning" - If you do not specify a window function, "Rectangle" will be assumed. - For WindowParameters, enter additional parameters for the "Exponential", "Force", and "Tapered Hanning" window functions. The second WindowParameter is currently not used.
IFFT(VibData)		By entering IFFT you calculate the inverse Fourier Transformation of the Vib data set. Note: With a measurement which is not triggered, the phase in the vibrometer channel is replaced with the differential phase to the reference channel. The calculation of the inverse FFT can provide unexpected results which are hard to interpret.
Integrate(VibData)		By entering Integrate you integrate the Vib data set. Requirement: The Vib data set must contain real data.

Function	Icon	Description
Differentiate(VibData)		By entering Differentiate you differentiate the Vib data set. Requirement: The Vib data set must contain real data.
Resample (VibData, Fnew)		By entering Resample you have the Vib data set sampled again at the new specified sample rate given in Hz (Fnew). Requirement: The Vib data set must contain real data.
Filter (VibData, Type, Quality, Flow, Fhigh)		By entering Filer you filter the Vib data set with real data. Parameter: - For Type select one of the character strings "Low Pass", "High Pass", "Band Pass" or "Notch". - For Quality select one of the character strings "Very Low", "Low", "Middle", "High" or "Very High". - For Flow select the cutoff frequency in Hz for Low Pass and High Pass or the lower cutoff frequency for Band Pass and Notch. - For Fhigh select the upper cutoff frequency in Hz for Band Pass and Notch.
FilterCoeff (Fsamp, Type, Quality, Flow, Fhigh)		By entering FilterCoeff you calculate the Vib data set which contains the filter kernel of the selected filter. Parameter: - For Fsamp enter the sample frequency in Hz. - For Type select one of the character strings "Low Pass", "High Pass", "Band Pass" or "Notch". - For Quality select one of the character strings "Very Low", "Low", "Middle", "High" or "Very High". - For Flow select the cutoff frequency in Hz for Low Pass and High Pass or the lower cutoff frequency for Band Pass and Notch. - For Fhigh select the upper cutoff frequency in Hz for Band Pass and Notch.
Transmission (Fsamp, Type, Quality, Flow, Fhigh, Count)		By entering Transmission you calculate the Vib data set which contains the frequency response function of the selected filter. Parameter: - For Fsamp enter the sample frequency in Hz. - For Type select one of the character strings "Low Pass", "High Pass", "Band Pass" or "Notch". - For Quality select one of the character strings "Very Low", "Low", "Middle", "High" or "Very High". - For Flow select the cutoff frequency in Hz for Low Pass and High Pass or the lower cutoff frequency for Band Pass and Notch. - For Fhigh select the upper cutoff frequency in Hz for Band Pass and Notch. - For Count enter the number of frequency lines for which the transmission function is to be calculated.
Min(VibData) Max(VibData) Mean(VibData) StdDev(VibData)		By entering Min, Max, Mean or StdDev, you respectively calculate the minimum, maximum, average or standard deviation of the Vib data set which contains an individual sample point. The return value is a scalar. You can calculate Mean for real or complex data. You can only apply all the other functions to real data.

Function	Icon	Description
StatMin(VibData) StatMax(VibData) StatMean(VibData) StatStdDev(VibData)		By entering StatMin, StatMean, StatMax or StatStdDev, you calculate the minimum, maximum, average or standard deviation of the Vib data set. The return value is a Vib data set with an individual sample point with the index 0. You can calculate StatMean for real or complex data. You can only apply all the other functions to real data. It can be meaningful to apply StatMean to a data set which only contains an individual sample point. This does not change the data set but means that the point index is set to 0. Points with the index 0 can be compared to all other indexed points.
Extract (VibData, xMin, xMax)	-	By entering Extract you extract the specified frequency line numbers or samples of the Vib data set. The default starting value is 0 and the full scale value is the maximum number –1. The full scale value must be larger than the starting value.
ExtractX (VibData, xMin, xMax)		ExtractX works in the same way as Extract. Here, you enter start and end of the range to be extracted in the physical quantity of the abscissa (Hz or s).
AP (VibData, Samplerate/ Bandwidth, Lines, Average, OverlapPercent, Window, WindowParameter, WindowParameter)		By entering AP you calculate the autopower spectrum of the Vib data set with time data. The description of the parameters are the same as for CP or COH (see below).
CP or COH (VibData, VibDataReference, Samplerate/Bandwidth, Lines, Average, OverlapPercent, Window, WindowParameter, WindowParameter)		By entering CP you calculate the crosspower spectrum of the Vib data set with time data. By entering COH you calculate the coherence of the Vib data set with time data. Parameter: - For VibDataReference, optionally specify a Vib data set that is to be used as a reference. - For Samplerate/Bandwidth enter the ratio of sample frequency to bandwidth (standard systems = 2.5; M-systems = 2.56; VibSoft-20 systems = 2.4). - For Lines enter the number of spectral lines which are to be calculated (e.g. 25, 50, 100, …). - For Average enter the number of averages for the spectral calculation. If you do not use the parameter, the software will try to use all available samples for the maximum number of averages. - For OverlapPercent enter the overlap in percent. - For Window, enter the window function used to multiply the time data before the transformation. If you do not specify a window function, "Rectangle" is assumed. - For WindowParameters, enter additional parameters for the "Exponential", "Force", and "Tapered Hanning" window functions. The second WindowParameter is currently not used.

Function	Icon	Description
AFFT(VibData, VibDataReference, Samplerate/Bandwidth, Lines, Average, ComplexAverage, OverlapPercent, Window, WindowParameter, WindowParameter)		By entering AFFT you calculate the averaged Fast Fourier Transformation of the Vib data set with time data. Parameter: - For VibDataReference, optionally specify a Vib data set that is to be used as a reference. - For Samplerate/Bandwidth enter the ratio of sample frequency to bandwidth (standard systems = 2.5; M-systems = 2.56; VibSoft-20 systems = 2.4). - For Lines enter the number of spectral lines which are to be calculated (e.g. 25, 50, 100, …). - For Average enter the number of averages for the spectral calculation. If you do not use the parameter, the software will try to use all available samples for the maximum number of averages. - For ComplexAverage, specify whether the result must be complex or not (1 = complex, 0 = not complex). - For OverlapPercent enter the overlap in percent. - For Window, enter the window function used to multiply the time data before the transformation. If you do not specify a window function, "Rectangle" is assumed. - For WindowParameters, enter additional parameters for the "Exponential", "Force", and "Tapered Hanning" window functions. The second WindowParameter is currently not used.
FRF (VibData, VibDataReference, Samplerate/Bandwidth, Lines, Window, WindowParameter, WindowParameter)		By entering FRF you calculate the frequency response of the Vib data set with time data. Parameter: - For VibDataReference, optionally specify a Vib data set that is to be used as a reference. - For Samplerate/Bandwidth enter the ratio of sample frequency to bandwidth (standard systems = 2.5; M-systems = 2.56; VibSoft-20 systems = 2.4). - For Lines enter the number of spectral lines which are to be calculated (e.g. 25, 50, 100, …). The number of spectral lines must be less than or equal to the number of samples of the Vib data set divided by Samplerate/Bandwidth. - For Window, enter the window function used to multiply the time data before the transformation. If you do not specify a window function, "Rectangle" is assumed. - For WindowParameters, enter additional parameters for the "Exponential", "Force", and "Tapered Hanning" window functions. The second WindowParameter is currently not used.

Function	Icon	Description
H1 or H2 (VibData, VibDataReference, Samplerate/Bandwidth, Lines, Average, OverlapPercent, Window, WindowParameter, WindowParameter)		By entering H1 or H2 you calculate the frequency response H1 or H2 of the Vib data set with time data. Parameter: - For VibDataReference, optionally specify a Vib data set that is to be used as a reference. - For Samplerate/Bandwidth enter the ratio of sample frequency to bandwidth (standard systems = 2.5; M-systems = 2.56; VibSoft-20 systems = 2.4). - For Lines enter the number of spectral lines which are to be calculated (e.g. 25, 50, 100, …). - For Average enter the number of averages for the spectral calculation. If you do not use the parameter, the software will try to use all available samples for the maximum number of averages. - For OverlapPercent enter the overlap in percent. - For Window, enter the window function used to multiply the time data before the transformation. If you do not specify a window function, "Rectangle" is assumed. - For WindowParameters, enter additional parameters for the "Exponential", "Force", and "Tapered Hanning" window functions. The second WindowParameter is currently not used.
Abscissa(VibData)	-	By entering Abscissa the Vib data set of a sample point with the index 0 is given for which the ordinate shows the same values as the abscissa.
NumValues(VibData)	-	By entering NumValues the number of samples or frequency lines of the Vib data set is given.
Resolution(VibData)	-	By entering Resolution the distance between the frequency lines or samples of the Vib data set is given.

Function	Icon	Description
ReIndex(VibData, IndexOrIndexSet, IndexOrIndexSet)	-	By entering ReIndex you change the indices of the sample points of the Vib data set. Parameter: - For IndexOrIndexSet, enter a single index or a range of indices. - The first index parameter is optional with Default {*} and the second optional with Default 0. - The intersection is formed from the first index parameter and the indices of the Vib data set. This intersection must have the same number of elements as the second index parameter, otherwise an error message appears. - The intersection of the second index parameter and the indices of the Vib data set must be empty. The sample points of the first index parameter get the indices from the second parameter, whereas their order is preserved. Example 1: ReIndex(VibData, {*}, 0) = ReIndex(VibData) - This changes the index to 0 for a Vib data set which contains an individual sample point. Example 2: VibData has the sample points 1…100; ReIndex(VibData, {21..30}, {101..110}) The result has the indices 1..20, 31..110.
Scale(VibData[, xMin, xMax, xName, xUnit, yMin, yMax, yName, yUnit, yScale, Name, 0dBReference, Power])	-	By entering Scale you modify scaling information of the Vib data set. This function is usually used to adapt the scaling after operations such as FFT or Integrate, as these operations change the dimensions of the axes. The changes you can make using the Scale command can be found in the following list. - VibData: Data set - xMin, xMax: Limits of the abscissa - xName: Name of abscissa - xUnit: Unit of abscissa - yMin, yMax: Limits of the ordinate - yName: Name of ordinate - yUnit: Unit of ordinate - yScale: The ordinate values are multiplied with yScale. - Name: Name of the data set - 0dBReference: Reference value of dB display - Power: 1 or 0 (Power = 1 shows that this is a power signal (10 dB corresponds to a tenfold increase). For other signals, set Power = 0 (20 dB correspond to a tenfold increase).

ⓘ In the table cells, you can display either the formula or the result. To toggle, select View > Formulae in the menu bar (key combination: Ctrl + #).

Conventions

You can also enter sets and complex numbers. You place sets in curly brackets and complex numbers as an option in round brackets. In TABLE 3.4 and TABLE 3.5 the input conventions are explained using examples.

Table 3.4: Examples for entering sets

Command	Meaning
{1,2}	1 and 2
{1:5} or {1..5}	1 to 5
{1,4,7..9}	1, 4 and 7 to 9
{}	Empty set
{*}	General set

Table 3.5: Examples for entering complex numbers

Command	Meaning
0;1 or (0;1)	i
–1.23E4;–5.67E8 or (–1.23E4;–5.67E8)	–1.23E4–5.67E8 · i

Cell links

You will find a description what kind of cell links you can generate in the signal processor in SECTION 1.5.5 under Calculation table. A distinction is made between relative and absolute cell links:

Relative Cell Links: If you create a formula, the references to cells are shown relative to the position of the cell which contains the formula.

For example the cell B6 contains the formula =A5. The signal processor searches the value in the cell which is located one cell above and one cell on the left of B6. References of this type are called relative cell links. If you copy a formula in which relative cell links are used, then the links were automatically adapted after inserting the formula to refer to other cells in relative to the position of the formula.

Absolute Cell Links: If you want to prevent that the links will be adapted when copying a formula in another cell, use an absolute cell link.

If, for example, a formula multiply the cell A5 by the cell C1 (=A5*C1) and is copied into another cell, both references will be adapted. You can generate an absolute cell link to the cell C1 by entering a dollar symbol ($) in front of the part of the link that should not be changed. To generate, for example, an absolute cell link to the cell C1 enter (=A5*C1) in the formula.

3.7.3 Calculation Examples for Functions

Differentiation in the frequency range

In the following table you will find an example for differentiation in the frequency range.

Cell	Function	Description
A1	= Complex spectrum	From measurement
B1	=IFFT(A1)	Transformation in the time domain
B2	=Differentiate(B1)	Differentiation in the time domain
A2	=2*Pi()*(0;1)*Abscissa(A1)	Multiplication with $2\pi \cdot i \cdot f$ in the frequency range corresponds to differentiation in the time domain.
A3	=A1*A2	Multiplication
B3	=IFFT(A3)	Transformation in the time domain

To calculate B2, first of all transform into the time domain and then differentiate there. To calculate B3, multiply in the frequency range with a function which corresponds to differentiation in the time domain. The result is the same apart from fringe effects and shifts in the order of magnitude of the sampling. Multiplication with $2\pi \cdot f$ corresponds to a high pass, multiplication with i to a phase shift by 90°.

Integration in the frequency range

In the following table you will find an example for the integration in the frequency range.

Cell	Function	Description
A1	= Complex spectrum	From measurement
B1	=IFFT(A1)	Transformation in the time domain
B2	=Integrate(B1)	Integration in the time domain
A2	=2*Pi()*(0;1)*Abscissa(A1)	Division by $2\pi \cdot i \cdot f$ in the frequency range corresponds to integration in the time domain.
A3	=A1/A2	Division
B3	=IFFT(A3)	Transformation in the time domain

To calculate B2 first of all transform into the time domain and then integrate there. To calculate B3, multiply in the frequency range with a function which corresponds to integration in the time domain. The result is the same as with differentiation apart from an additive constant.

Average over partition

In the following table you will find an example for creating the average over a partition.

Cell	Function	Description
A1	= Single sample point	From measurement
A2	=(sign(Abscissa(A1)–4000)–sign(Abscissa(A1)–6000))/2	Function which has the value 1 in the interval [4 kHz…6 kHz] and has the value 0 outside
A3	=A1*A2	Values outside the interval [4 kHz…6 kHz] are set to 0.
A4	=Mean(A3)/Mean(A2)	Average in the range [4 kHz…6 kHz]

Power spectral density

In the following table you will find an example for calculating the power spectral density.

Cell	Function	Description
A1	= Complex spectrum	From measurement
A2	=abs(A1)^2/Resolution(A1)	Power spectral density

Calculation of MAC values

MAC values (Modal Assurance Criterion) are used to compare band data (Operational Deflection Shapes) or modal deflection shapes (Mode Shapes) and have values between 0 and 1. Here, a MAC value of < 0.9 means that the shapes are different, a MAC value of > 0.9 means that the shapes are similar and a MAC value of 1 means that the shapes are the same. In the following table you will find an example of calculating a MAC value from two bands.

Cell	Function	Description
A1	= Band data	From measurement or imported calculation. Display a band in a measurement file and select Copy Selected Band in the context menu. Insert the band in the cell A1.
B1	= Band data	The band to be compared
B2	=Re(B1)–(0;1)*Im(B1)	Calculates the conjugated complexes of cell B1
B3	=StatMean(A1)*StatMean(B2)	Calculated the cross correlation between the bands. StatMean is used to give both bands the band index 0 and can then be multiplied with each other.
C1	=Abs(Mean(B3))^2/(Mean(Abs(A1)^2)*Mean(Abs(B1)^2))	Calculating the MAC value

3.7.4 Setting up Signal Description

You can adapt the presentation of the recalculated measurement data in the analyzer. To do so, proceed as follows:

1. Use the right mouse button to click the cell which contains the recalculated measurement data and select Signal Description in the context menu.

   The Signal Description dialog appears.

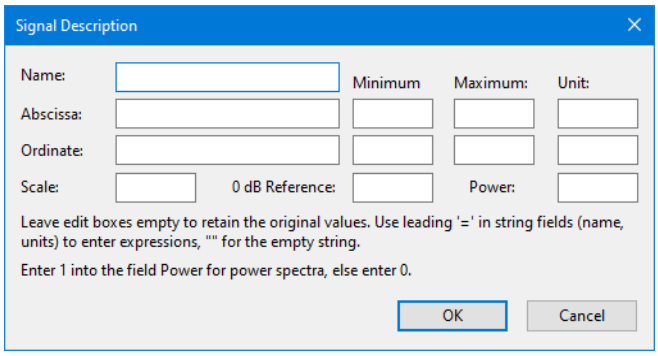


Figure 3.30: Signal Description dialog in the signal processor

2. Fill in the fields as described in the following.
3. Click OK.

ⓘ Entries in this dialog are saved and shown in any case, even if the input data for the signal calculation changes. If you do not enter anything in a field, the original value from the input data is retained or adapted.

Dialog description

Name: Enter the description of the recalculated measurement data. This name is shown in the respective cell. After saving the measurement data it will also appear on the list of available signals in the analyzer. Refer also to SECTION 3.7.5.

Abscissa: Enter the description for the x-axis.

Ordinate: Enter the description for the y-axis.

Minimum: Enter the lower value for the scale of the x- or y-axis.

Maximum: Enter the upper value for the scale of the x- or y-axis.

Unit: Enter the abbreviation for the unit of the x- or y-axis.

Scale: Enter the factor by which the presentation of the measurement values is to be magnified or reduced.

0 dB Reference: If you have selected a view with [dB], here you enter the value which corresponds to 0 dB (reference value).

Power: If for example the signal includes a cross power spectrum, enter the value 1. If the signal is not a power signal, enter the value 0 (default).

3.7.5 Saving Calculated Data (User Defined Data Sets or Bands)

You can add the measurement data which you calculated in the signal processor to the original file as user defined data sets or bands.

ⓘ The indices of the scan points in the original file have to match the indices of the data set. If not, only the data with corresponding indices is applied. The geometry of the objects, i.e. the number of scan points should be the same. If not, a message will appear and you will be asked whether you want to apply the data sets.

Add measurement data

You have the following options for adding the calculated measurement data:

1. Mark the cell which contains the recalculated measurement data in the signal processor.
2. Click the cell with the right mouse button and select Copy in the context menu.
3. Then use the right mouse button to click the name of the original file in the project browser and select Paste in the context menu.

Or:

4. Mark the cell which contains the recalculated measurement data in the signal processor.
5. Hold the left mouse button pressed and drag the cell to the analyzer.

Or:

6. Mark the cell which contains the recalculated measurement data in the signal processor.
7. Hold the left mouse button pressed and drag the cell to the name of the original file in the project browser.

In the Object view, you will now immediately have another channel available with the name Usr, Usr X, Usr Y, Usr Z or Usr 3D. In the list of signals you will find the recalculated measurement data listed with the name which you entered in the Signal Description dialog.

ⓘ You can also copy original signals or original bands respectively and can insert them as user defined signals or bands respectively.

Display properties

You can display the properties of user defined signals and change or delete them. To do so, proceed as follows:

1. Select Analyzer > Channel in the menu bar (toolbar of the analyzer: ) and select the Usr, Usr X, Usr Y, Usr Z or Usr 3D channel in the context menu or pull-down menu.
2. Select Analyzer > Signal in the menu bar (toolbar of the analyzer: ) and select Properties in the context menu or pull-down menu.

   The Properties of User Defined Signals dialog appears.

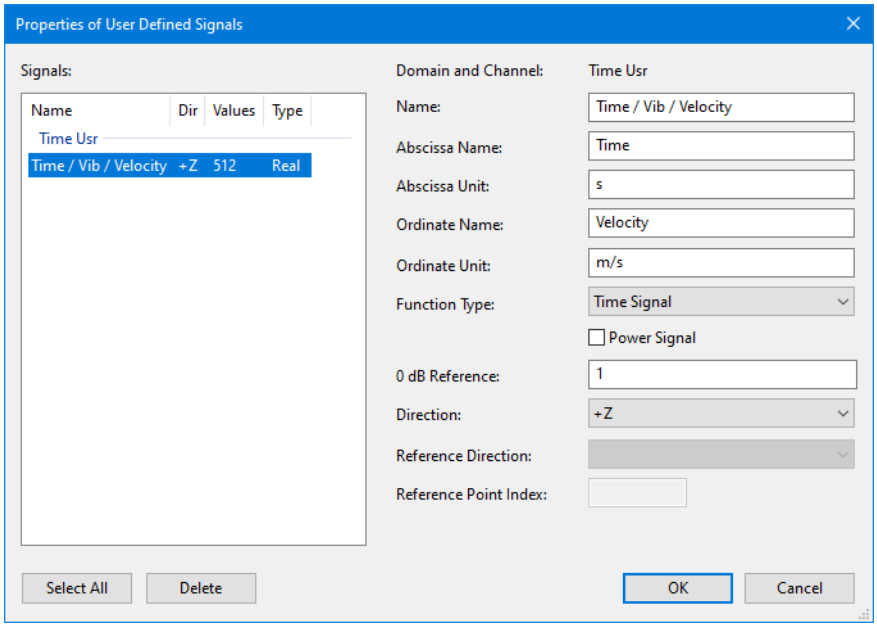


Figure 3.31: Properties of User Defined Signals dialog

Dialog description

Signals or Band Signals: On the left you will see the user defined signals or band signals with the name you assigned in the Signal Description dialog.

Domain and Channel: The domain, the channel as well as the signal name and other properties are shown. You can change these properties for every signal. If you have inserted user defined bands, you will see Band Definitions right at the bottom.

Select All: Click here if you want to assign a property to all user-defined signals or bands (see below).

Delete: You can delete user defined signals or bands. To do so, mark the signal or band and click Delete. The user defined data set is deleted from the file.

Equalize properties

To set the properties of several user defined signals to the same value, proceed as follows:

1. Mark the respective signals in the list. To do so, hold the Control key pressed and click the signals.
2. Enter the required common property on the right.
3. As soon as you now change the signal selection or click OK, the changes are applied to the signals.

3.7.6 Displaying Measurement Results

Analyzer in the signal processor

At the same time as recalculating the data, the updated graph is shown in the analyzer. The analyzer of the signal processor has the functions described in SECTION 1.5.3. If you mark several table cells, then you will be shown the respective frames for selection. If you select several frames, you will be shown the respective graphs in parallel.

Example

ⓘ Band data is plotted in the analyzer using the scan point number. The scan point number does not necessarily correspond to the scan point index shown in the presentation window.

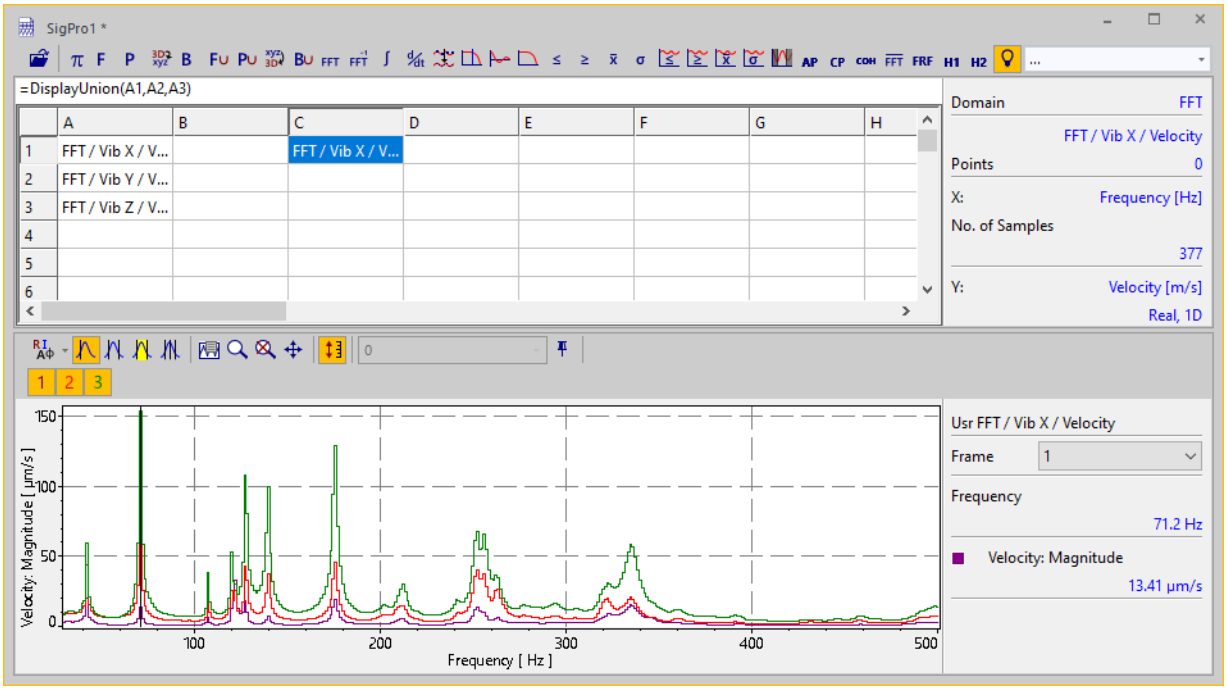


Figure 3.32: Displaying several graphs in the analyzer of the signal processor

Fix the view style

If you would like to stop updating the graph at a certain point in the arithmetic chain, click (Fix). The presentation then corresponds to the contents of the table cell selected at that moment in time. All subsequent arithmetic steps are carried out anyway, but do not have any effect on the presentation of the graph. However, changes in the formulae before the position marked with still have an effect on the adaptation of the presentation up to this point.

3.7.7 Saving Formulae

To save formulae after you have recalculated the measurement data, proceed as follows:

1. In the menu bar, select File > Save As (toolbar: ).

   The Save As dialog appears.

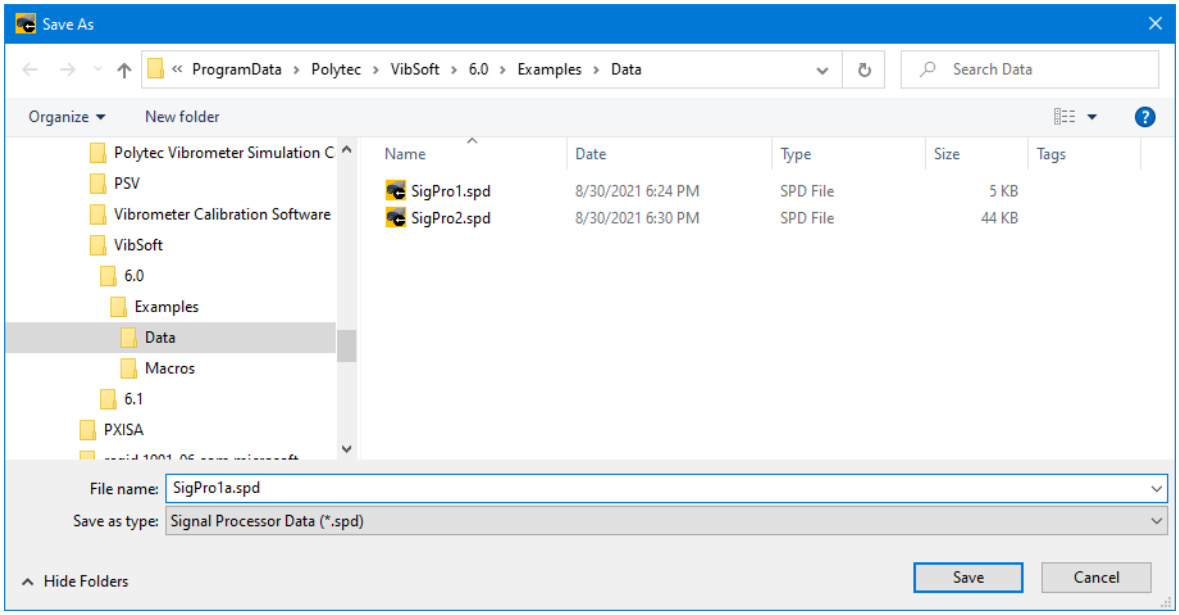


Figure 3.33: Save As dialog for signal processor files

2. Navigate to the saving location and enter the file name.
3. Click Save.

   The file saved in this way does not contain any measurement data itself, but only contains references to data. The reference among other things includes the full path name of the original file. If you move or rename the measurement, then the references to the measurement data become invalid. To permanently save your results, you have to add them to the original file.

ⓘ A file in Signal Processor Data (*.spd) format does not contain any data, only calculation specifications!

3.7.8 Finding and Replacing in Formulae

To search for functions or expressions in formulae and replace them accordingly, proceed as follows:

1. Click the desired cell with the right mouse button and select Find and Replace in Formulae in the context menu.

   The Find and Replace in Formulae dialog appears.

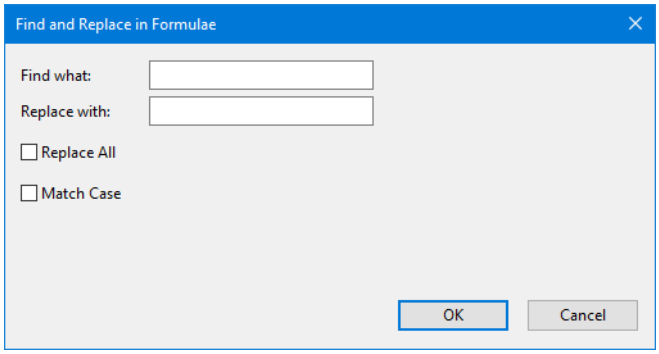


Figure 3.34: Find and Replace in Formulae dialog in the signal processor

2. Enter the expression you want to replace in the Find what field.

ⓘ If you want to specify the search, tick the Match Case check box.

3. Enter the new expression in the Replace with field.
4. If you want to replace all expressions found in this way at once, additionally tick the Replace All check box.
5. Click OK.

3.7.9 Selecting Files

You can directly select a file in the signal processor to replace the copied measurement data (CopyVibData function). The following file formats are available, depending on your measurement system:

- Scan Data (*.svd)
- MPV Data (*.mvd)
- Single Point Data (*.pvd)

You have the following options to select a file for the data in the signal processor:

- Context menu of the right mouse button
- Icons in the toolbar of the signal processor
- Project browser with drag & drop (only in Presentation)

You can undo the selection.

Context menu

For selection via the context menu, proceed as follows:

1. Click one or several selected cells with the right mouse button and select Select File in the context menu.

   The Open dialog appears.
2. Navigate to the saving location and select Files of type and File name.
3. Click Open.

   The following options are available:

Selection	Content	Effect
One cell	Without CopyVibData function	The relative path name is transferred to the selected cell as a string (RelativePath). If it is not possible to use the relative path, the absolute path will be used.
One cell	With CopyVibData function with path name	In the selected cell, the FilePath (pathname) parameter of the CopyVibData function will be replaced with the new path name.
Several cells	Without CopyVibData function	None
Several cells	With CopyVibData function with a path name	In the selected cells, the FilePath (pathname) parameter of the CopyVibData function will be replaced by the new path name.

Toolbar

To use the icon in the toolbar, proceed as follows:

1. In the toolbar of the signal processor, click  .

   The Open dialog appears.
2. Navigate to the saving location and select Files of type and File name.
3. Click Open.

   In all cells with CopyVibData function, the FilePath (path name) parameter will be replaced by the new path name.

Project browser

To use drag & drop in the project browser, proceed as follows (only available in Presentation):

1. Open the scan file, MPV file, or a Single Point Data file, which contains the respective data.
2. Drag the signal processor document from the project browser onto the open file.

   The signal processor document opens, and in all cells with CopyVibData function, the FilePath (path name) parameter will be replaced by the new path name.

3.7.10 Application Examples for Processing Measurement Data

You will find more information on various calculations of measurement data using the following examples:

- The first example shows how to use the signal processor in presentation mode to help you calculate and save the complex averaged spectrum across all scan points so that you have access to it at all times. See SECTION 3.7.10.1 on this.
- The second example shows how you can use the signal processor to multiply an FRF by the spectrum of the excitation signal averaged over all scan points and save and display the result in the original file. See SECTION 3.7.10.2 on this.
- The third example shows the analysis of a time series of the response of a loudspeaker to a rectangular excitation signal. See SECTION 3.7.10.3 on this.
- You can also switch on a function wizard for the functions which are offered as icons in the signal processor. See SECTION 3.7.10.4 on this.

3.7.10.1 Complex Average Spectrum across All Scan Points

During data acquisition, the software calculates an average spectrum across all scan points. However, it is not complex averaging but the amplitudes of the individual spectra are calculated and these amplitudes are averaged.

This example shows how to use the signal processor in presentation mode to help you calculate and save the complex averaged spectrum across all scan points so that you have access to it at all times.

1. To create a new signal processor document, select File > New > Signal Processor in the menu bar (toolbar:  > Signal Processor).
2. Open the EXAMPLE2.SVD file.
3. In the toolbar of the presentation window, click  and select the Point Data view from the pull-down menu.

   In the list for frequency bands, the Cursor entry is displayed.
4. In the FFT domain, display the Vib & Ref1 channel and the H1 Velocity/ Voltage signal.
5. In the presentation window, click the object with the right mouse button and select Copy All Points in the context menu.
6. Change to the signal processor, click cell A1 with the right mouse button and select Paste in the context menu.

   The data is available in the signal processor document.
7. Click the cell A2, enter the =StatMean(A1) command and confirm with Enter.

   The signal processor calculates the complex average across all scan points and shows this in the analyzer.

ⓘ The average across all scan points is given the index 0.

8. To scale the data in the analyzer, click  in the toolbar of the analyzer.

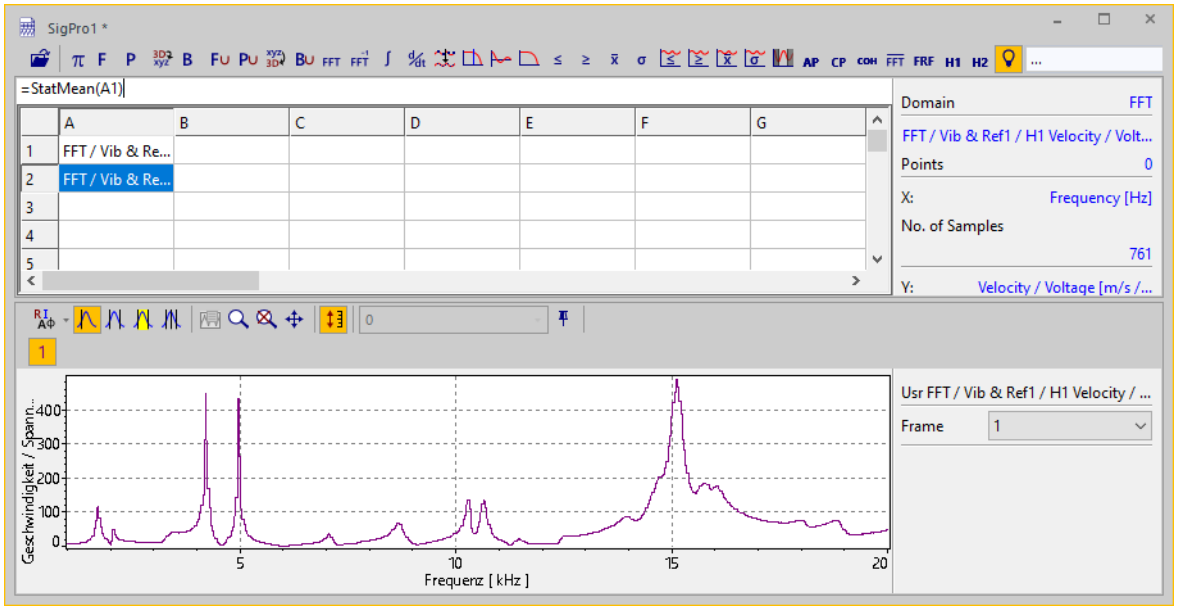


Figure 3.35: Displaying the complex average across all scan points

9. In the toolbar of the analyzer, click  and select the Mag. [dB] & Phase view from the pull-down menu.

   Both the magnitude and the phase of the averaged spectrum are displayed.

10. Click the cell A2 with the right mouse button and select Signal Description in the context menu.

    The Signal Description dialog appears.

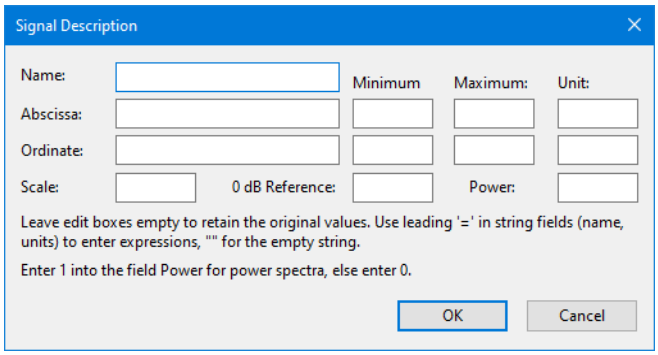


Figure 3.36: Signal Description dialog in the signal processor

11. In the Name field, enter the description Complex Average Spectrum.
12. To close the dialog, click OK.
13. Arrange the signal processor and the presentation window so that they are both completely visible. To do so, select Window > Tile Horizontally in the menu bar.

14. Click the cell A2 in the signal processor.
15. Click the cell A2 again and holding the left mouse button pressed, drag the mouse pointer across to the presentation window. Release the mouse button there.

    This then means that the data in the cell A2 is added to the Usr channel as a user defined signal Complex Average Spectrum.

ⓘ You can also click the cell A2 with the right mouse button and select Copy in the context menu. Change to the presentation window and select the Average Spectrum view here. Select Edit > Paste. The complex average spectrum is added to the EXAMPLE2.SVD document as a user defined signal Usr.

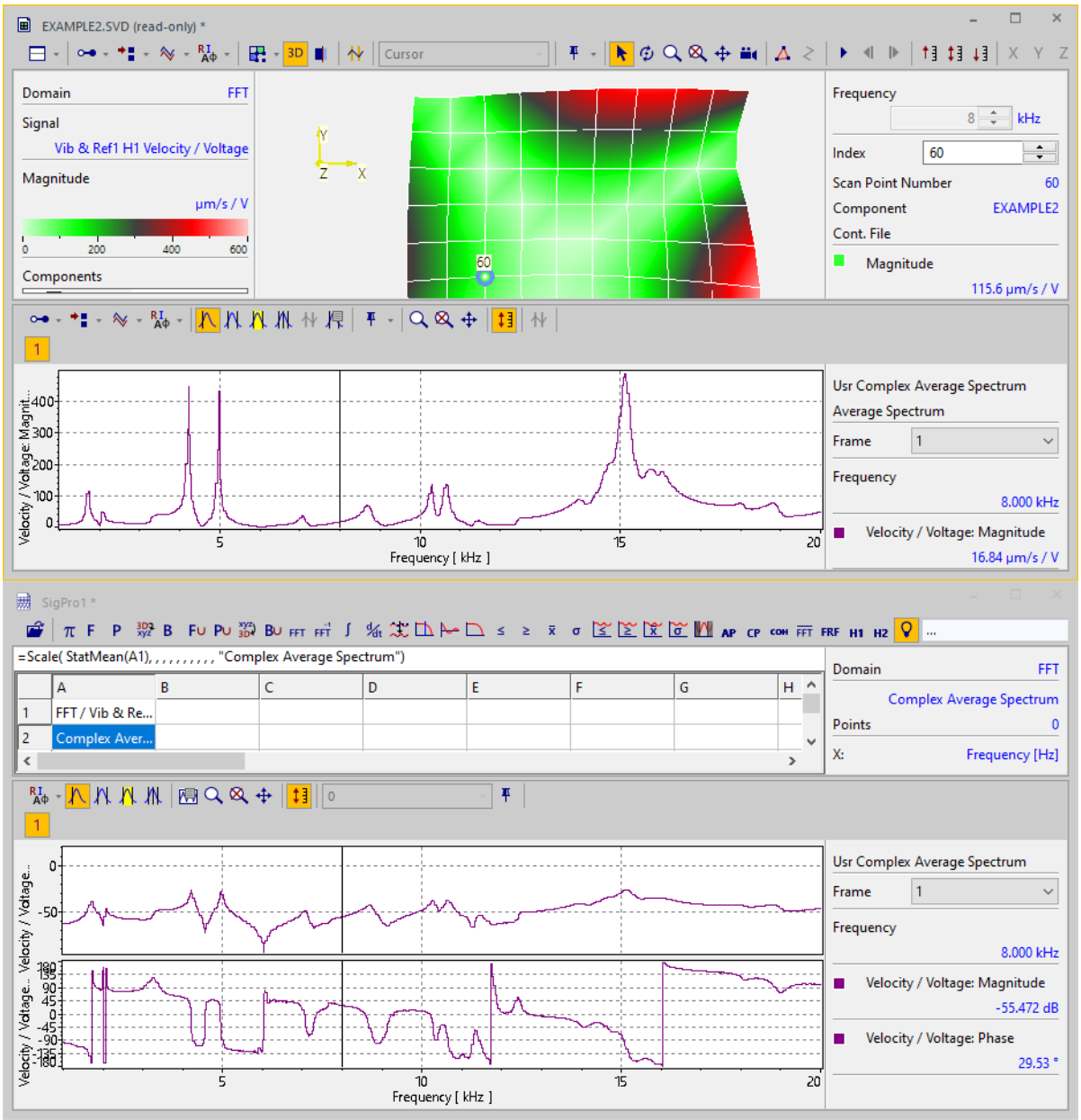


Figure 3.37: Display of the complex average spectrum

16. Save this file with a different name.

    You can now display the complex average spectrum again at any time. To do so, open the file you have saved and select the Average Spectrum view in the presentation window, the Usr channel in the analyzer toolbar, and the Complex Average Spectrum signal in the analyzer toolbar.

ⓘ You can display the properties of the user defined data sets as described in SECTION 3.7.5.

3.7.10.2 Multiplication of an FRF with the Reference Signal

An FRF (Frequency Response Function) is generated by dividing an answer signal (vibrometer signal) by a reference signal (excitation signal). The same applies to the H1 and H2 signals. This means that, for example, the FRF has the unit Velocity /Voltage. The advantage of this process is that fluctuations in the excitation over the duration of the scan do not have any great effect because the measurement at every scan point relates to the excitation measured at this scan point. However, it is often desirable to show the answer in its original unit.

This example shows how you can use the signal processor to multiply an FRF by the spectrum of the excitation signal averaged over all scan points and can save and display the result in the original file.

1. To create a new signal processor document, select File > New > Signal Processor in the menu bar (toolbar:  > Signal Processor).
2. Open the EXAMPLE2.SVD file.
3. In the toolbar of the presentation window, click  and select the Point Data view from the pull-down menu.

   In the list for frequency bands, the Cursor entry is displayed.
4. In the FFT domain, display the Vib & Ref1 channel and the H1 Velocity/ Voltage signal.
5. In the presentation window, click the object with the right mouse button and select Copy All Points in the context menu.
6. Change to the signal processor, click cell A1 with the right mouse button and select Paste in the context menu.

   The data is available in the signal processor document.
7. Change to the presentation window and select the Voltage signal of the Ref1 channel.
8. In the presentation window, click the video image with the right mouse button and select Copy All Points in the context menu.
9. Change to the signal processor, click cell A2 with the right mouse button and select Paste in the context menu.

   The data is available in the signal processor document.

10. Click the cell A3, enter the =A1*StatMean(A2) command and confirm with Enter.

    The signal processor first of all calculates the average of the spectra of the excitation signal across all scan points (StatMean(A2)) and multiplies the result at every spectrum of the signal H1, which is taken from the cell A1. The result is shown in the analyzer.

11. To scale the data in the analyzer, click  in the toolbar of the analyzer.

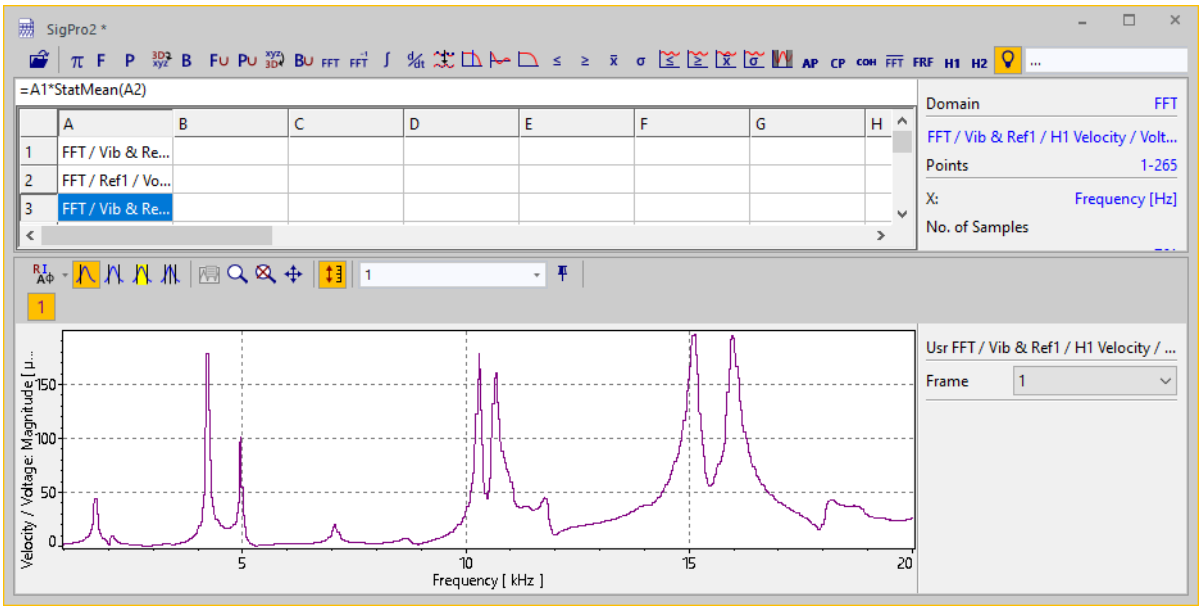


Figure 3.38: Displaying the multiplication of the averaged spectrum of the excitation signal with the signal H1

12. Click the cell A3 with the right mouse button and select Signal Description in the context menu.

    The Signal Description dialog appears.

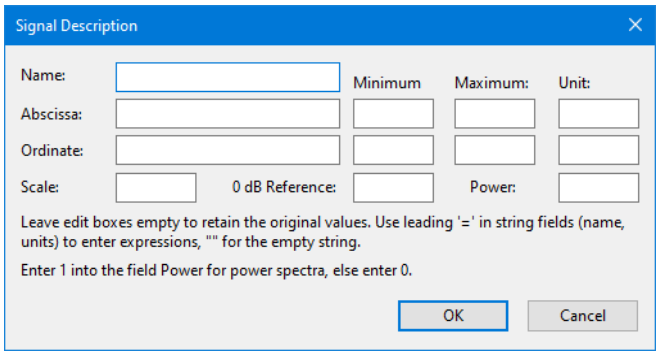


Figure 3.39: Signal Description dialog in the signal processor

13. In the Name field, enter the description H1 Vib*Ref1 Velocity.

ⓘ For the Integrate, Differentiate, FFT and IFFT functions, the software automatically adapts the axis name and the unit. For the other functions, enter the name in the dialog.

14. In the Ordinate field, enter the description Velocity.
15. In the Unit field of the ordinate, enter the unit m/s.

16. To close the dialog, click OK.
17. Arrange the signal processor and the presentation window so that they are both completely visible. To do so, select Window > Tile Horizontally in the menu bar.
18. Click the cell A3 in the signal processor.
19. Click the cell A3 again and holding the left mouse button pressed, drag the mouse pointer across to the presentation window. Release the mouse button there.

    The data in the cell A3 is then added to the Usr channel as a user defined signal H1 Vib*Ref1 Velocity.

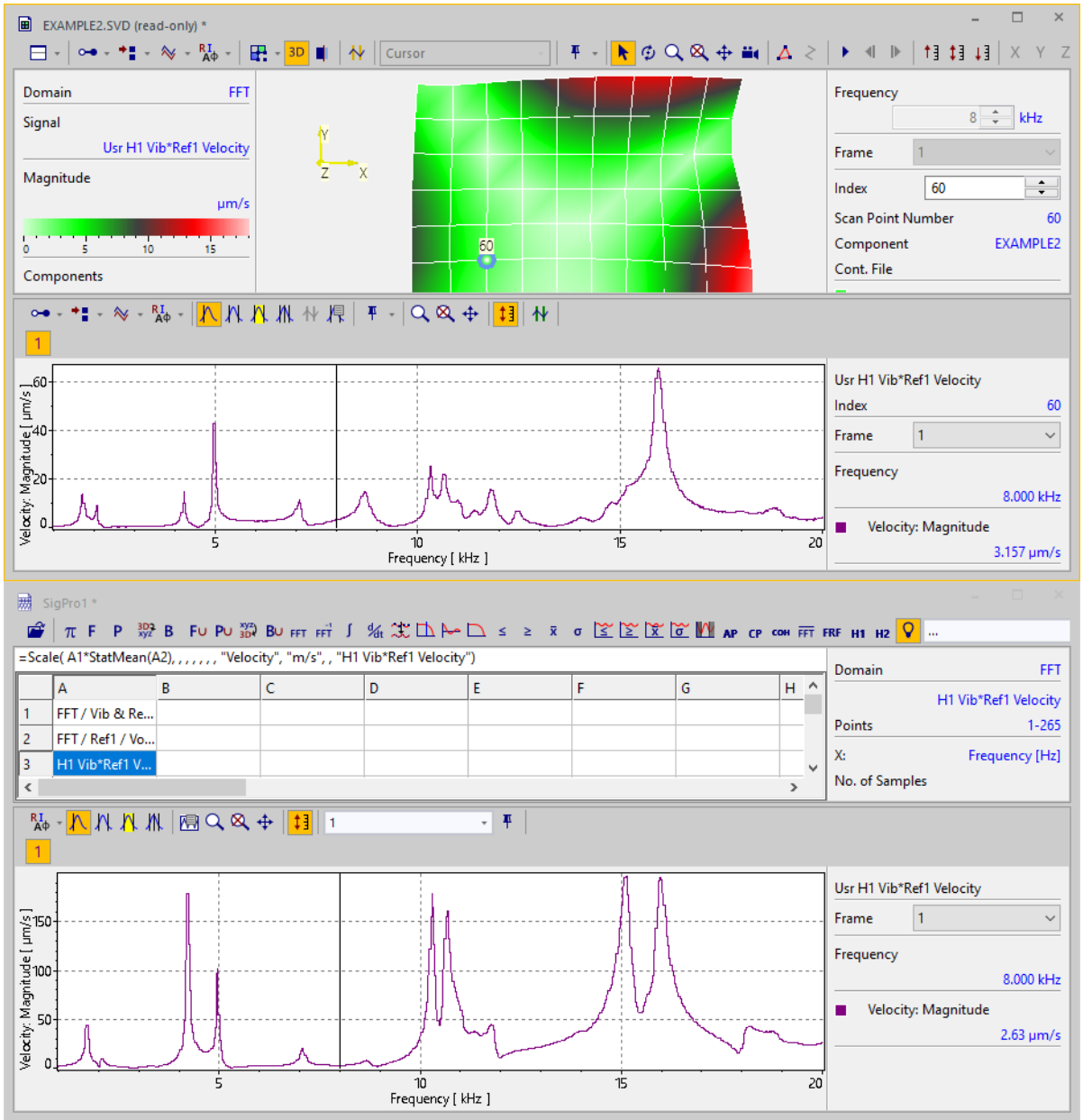


Figure 3.40: Displaying the multiplication of FRF and reference signal

20. Save this file with a different name.

    From now on you can display the signal H1 multiplied with the average spectrum of the reference channel at any time. To do so, open the file you saved and select the Usr channel and the H1 Vib*Ref1 Velocity signal in the presentation window.

ⓘ You can display the properties of the user defined data sets as described in SECTION 3.7.5.

3.7.10.3 Post Processing Time Series

With the Time Domain Data software option, data can be captured in the Time domain. This data can be evaluated using the signal processor.

The following example shows the analysis of a time series of the response of a loudspeaker to a rectangular excitation signal.

1. To create a new signal processor document, select File > New > Signal Processor in the menu bar (toolbar:  > Signal Processor).
2. Open the file Ex Time.pvd.
3. In the Time domain, display the Vib channel and the Velocity signal.
4. Click the analyzer with the right mouse button and select Copy in the context menu.

   The Copy dialog appears.

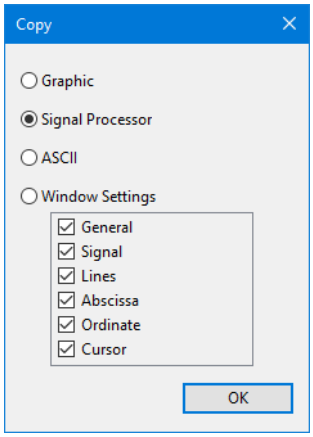


Figure 3.41: Copy dialog for signal processor

5. Select Signal Processor and click OK.
6. Change to the signal processor, click cell A1 with the right mouse button and select Paste in the context menu.

   The data is available in the signal processor document. In the analyzer of the signal processor you will see a graphic representation of the time series measured.

7. Click the cell A2, enter the =Integrate(A1) command and confirm with Enter.

8. Click the cell A2 with the right mouse button and select Signal Description in the context menu.

   The Signal Description dialog appears.

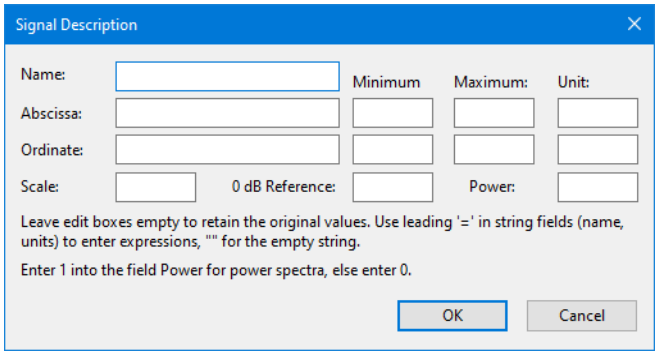


Figure 3.42: Signal Description dialog in the signal processor

9. In the Name field, enter the description Integrated to Displacement.

ⓘ For the Integrate, Differentiate, FFT and IFFT functions, the software automatically adapts the axis name and the unit. For the other functions, enter the name in the dialog.

10. In the Ordinate field, enter the description Displacement.
11. In the Minimum field of the ordinate, enter the value –10e–6.
12. In the Maximum field of the ordinate, enter the value 10e–6.
13. In the Unit field of the ordinate, enter the unit m.
14. To close the dialog, click OK.
15. To scale the data in the analyzer, click  in the toolbar of the analyzer.

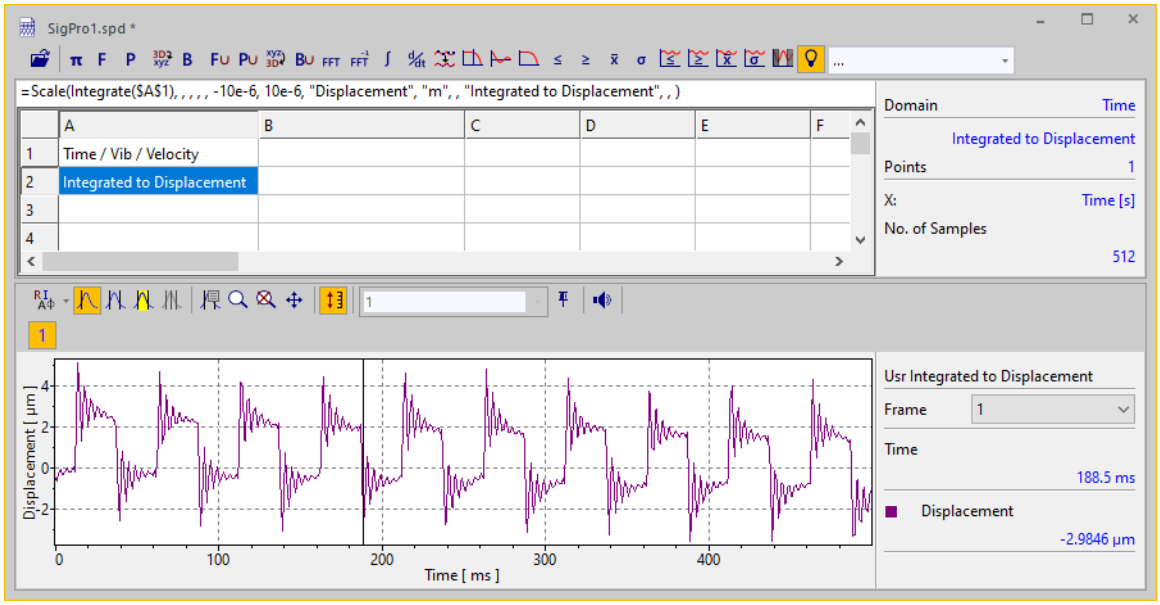


Figure 3.43: Displaying the displacement signal attained through integration

The analyzer will now display the displacement signal gained by integrating the velocity signal. Along the time axis you will be able to see a linear decay of the signal. This decay is caused by a negative offset on the velocity signal which has an increased effect due to insufficient amplitude level of the data acquisition board when acquiring the velocity signal.

16. Copy the contents of cell A2 into cell B2. To do so, select cell A2 and press the Strg + C key combination. Then select cell B2 and press the Strg + V key combination.
17. Click in the formulae editor where the formula from cell B2 is displayed.
18. There replace the (A1) parameter of the Integrate function by entering (A1-Mean(A1)) and confirm the change with Enter.

    This changing to the formula means that the offset from the time series is subtracted in cell A1. For this purpose, first of all the average of the time series is determined and this average is subtracted from every time value in the time series. As the time series is periodic in the time window measured, this has the effect of completely correcting for the offset. Then the integration is calculated.
19. Click the cell B2 and also select the cell A2. To do so, hold the Control key pressed and click the cell A2.

    The analyzer in the signal processor now shows two graphs. One corresponds to the signal in cell B2 and one to the signal in cell A2. Through a direct comparison of the graphs, you can clearly see the disappearance of the linear decay by correction of the offset.

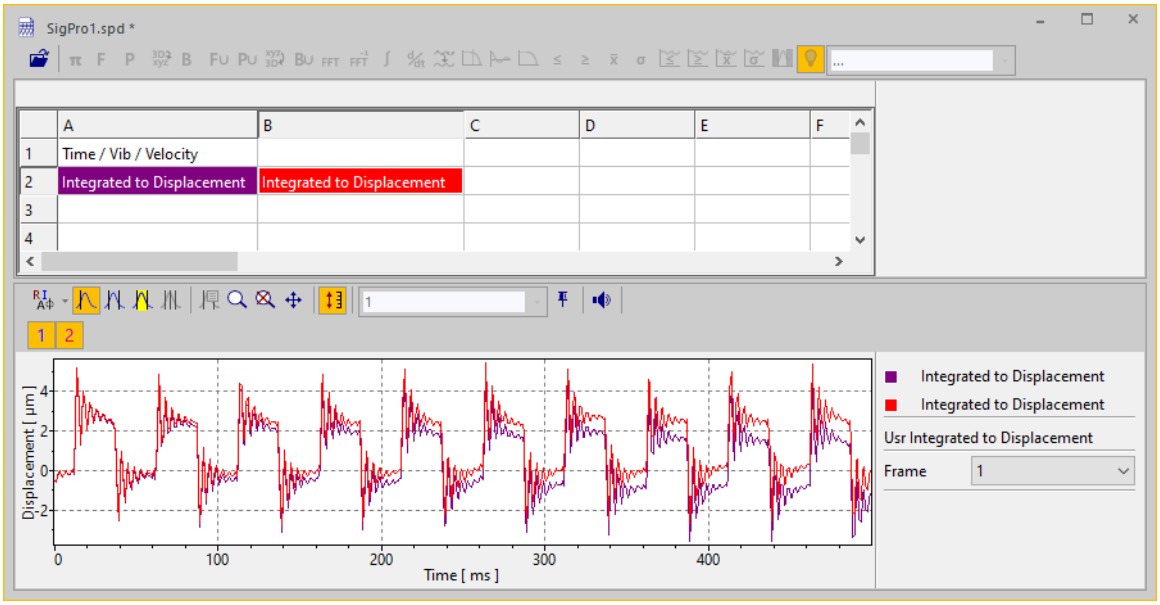


Figure 3.44: Displaying two graphs in the analyzer of the signal processor

20. Click the cell B3 and activate the input line.

21. In the toolbar of the signal processor, click FFT.

    This command inserts a template for the FFT function into the input field and the formulae editor.

22. Replace the descriptions VibData with B2, Lines with 256 and Window with "Rectangle" and delete the two WindowParameter descriptions. When doing so, make sure that you put the value "Rectangle" in quotation marks.

23. Confirm the change with Enter.

ⓘ You can also use the function wizard. See SECTION 3.7.10.4 on this.

24. Click the cell B3 with the right mouse button and select Signal Description in the context menu.

    The Signal Description dialog appears.

25. In the Name field, enter the description Displacement FFT.

26. In the Abscissa field, enter the description Frequency.

27. In the Unit field of the abscissa, enter the unit Hz.

28. To close the dialog, click OK.

    From the 512 samples of the integrated time series in cell B2, the signal processor will now calculate an FFT with 256 lines and will display this in the analyzer.

29. Arrange the signal processor and the presentation window so that they are both completely visible. To do so, select Window > Tile Horizontally in the menu bar.

30. Click the cell B2 in the signal processor.

31. Click the cell B2 again and holding the left mouse button pressed, drag the mouse pointer to the analyzer. Release the mouse button there.

    This then means that the data in the cell B2 is added to the Usr channel as a user defined signal Integrated to Displacement.

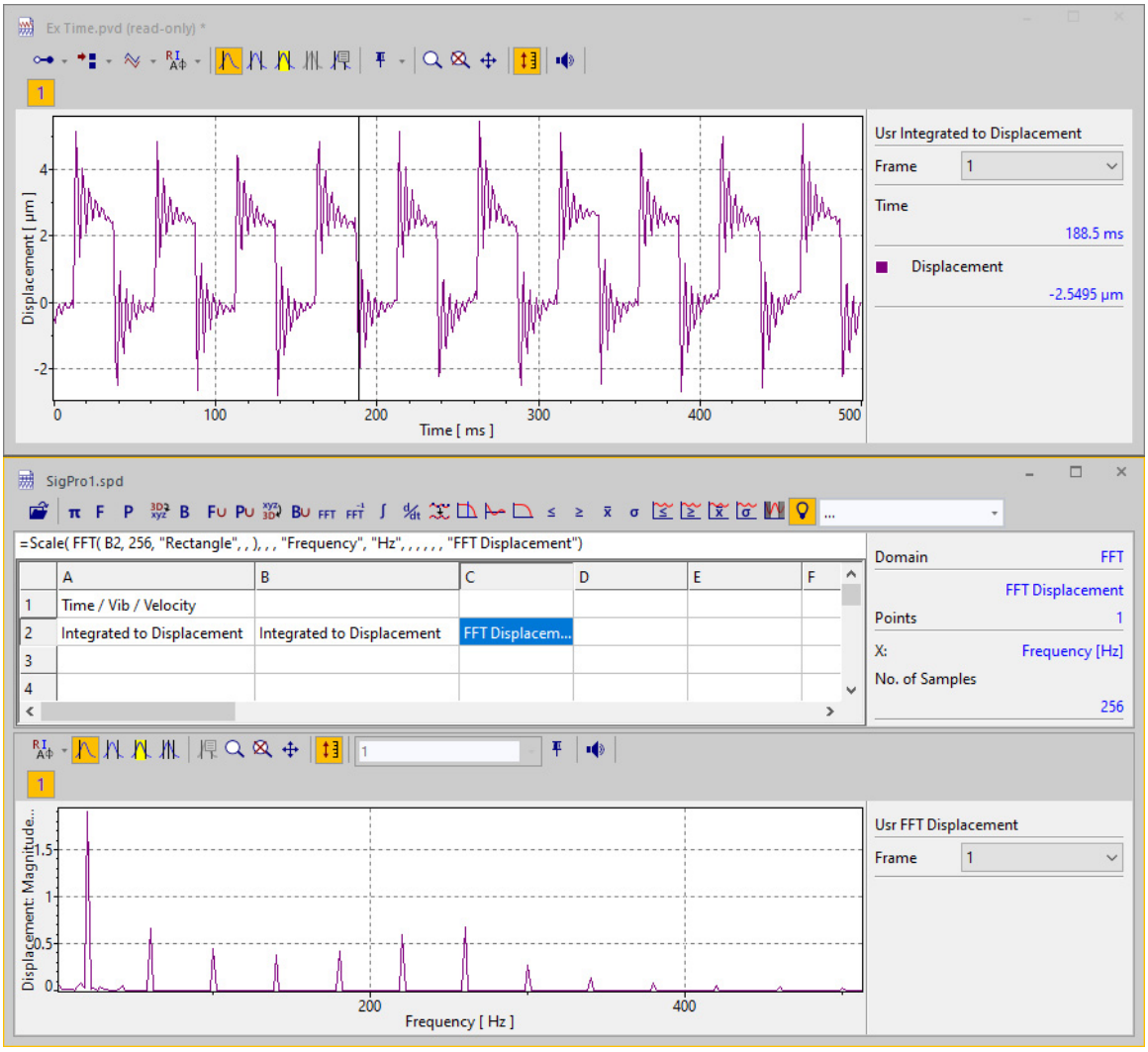


Figure 3.45: Display of post processing time series

32. Repeat steps 30 and 31 for the cell B3.

    Thus you are adding the contents of the cell as a user defined signal FFT Displacement.

33. Save this file with a different name.

    You can now display the integrated time series and the FFT calculated from it at any time. To do so, open the file you saved and select the Usr channel in the FFT or Time domain.

ⓘ You can display the properties of the user defined data sets as described in SECTION 3.7.5.

3.7.10.4 Example of the Function Wizard

You can also switch on a function wizard for the arithmetic operations which are offered as icons in the signal processor. To do so, click  in the toolbar of the signal processor.

As an example you replace steps 20 and 21 from SECTION 3.7.10.3 with the following instruction:

1. In the toolbar of the signal processor, click  .

   The Function Wizard dialog appears.

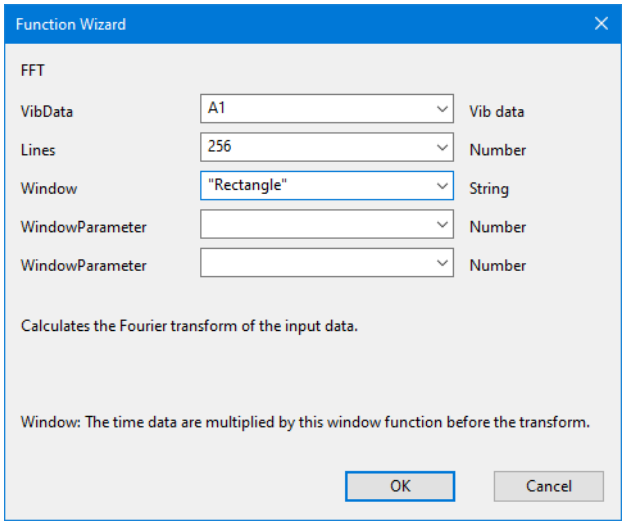


Figure 3.46: Function Wizard dialog for the signal processor

2. For VibData enter the value B2, for Lines the value 256 and for Window select the entry "Rectangle" from the list.
3. Click OK.
4. Then proceed with step 22.

3.8 Creating a New Scan File

If you want to add user defined data sets to 3D or 2D geometries which have already been acquired, you will need to generate a new scan file. To do so, proceed as follows:

You can generate a new scan file from the following file formats:

- Scan Data (*.svd)
- Settings Files (*.set)
- ASCII Files (*.txt)
- PLY Files (*.ply)
- OBJ Files (*.obj)
- STL Files (*.stl)
- ASAM ODS ATFX Files (*.atfx)
- ME'Scope5 Project Files (*.vtprj)
- ME'Scope Structure Files (*.str)
- Universal Files (*.unv, *.uff, *.asc)

1. Select File > New > Scan File in the menu bar (toolbar: > Scan File).

ⓘ You can also use the command Create Scan File in the project browser. See SECTION 1.5.4 on this as well.

The Open dialog appears.

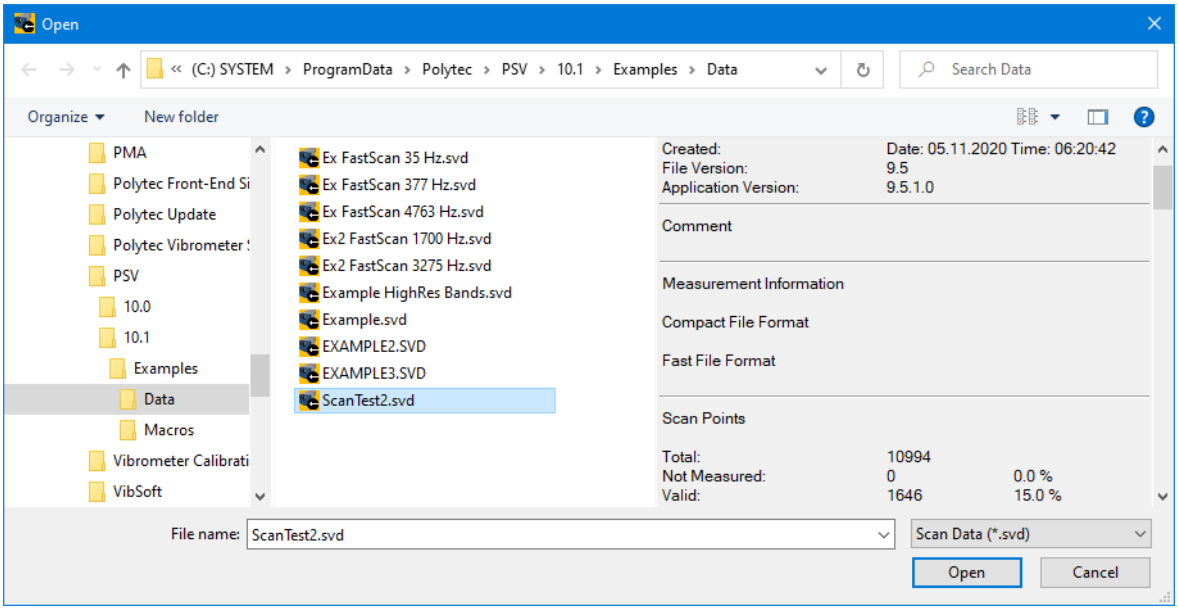


Figure 3.47: Open dialog to generate a new scan file

2. Navigate to the saving location and select Files of type and File name.
3. To identify the files more easily, show the preview of the meta information by clicking .

You will receive information on the project, on scan points, the hardware, and the setting used to measure the data.

4. Click Open.

   In the new scan file, only the 3D or 2D geometry respectively is applied. If the selected file does not include any geometry, you will receive an error message.

5. You can now add user defined data sets and save the file.

4 Evaluating Measurement Data

The software offers numerous possibilities to evaluate measurement data. To do this, first open the measurement file as described in SECTION 5.1.2. Depending on which measurement mode you used to acquire the data, you will see the results of your measurement in the time or frequency range.

You can see which mode a measurement was made in by opening the additional information in the dialog at the bottom of a file. Read SECTION 5.1.1 on this.

If you have opened a measurement file, you will see in the status bar which mode the measurement was made in.

When you open a measurement file for the first time in presentation mode which was generated in FFT measurement mode, you will see the root mean square of the amplitude over the entire bandwidth. You can apply certain evaluation functions directly to the root mean square. See SECTION 4.1 and SECTION 4.3 on this.

You can also evaluate sections of the measured spectrum – for example, in the proximity of resonance or 1/3 octave bands. See SECTION 4.2 on this. You first define these frequency bands as described in SECTION 4.2.1. At the peak frequencies of the frequency bands you can also animate the vibration, see SECTION 4.4 on this. You can also evaluate the spectrum with a single cursor, see SECTION 4.2.2 on this.

In the FastScan measurement mode, you only measure a certain individual frequency. Correspondingly you will only see a surface-like presentation of the data of this frequency if you open the measurement file in presentation mode.

When you open a measurement file in presentation mode for the first time which had been generated in Time measurement mode, you will see the root mean square of the amplitude over the whole sample time. With the time data animation you can present the momentary amplitudes of the sample time for selected ranges.

You can retrospectively edit scan points in presentation mode. See SECTION 4.5 on this.

4.1 Processing Data

During the measurement, the scan points are given a status. Explanations for the different statuses can be found in SECTION 4.1.1.

You can use the following commands in the context menu or in the Presentation menu to prepare measurement data for display:

- You can smooth the presentation of areal measurement data with a Filter. See SECTION 4.1.2 on this.
- In principle, you can switch on the function to Interpolate scan points. See SECTION 4.1.3 on this.
- You can still retrospectively change the direction of the vibration in the presentation window. See SECTION 4.1.4 on this.

4.1.1 Status of Scan Points

During the scan, the scan points are given a status. There are different views for the status which are described in the following:

- Scan
- Geometry
- Laser Focus
- Interpolate - See SECTION 4.1.3 on this.

ⓘ You can display the explanations for each status mentioned below in the software as well. To do so, leave the mouse pointer on the name of the status until the message text is displayed.

Select scan points

ⓘ Regardless of the view style of the data, you can show/hide a legend for the status of the scan points on the left of the object by selecting Presentation > Status in the menu bar. If the legend on the left of the presentation window is not visible, previously select Presentation > Legend to display it.

You can display the scan points of a certain status on the object. To do so, tick the corresponding check box of the point status. The other way round, you can select scan points on the object using the mouse, and in the legend the statuses of the selected scan points are displayed.

Scan status

To change the views, select the required view in the Status list.

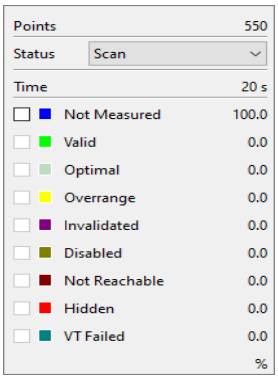


Figure 4.1: Display of the Scan status

Not Measured: Before a scan, all scan points have the Not Measured status.

Valid: When scanning, the software gives every scan point the Valid, Optimal or Overrange status.

Optimal: The Optimal status can only occur if Signal Enhancement is active for at least one channel. See the acquisition mode on this. At these scan points, the signal-to-noise ratio is relatively high.

ⓘ If Signal Enhancement is active, scan points can change their status between Valid and Optimal at the end of the scan.

Overrange: At scan points with Overrange status, the input range of the data acquisition board was exceeded on at least one channel. However, the PSV or MSA-100-3D works with an amplitude level reserve of 25% (M-/V systems) or of 100% (A-/B-/H systems) respectively. The measurement signals can exceed the nominal input range by the specified amounts (is detected as Overrange), without them being limited.

ⓘ The software may determine the Overrange scan status, even though the resulting time signal is displayed unaffected. This is due to the fact that, depending on the measurement settings, you are working with a higher sample rate than actually visible in the resulting time signal (Oversampling). If a brief voltage peak causes overloading, the scan point is given the Overrange status, even though the decimated time signal is displayed correctly.

Invalidated: After the scan, you can manually assign the scan points the Invalidated status. For details, see SECTION 4.1.3.

Disabled: If you do not want to scan single scan points, you can assign them the Disabled status. See the acquisition mode on this.

Not Reachable: If you define the scan points, then scan points which are outside the scan area of at least one scanning head are given the Not Reachable status.

Hidden: If you are working with 3D geometries and parts of the object hide scan points so that they cannot be reached by at least one laser beam, then these scan points are identified as Hidden. Prerequisite of this is that in the Preferences dialog on the Geometry page, you have activated Hidden Points Calculation. See the acquisition mode on this.

VT Failed: PSV measurement systems only: If VideoTriangulation fails, the geometry status remains unchanged, then no vibration measurement data is recorded, the scan points are given the VT Failed scan status.

Geometry status To change the views, select the required view in the Status list.

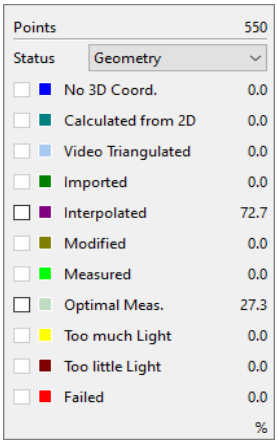


Figure 4.2: Display of the Geometry status

No 3D Coord.: If no 3D coordinates are available, the scan points are given the No 3D Coordinates status.

Calculated from 2D, Triangulated: PSV measurement system: If you are working with a 3D geometry or a distance measurement, you can see whether scan points have been calculated from the 2D geometry (Calculated from 2D, PSV-1D) or by using triangulation of the three laser beams (Triangulated, PSV-3D).

MSA/MSA-3D/UHF measurement system: If you are working with a 3D geometry, you can see whether scan points have been calculated from the 2D geometry (Calculated from 2D).

Video Triangulated: Only PSV measurement system: If VideoTriangulation is successful, the scan points are given the Video Triangulated status. See data acquisition and your PSV Theory Manual on this.

Imported: If you import the geometry from ASCII, Universal File, ME'Scope or ASAM ODS ATFX files using the optional geometry import, the scan points are given the Imported status. See the acquisition mode on this.

Interpolated: If 3D coordinates were interpolated from the coordinates of the neighboring scan points that were not calculated or were calculated incorrectly, they are given the Interpolated status.

Modified: If the coordinates of the scan points are retrospectively changed, they are given the Modified status. See the acquisition mode on this.

Measured, Optimal Meas.: Only PSV/MSA measurement system: If you are working with a 3D geometry or distance measurement, you can see if scan points will be measured with the geometry scan unit or resp. the MSA geometry unit (Measured or Optimal Measured status).

Too much Light, Too little Light: Only PSV measurement system: If the signal strength of the geometry scan unit being too high or too low is the reason for the measurement failing, then you will see this from the Too much Light or respectively Too little Light status.

ⓘ In the cases of Too much Light, Too little Light and Failed, it means that these scan points do not have any valid 3D coordinates.

Failed: Only PSV/MSA measurement system: If the distance measurement has failed, the respective scan points are given the Failed status.

Laser Focus status

To change the views, select the required view in the Status list.

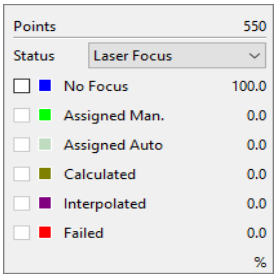


Figure 4.3: Display of the Laser Focus status

No Focus: If no focus value has been assigned, the scan points are given the No Focus status.

Assigned Man.: If you have allocated focus values using the Assign Focus Manually method, the scan points are given the Assigned Manually status. See Acquisition on how to assign focus values.

Assigned Auto: Only PSV/MSA measurement system: If you have allocated focus values using the Assign Focus Automatically method, the scan points are given the Assigned Auto status. See Acquisition on how to assign focus values.

Calculated: PSV/MSA measurement system: If you have ticked the Calculate Focus Values Automatically check box in the Preferences dialog on the Geometry page, and the selected scan points have 3D coordinates, the software will calculate the focus values automatically. These scan points are given the Calculated status.

MSA-3D/UHF measurement system: If you have allocated focus values using the Calculate Focus Values method, the scan points are given the Calculated status.

Interpolated: If the software interpolates the focus value of single scan points from those of the neighboring points, these scan points are given the Interpolated status.

Failed: Only PSV/MSA measurement system: Scan points for which no focus value could be determined using the Assign Focus Automatically method are identified as Failed.

4.1.2 Filtering Data

You can smooth the presentation of area-based measurement data with a filter. Filtering is done on two levels. First of all median averaging and then average filtering of neighboring scan points is carried out.

To smooth the display, select Presentation > Filter Data in the menu bar.

To undo the command, select Presentation > Filter Data agin.

The commands Interpolate Data and Filter Data are measures you can take to remove interference signals and noise levels from the measurement data. When defining the quantity of scan points, already ensure that there is a sufficient scan point density, depending on the maximum vibration frequency of the object to be measured. Careful scan point definition also helps avoiding interference signals caused by unsuitable selection of scan points, e.g. glancing incidence at the edge of an object or undesirable measuring in object openings.

If Filter Data is active, then in ASCII export the smoothened values are saved. If you export as Universal File, ME'Scope and ASAM ODS ATFX or access via Polytec File Access, you will obtain the original data.

4.1.3 Interpolating Points

In principle, you can switch on interpolating scan points by selecting Presentation > Interpolate Points in the menu bar. As a consequence, points with the Interpolate Valid or Interpolate Invalid interpolate status will be interpolated automatically.

To switch off interpolating, select Presentation > Interpolate Points again.

By default, points with the Overrange, Invalidated or VT Failed scan status have the Interpolate Invalid status. All other points will not be interpolated. The various statuses of the single scan points are displayed in the readout of the presentation window.

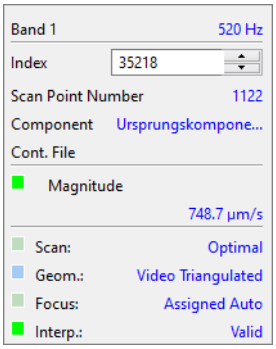


Figure 4.4: Status display in the readout

For an explanation of the Scan status, Geometry status and Laser Focus status read SECTION 4.1.1. For an explanation of the Interpolate status read further below.

If Interpolate Points is active, then the interpolated data is saved during ASCII export. If you export as Universal File, ME'Scope and ASAM ODS ATFX or access via Polytec File Access, you will obtain the original data.

Invalidated

In presentation mode you can mark the scan points you have already measured to be remeasured. To do so, assign the Invalidated scan status to the points as described in the following. Then change back into acquisition mode and open the file to remeasure.

In principle you can assign the Invalidated scan status to all scan points which have the Valid or Optimal scan status. Normally, this is only necessary for those points that did not provide a usable measurement result (e.g. such with a bad signal-to-noise ratio). For scan points with the Invalidated scan status the software does not show measurement data but, if applicable, data interpolated from neighbor points. The averaged spectrum over all scan points is corrected correspondingly.

ⓘ If you have user defined data sets available, you can invalidate, interpolate and display the points for every data set separately. The original data remains unaffected by this.

To assign the Invalidated scan point status to one or more scan point, proceed as follows:

1. Place a cursor at the scan point or scan points. See SECTION 3.3.3 on this.
2. Select Edit > Modify Selected Points > Invalidated in the menu bar or click the scan point with the right mouse button and select Invalidated in the context menu.

   The scan points that you have assigned the Invalidated scan status to are given automatically the Interpolate Invalid interpolate status. This means that these scan points will be interpolated if you have activated Interpolate Points.

Valid

To restore the original scan status of a scan point, proceed as follows:

1. Select Edit > Modify Selected Points > Valid in the menu bar or click the scan point with the right mouse button and select Valid in the context menu.

   The scan points that you have assigned the Valid scan status to are given automatically the Valid interpolate status. This means that these scan points will not be interpolated.

Validate All Points

To allocate their original scan point status to all scan points again, proceed as follows:

1. Select Edit > Validate All Points in the menu bar or click the presentation window (but not a scan point) with the right muse button and select Validate All Points in the context menu.

   The scan points that you have assigned the Valid scan status to are given automatically the Valid interpolate status. This means that these scan points will not be interpolated.

Invalidate Not Optimal Points

This command is only active if the following conditions are met:

- Signal Enhancement was active during scanning. See the acquisition mode on this as well.
- Not all scan points have the Optimal scan status.

To assign the Invalidated scan status to all scan points which do not have the Optimal scan status but the Valid scan status, proceed as follows:

1. Select Edit > Invalidate Not Optimal Points in the menu bar or click the presentation window (but not a scan point) with the right mouse button and select Invalidate Not Optimal Points in the context menu.

   All scan points which are not Optimal but Valid are given the Invalidated scan status. The scan points that you have assigned the Invalidated scan status to are given automatically the Interpolate Invalid interpolate status. This means that these scan points will be interpolated if you have activated Interpolate Points.

2. You can restore the original scan status of the scan points. To do so, select Edit > Invalidate Not Optimal Points in the menu bar again or click the presentation window (but not a scan point) with the right mouse button and select Invalidate Not Optimal Points in the context menu again.

ⓘ If you had changed the status of single scan points with the command Invalidate (see above), these scan points are also allocated their original status.

The scan points that you have assigned the Valid scan status to are given automatically the Valid interpolate status. This means that these scan points will not be interpolated.

Interpolate Disabled Points

For scan points with the Disabled scan status the software does not display any measurement data. You can interpolate data for these scan points from neighboring scan points and display it.

ⓘ You can manually assign the scan points the Disabled scan status before the scan. See the acquisition mode on this.

To interpolate disabled points, proceed as follows:

1. Select Edit > Interpolate Disabled Points in the menu bar or click the presentation window (but not a scan point) with the right mouse button and select Interpolate Disabled Points in the context menu.
2. You can cancel the interpolation of the disabled points. To do so, select Edit > Interpolate Disabled Points in the menu bar again or click the presentation window (but not a scan point) with the right mouse button and select Interpolate Disabled Points in the context menu again.

Interpolate

You can interpolate data for scan points from neighboring scan points and display it. If you have user defined data sets available, you can interpolate and display the points for every data set separately. The original data remains unaffected by this.

ⓘ Scan points without valid data are not displayed primarily in the presentation window. Change to the Status Points view style to display these points. Then you can select and interpolate the points.

To interpolate scan points, proceed as follows:

1. Select Edit > Modify Selected Points > Interpolate in the menu bar or click the scan point with the right mouse button and select Interpolate in the context menu.
2. You can undo the interpolation. To do so, select Edit > Modify Selected Points > Clear Interpolate or click the scan point with the right mouse button and select Clear Interpolate in the context menu.

Reset interpolation

You can reset interpolation of all scan points to default behavior. To do so, select Edit > Reset Interpolation in the menu bar or click the presentation window (but not a scan point) and select Reset Interpolation in the context menu.

Thereby the following statuses are made:

- Scan points with the Overrange, Invalidated and VT Failed scan status are given the Interpolate Invalid interpolate status and will be interpolated.
- Scan points with the Not Measured, Disabled, Not Reachable or Hidden scan status will given the Invalid interpolate status and will not be interpolated.
- Scan points with the Valid and Optimal scan status are given the Valid interpolate status and will not be interpolated.

Interpolate status

In the Status Points view style on the left of the object you will see the legend for the status display of the single scan points. To change the views, select the required view in the Status list. Select the Interpolate status.

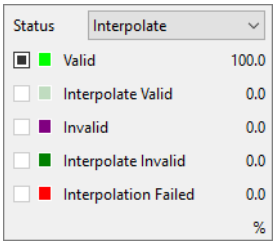


Figure 4.5: Display of the Interpolate status

Valid: Scan points with the Valid or Optimal scan status that are not interpolated will be given the Valid interpolate status.

Interpolate Valid: Scan points that you have selected for interpolating although valid measurement data is available (i.e. scan status Valid or Optimal) will be given the Interpolate Valid status.

Invalid: Scan points with the Not Measured, Overrange, Invalidated, Disabled, Not Reachable, Hidden or VT Failed scan status that are not interpolated are given the Invalid interpolate status.

Interpolate Invalid: Scan points with the Not Measured, Overrange, Invalidated, Disabled, Not Reachable, Hidden or VT Failed scan status that are interpolated are given the Interpolate Invalid interpolate status.

Interpolation Failed: If interpolation fails, the scan points are given the Interpolation Failed interpolate status. This can happen if the next scan points with valid data are too far away or no connecting lines are defined between the scan points.

4.1.4 Changing Vibrational Direction

You can still retrospectively change the direction of the vibration in the presentation window. The vibrational direction shown in the readout is the one you selected in the Acquisition Settings dialog on the Channels page. See the acquisition mode on this as well. Select the desired direction of vibration in the Direction list.

ⓘ You cannot change the direction of the vibration for 3D data.

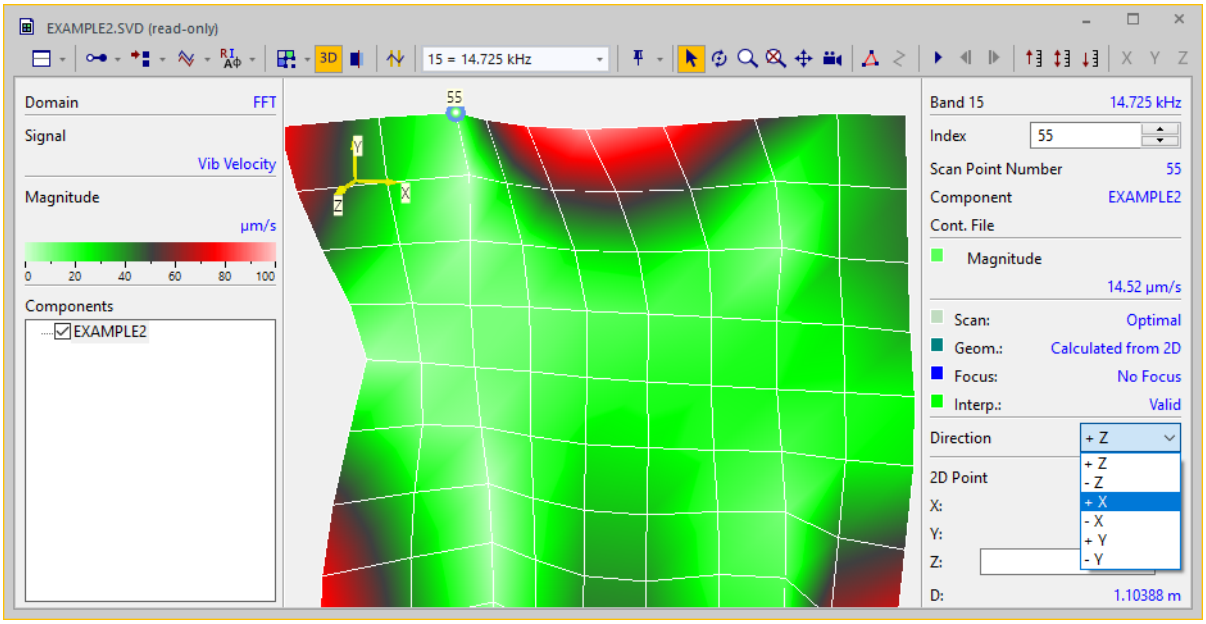


Figure 4.6: Changing the vibrational direction

With 2D geometries the direction of the vibration defines the scanning head coordinate system. Read the data acquisition under Setting the Optics (Enter axes coordinates) and your PSV theory manual on this.

4.2 Evaluating Sections of a Spectrum

If you have opened a scan in the presentation window, the Entire Bandwidth is displayed (RMS domain). You can evaluate sections of the measured spectrum. To do so, you have the following options:

- Evaluate band data to analyze the spectrum with the aid of frequency bands (not in Time measurement mode) - See SECTION 4.2.1 on this.
- Evaluate point data to set cursors and evaluate the frequency range (Cursor Mode) - See SECTION 4.2.2 on this.

4.2.1 Evaluating Band Data (not in Time Measurement Mode)

To analyze the spectrum using frequency bands, you have to display the band data. To do so, select Presentation > View > Band Data in the menu bar (toolbar of the presentation window:  > Band Data).

Manually define the frequency bands by entering the start and end frequencies of the frequency bands into the frequency band definition. The software determines the peak frequency and the bandwidth of the frequency bands. See SECTION 4.2.1.1 on this.

Automatically define frequency bands based on certain properties, as described in SECTION 4.2.1.2.

Select frequency bands and edit them as described in SECTION 4.2.1.3.

Requirement

You can define frequency bands for scan files which contain a complete spectrum. Scans with the following properties do not contain a full spectrum:

- Only frequency bands are saved in the file. See SECTION 5.2.1 on this as well.
- The file was measured in FastScan measurement mode.
- The file was measured in Time measurement mode.

Save and open

The software saves the frequency band definition in the file as soon as you save the file. Read SECTION 5.2.1 on this.

The software loads the frequency band definition when you open the file again.

You can also save the frequency band definition separately as an ASCII file and load it into several files. See SECTION 5.2.3 and SECTION 5.1.3 on this.

4.2.1.1 Manually Defining Frequency Bands

Select a view

You can only define frequency bands in Single Scan Point or Average Spectrum view. Proceed as follows:

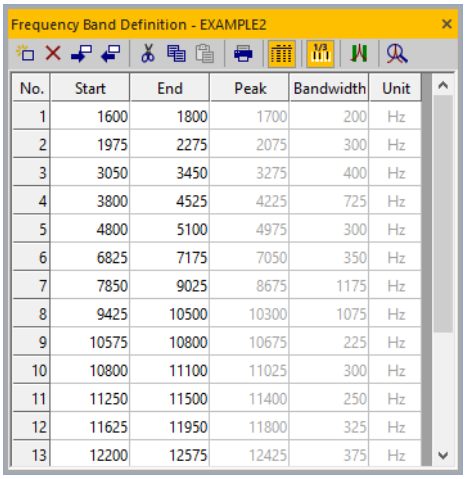
1. Select Presentation > View in the menu bar and then the respective view in the context menu (toolbar of the presentation window:).

Define frequency band

To manually define frequency bands, proceed as follows:

2. Open the frequency band definition by selecting Setup > Frequency Bands in the menu bar (toolbar of the presentation window:).

   The Frequency Band Definition dialog appears.



No.	Start	End	Peak	Bandwidth	Unit
1	1600	1800	1700	200	Hz
2	1975	2275	2075	300	Hz
3	3050	3450	3275	400	Hz
4	3800	4525	4225	725	Hz
5	4800	5100	4975	300	Hz
6	6825	7175	7050	350	Hz
7	7850	9025	8675	1175	Hz
8	9425	10500	10300	1075	Hz
9	10575	10800	10675	225	Hz
10	10800	11100	11025	300	Hz
11	11250	11500	11400	250	Hz
12	11625	11950	11800	325	Hz
13	12200	12575	12425	375	Hz

Figure 4.7: Example of the Frequency Band Definition dialog

3. Enter the start and end frequencies of the frequency bands. You have two options for this, as also described below: directly in the analyzer or in the Frequency Band Definition dialog.

   The software determines the peak frequencies of the frequency bands for the data set displayed in the view set.

4. You can change between different views or display other data sets without closing the frequency band definition.

   The software does not determine the peak frequencies again at that point.

ⓘ If you want to determine the peak frequencies again, click in the Frequency Band Definition dialog.

5. Once you have defined all frequency bands, close the frequency band definition by selecting Setup > Frequency Bands in the menu bar (toolbar of the presentation window:).

   The software calculates the RMS signal of the frequency bands and saves them in the file.

Entry in the analyzer

You can enter the start and end frequency of a frequency band directly in the analyzer. To do so, proceed as follows:

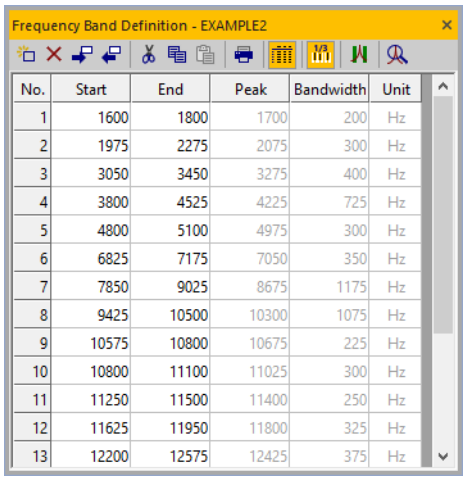
1. In the menu bar, select Analyzer > Show Bands (toolbar of the analyzer: ).
2. Use the mouse pointer to point at the diagram (not at the edge of the diagram) on the starting point of the required frequency band.
3. Press the mouse button and then drag right or left to the end point.

   The software displays the frequency band in color and enters it in the Frequency Band Definition dialog (see below).

Entry in the dialog

You can enter the start and end frequency of a frequency band in the Frequency Band Definition dialog. To do so, proceed as follows:

1. The Frequency Band Definition dialog consists of a toolbar and a table, the frequency band editor. If you do not see the table, click  in the toolbar of the frequency band definition to display it.



No.	Start	End	Peak	Bandwidth	Unit
1	1600	1800	1700	200	Hz
2	1975	2275	2075	300	Hz
3	3050	3450	3275	400	Hz
4	3800	4525	4225	725	Hz
5	4800	5100	4975	300	Hz
6	6825	7175	7050	350	Hz
7	7850	9025	8675	1175	Hz
8	9425	10500	10300	1075	Hz
9	10575	10800	10675	225	Hz
10	10800	11100	11025	300	Hz
11	11250	11500	11400	250	Hz
12	11625	11950	11800	325	Hz
13	12200	12575	12425	375	Hz

Figure 4.8: Example of the Frequency Band Definition dialog

2. Click  in the toolbar.

   A new line is added to the bottom of the table.
3. Enter the start frequency in hertz in the Start column.

   The software replaces the value entered by the next possible one.

ⓘ The start and end frequency must be on FFT lines.

4. Press the tabulator key or click the corresponding cell in the End column and enter the end frequency in hertz.
5. Press Enter.

   The software enters the peak frequency and the bandwidth in the frequency band editor and displays the frequency band in the analyzer in color.

ⓘ If you have defined a new frequency band and close the frequency band definition, the new frequency band remains selected.

4.2.1.2 Automatically Defining Frequency Bands

Select a view

You can only define frequency bands in Single Scan Point or Average Spectrum view. Proceed as follows:

1. Select Presentation > View in the menu bar and then the respective view in the context menu (toolbar of the presentation window:).

Define frequency bands

To automatically define the frequency bands based on certain properties, proceed as follows:

2. Open the frequency band definition by selecting Setup > Frequency Bands in the menu bar (toolbar of the presentation window:).

   The Frequency Band Definition dialog appears.

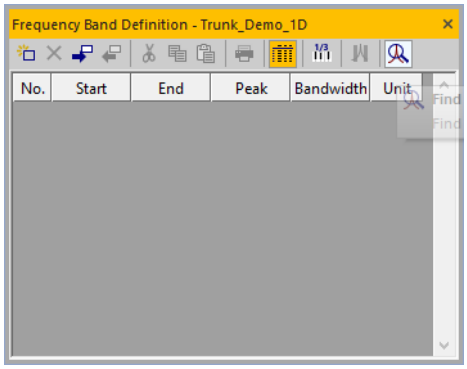


Figure 4.9: Frequency Band Definition dialog before searching

3. Click in the toolbar.

   The Find Peaks dialog appears.

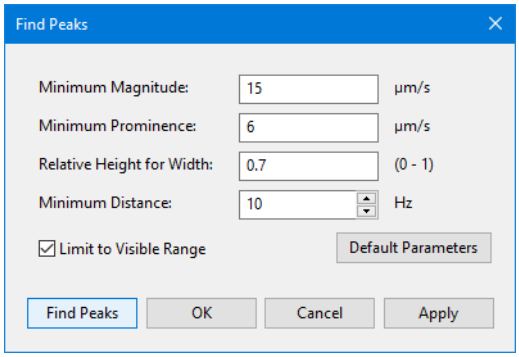
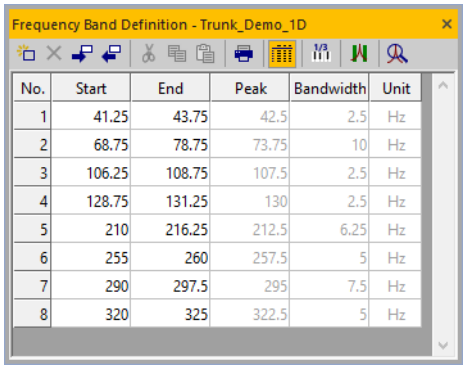


Figure 4.10: Find Peaks dialog

4. Enter the values. The dialog description can be found below.
5. Click Find Peaks.

   The software determines the peaks according to the inputs and defines the frequency bands at the positions, enters the values into the frequency band editor and displays the frequency bands in color in the analyzer.



No.	Start	End	Peak	Bandwidth	Unit
1	41.25	43.75	42.5	2.5	Hz
2	68.75	78.75	73.75	10	Hz
3	106.25	108.75	107.5	2.5	Hz
4	128.75	131.25	130	2.5	Hz
5	210	216.25	212.5	6.25	Hz
6	255	260	257.5	5	Hz
7	290	297.5	295	7.5	Hz
8	320	325	322.5	5	Hz

Figure 4.11: Frequency Band Definition dialog after searching successfully

6. You can make changes to the values in the Find Peaks dialog to optimize the frequency band definition.
7. If you want to define the frequency bands only in the visible range of the analyzer, tick the the Limit to Visible Range check box.
8. To save the frequency band definition in the visible area of the analyzer, click Apply.

   You can then scroll the abscissa and define more frequency bands using Find Peaks.
9. To close the Find Peaks dialog, click OK.

Dialog description

Minimum Magnitude: Enter the minimum magnitude above which a peak should be detected.

Maximum Prominence: Enter the value by how much higher the magnitude of the peak must be than the magnitude of the minima to the right and left of it (peak height).

Relative Height to Width: Enter the fraction of the prominence at which the width of the peak is determined.

Minimum Distance: Enter the minimum distance in the unit of the abscissa for the distance between two peaks.

Limit to Visible Range: If you want to limit the determination of the frequency bands to the visible range in the analyzer, tick the check box.

Default Parameters: If you want to set the default parameters of the software, click the button. This also adjusts the units to the current setting in the analyzer.

Find Peaks: If you have entered or changed the parameters for the automatic frequency band definition, click the button to update the display. Frequency bands that have already been determined are overwritten.

Apply: If you have automatically defined frequency bands (e.g. in the visible area of the analyzer), click the button to save the frequency bands. You can then scroll the abscissa and automatically define more frequency bands.

4.2.1.3 Selecting and Editing Frequency Bands

Determine peak frequency again

If you have switched between different views or displayed other data sets without closing the frequency band dialog, the peak frequencies are not determined again.

To determine the peak frequencies again, you have the following options:

- Click  in the Frequency Band Definition dialog.
- Close the Frequency Band Definition dialog by clicking  in the toolbar of the presentation window. Confirm the query whether you want to recalculate the frequency bands with Yes.

Calculate 1/3 octave bands

If you want the software to calculate 1/3 octave bands, proceed as follows:

1. Click  in the Frequency Band Definition dialog.
2. Close the Frequency Band Definition dialog by clicking  in the toolbar of the presentation window.
3. Confirm the query whether you want to recalculate the frequency bands with Yes.

   Additionally you will be given the 1/3 Octave domain for selecting.

Select frequency band

To delete or copy frequency bands from the definition, you have to select them first. To do so, proceed as follows:

1. Click the frequency band in the analyzer or its line number in the frequency band editor.
2. To select further frequency bands, press the Control key and click the number of the line in the frequency band editor.

ⓘ To select all frequency bands or to undo a selection, click the top left of the table.

Delete frequency band

You can delete frequency bands from a definition. To do so, proceed as follows:

1. Select the frequency bands as described above.
2. Press the Delete key or click  in the Frequency Band Definition dialog.

Cut or copy frequency band

You can cut a frequency band definition and paste or copy it into another file via the clipboard. To do so, proceed as follows:

1. Select the frequency bands as described above.
2. In the Frequency Band Definition dialog, click (Cut) or (Copy).
3. Close the current frequency band definition by selecting Setup > Frequency Bands in the menu bar (toolbar of the presentation window:).
4. Open the file which you want to paste the frequency bands into.
5. Open the frequency band definition by selecting Setup > Frequency Bands in the menu bar (toolbar of the presentation window:).
6. Click (Paste) in the Frequency Band Definition dialog.

   The cut out or copied frequency band is inserted.

Drag with the mouse in the project browser

You can also apply frequency bands to another scan in the project browser by dragging with the mouse. To do so, proceed as follows:

1. Holding the left mouse button pressed, drag the Frequency Band entry from the project browser to the frequency band definition.

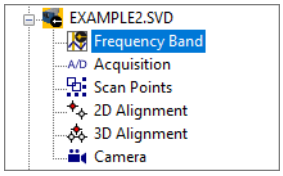


Figure 4.12: Frequency Band entry in the project browser

2. Close the frequency band definition. A message box appears with the question whether the frequency bands are to be recalculated.
3. Click Yes.

ⓘ If you have defined a new frequency band and close the frequency band definition, the new frequency band remains selected.

4.2.2 Evaluating Point Data (Cursor Mode)

Requirement

In the Point Data view, you can set single cursors and evaluate the frequency range. With a single cursor you can evaluate spectra for scans which contain a complete spectrum.

The following scans do not contain a complete spectrum:

- Scans which only have frequency bands saved in them. See SECTION 5.2.1 on this as well.
- Scans measured in FastScan measurement mode.
- Scans measured in Time measurement mode.

Approach

For evaluation, proceed as follows:

1. Select Presentation > View > Point Data in the menu bar (toolbar of the presentation window:  > Point Data).

   The domain changes to FFT and in the Frequency Band list the Cursor entry is displayed.

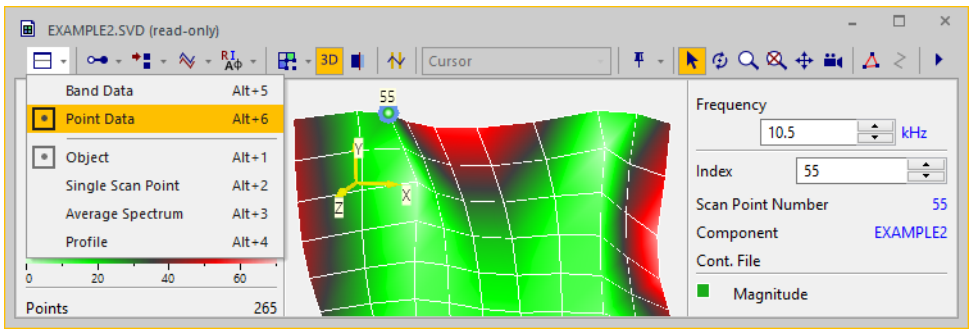


Figure 4.13: Selecting point data and Cursor entry in the frequency band list (FFT)

Depending on the size of the file, the following message may appear:

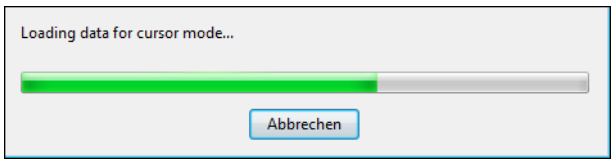


Figure 4.14: Loading data for cursor mode

ⓘ If you click Cancel during loading, the software shows the band data again.

2. If you do not have enough memory available, you will get the message that you need to reduce the amount of data. We recommend using the suggested value or entering a higher one.

   Thus you reduce the amount of data which is loaded. This means that only every n-th FFT line or respectively only the given number of frames is used. Below in the dialog you will get information on the memory.

3. Click OK.

   As a result the data is reloaded.

ⓘ If you click Cancel, the software shows the band data again.

4. Select Presentation > View in the menu bar and select Single Scan Point or Average Spectrum in the context menu (toolbar of the presentation window: ).

   The analyzer is displayed at the bottom in the presentation window.

5. Set a single cursor in the analyzer as described in SECTION 3.2.1.

   The frequency and the magnitude at the current cursor position can be seen in the readout on the right. On top in the presentation window you will see the vibration position which the object under investigation takes at the selected frequency.

ⓘ If the readout is not visible on the right in the analyzer, select Analyzer > Readout in the menu bar to display it.

6. Drag the cursor while holding the left mouse button pressed to the left or right.

   The readout and the data in the presentation window above are updated simultaneously.

ⓘ If you have set a band cursor and the fitting is successful, i.e. the fitted curve is displayed, the frequency for the surface view is determined from the peak frequency otherwise from the frequency of the maximum.

ⓘ If you do not set a cursor or delete the cursor you had set, you can also enter the required frequency directly in the readout of the presentation window.

User defined data sets

If you are working with user defined data sets, you can also evaluate them. You will find information on user defined data sets in SECTION 3.7.

For evaluation, click  and select the Usr channel in the pull-down menu. It is also necessary to load the data on this. If you do not have enough memory available, you will get the message that you need to reduce the amount of data.

We recommend that you use the suggested value or enter a higher one. Thus you reduce the amount of data which is loaded. This means that only every n-th FFT line or respectively only the given number of frames is used. Below in the dialog you will get information on the memory.

4.3 Displaying Profiles

You can display profiles of the data. To do this, you first draw profile lines on the object. You will see the profiles at the bottom in the analyzer. The software automatically saves the profile lines in the scan and loads them when you reopen the scan.

Draw profile line

To draw a profile line on the object, proceed as follows:

1. Display the Profile view by clicking  in the toolbar of the presentation window and selecting Profile in the pull-down menu.

   The software switches to 2D view.
2. Display the Surface, Wireframe, Isolines or Scan Points view style by clicking  in the toolbar of the presentation window and selecting the view style in the pull-down menu.
3. Click  .
4. Point with the mouse at the object and click the starting point of the profile line.

   The mouse pointer becomes an arrow symbol  .
5. Click as many other vertex points as you like.

   You can see the profile at the bottom in the analyzer.
6. To define the end point of the profile line, double-click the last vertex point.

Select profile line

First of all select the profile line for editing, cutting or copying. To do so, proceed as follows:

1. Click  .
2. Point with the mouse at a profile line.

   The mouse pointer becomes an arrow cross icon  .
3. Click the profile line.

   The software marks the vertex points of the profile line with white markers.
4. To select or deselect more profile lines, click them while holding the Control key pressed.

ⓘ You can select all profiles lines by selecting Edit > Select All in the menu bar (key combination: Ctrl + A).

Move profile line

To move profile lines, proceed as follows:

1. Select the profile lines as described above.
2. Point with the mouse at one of the lines (not at a vertex point).

   The mouse pointer becomes an arrow cross icon  .
3. Press the mouse button and drag in the direction required.

Move vertex point

To change the shape of profile lines by moving individual vertex points, proceed as follows:

1. Select the profile lines as described above.
2. Point with the mouse at a vertex point.

   The mouse pointer becomes an arrow symbol  .
3. Press the mouse button and drag in the direction required.

Duplicate profile line

To duplicate profile lines, proceed as follows:

1. Select the profile lines as described above.
2. Select Edit > Copy in the menu bar (toolbar: ).

   The profile lines are copied onto the clipboard.
3. Open a measurement file.
4. Select Edit > Paste in the menu bar (toolbar: ).

   The profile lines are pasted from the clipboard.

ⓘ If you have not selected a profile line, you can save data onto the clipboard with Edit > Copy as described in SECTION 5.2.6.

Cut profile line

To cut profile lines and paste them into another scan file, proceed as follows:

1. Select the profile lines as described above.
2. Select Edit > Cut in the menu bar (toolbar: ).

   The profile lines will be removed and saved to the clipboard.
3. Open a measurement file.
4. Select Edit > Paste in the menu bar (toolbar: ).

   The profile lines are pasted from the clipboard.

Delete profile line

To delete profile lines, proceed as follows:

1. Select the profile lines as described above.
2. Select Edit > Delete in the menu bar (toolbar:  , Delete key).

Animate vibration

You can animate the vibrations of the profiles. See SECTION 4.4 on this.

Undo and redo

You can undo up to 50 operations and redo them. In the menu bar, select Edit > Undo or Edit > Redo (toolbar:  or ).

4.4 Animating Vibration

For you to be able to animate the measured vibration, during the measurement either the signal has to be triggered (phase-related measurement) or a reference signal has to have been captured.

ⓘ In Time measurement mode, you cannot capture a reference signal.

You can animate vibrations in different measurement modes:

- FFT, FastScan or MultiFrame measurement mode - See SECTION 4.4.1 on this.
- Time measurement mode - See SECTION 4.4.2 on this.

4.4.1 Animation in FFT, FastScan or MultiFrame Measurement Mode

If you have carried out a measurement in one of the above mentioned modes, you can set how many images per animation cycle are shown.

To animate a vibration, proceed as follows:

Prepare

1. Define frequency bands as described in SECTION 4.2.1.
2. Display the frequency band with the peak frequency you want to animate. You can also change to cursor mode. See SECTION 4.2.2 on this.
3. Display the FFT domain (FFT spectrum) by clicking  and selecting FFT in the pull-down menu.
4. Display a complex signal.

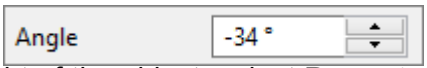
ⓘ For complex signals, the Phase display type is available.

5. Display the 3D view by clicking  .

ⓘ You can animate in all view styles but not in the Status Points view style.

Start animation

You have the following options to start an animation:

- You start the animation of an active presentation window by selecting Animation > Start in the menu bar (toolbar of the presentation window: ).
  The software displays the signal in the Inst. Value display type and animates the vibration. You can see the current angle in the readout in the Angle 

  field. If the readout is not visible on the right of the object, select Presentation > Readout in the menu bar to display it.
- You start animations simultaneously in all open presentation windows by selecting Animation > Start All in the menu bar.

ⓘ You can increase the speed of the animation using the + key or decrease it respectively using the – key.

Stop animation

You have the following options to stop an animation:

- You stop the animation of an active presentation window by selecting Animation > Stop in the menu bar (toolbar of the presentation window: ).
- You stop all running animations by selecting Animation > Stop All in the menu bar.

Animation forwards or backwards

You can make an animation run forwards or backwards image by image. To do so, you have the following options:

- Step Forward: Select Animation > Step Forward in the menu bar (toolbar of the presentation window: ). You will see the next animation frame.
- Step Backward: Select Animation > Step Backward in the menu bar (toolbar of the presentation window: ). You will see the previous animation frame.

Animation faster or slower

You can make an animation run faster or slower. To do so, you have the following options:

- Faster: Select Animation > Increase Speed in the menu bar (+ key). The animation runs faster forwards or backwards, depending on your selection.
- Slower: Select Animation > Decrease Speed in the menu bar (– key). The animation runs slower forwards or backwards, depending on your selection.

Animated spectrum

You can set the number of frames per second that are displayed during animation in the Preferences dialog. To open the dialog, select Setup > Preferences in the menu bar. Display the Display page.

ⓘ This setting has no influence on measurements that were made in Time measurement mode.

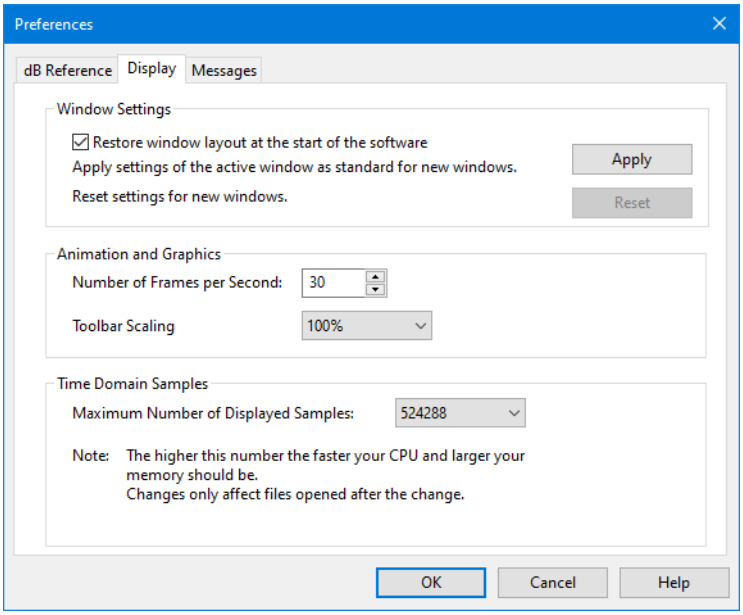


Figure 4.15: Display page in the Preferences dialog in presentation mode

Number of Frames per Second: Enter the frame rate for animations, i.e. the number of frames per second that will be shown during animation. The minimum setting is 10, the default setting is 30. The frame rate is also used for stored animations.

Toolbar Scaling: Select the scaling for the display of the toolbars in the software. The selection of the scaling affects the display in Presentation mode, and at the next start also in Acquisition mode.

Maximum Number of Displayed Samples: Enter an upper limit for the number of samples to be shown simultaneously in the Time domain.

Scale the deflection

To change the degree of deflection in animation of non-scalar-data, you have the following options:

- You display the deflection intensified by selecting Presentation > Scale Up on the menu bar (toolbar: ).
- You display the deflection with default settings by selecting Presentation > Auto Scale in the menu bar (toolbar: ).
- You display the deflection attenuated by selecting Presentation > Scale Down in the menu bar (toolbar: ).

ⓘ Hold the Shift key pressed during scaling and the scaling is immediately changed by a factor of 20.

If you link windows, the scaling factor will be also transferred with the View setting.

Display vibrational direction (only PSV-3D)

Once you have made a measurement with the PSV-3D, you can show the vibrations of the object individually in x-, y- and z-direction. You can also combine displaying several vibrational directions together. To show/hide the vibrations in the various directions, click X, Y and Z in the toolbar.

If the respective icon is activated, then the vibration is being shown in the respective direction. If the icon is deactivated, then the vibration is not being shown in the respective direction.

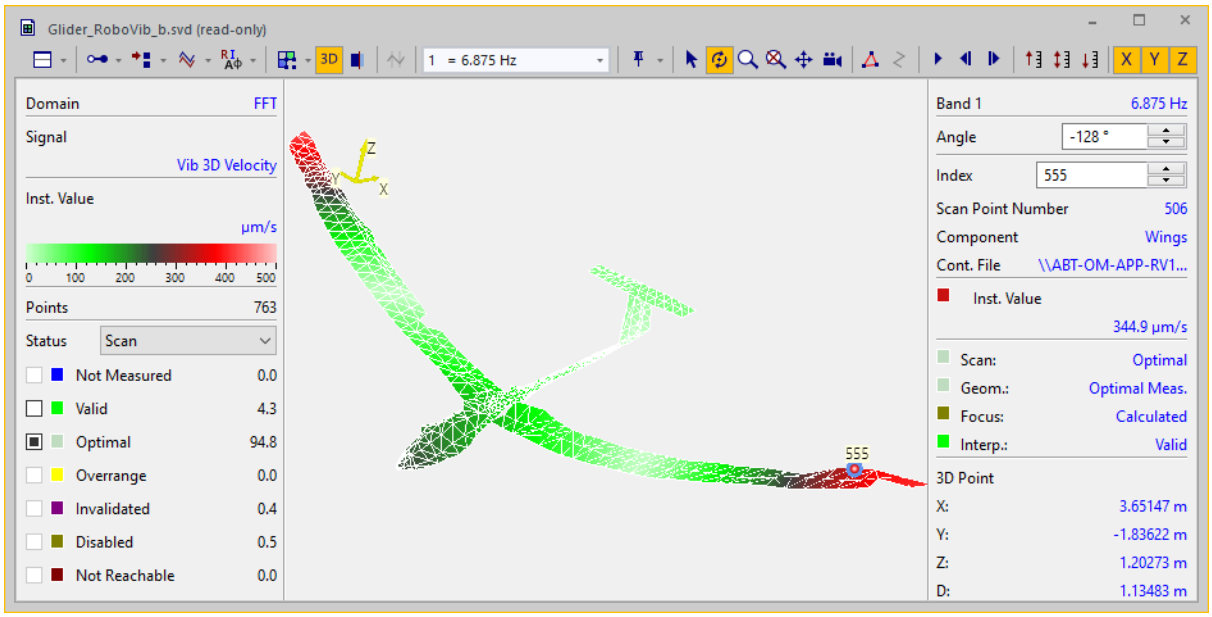


Figure 4.16: Display of vibrational direction in the Object view

Save animation

You can save animations in the AVI (*.avi) multimedia format or as an animated graphics in the GIF (*.gif) format. See SECTION 5.2.5 on this.

4.4.2 Animation in Time Measurement Mode

If you have made the measurement in Time measurement mode, you will see the animation in the time domain.

To animate a vibration, proceed as follows:

Prepare

1. Select Presentation > View > Point Data in the menu bar (toolbar of the presentation window: > Point Data).

   The domain changes to Time and in the Frequency Band list the Cursor entry is displayed.

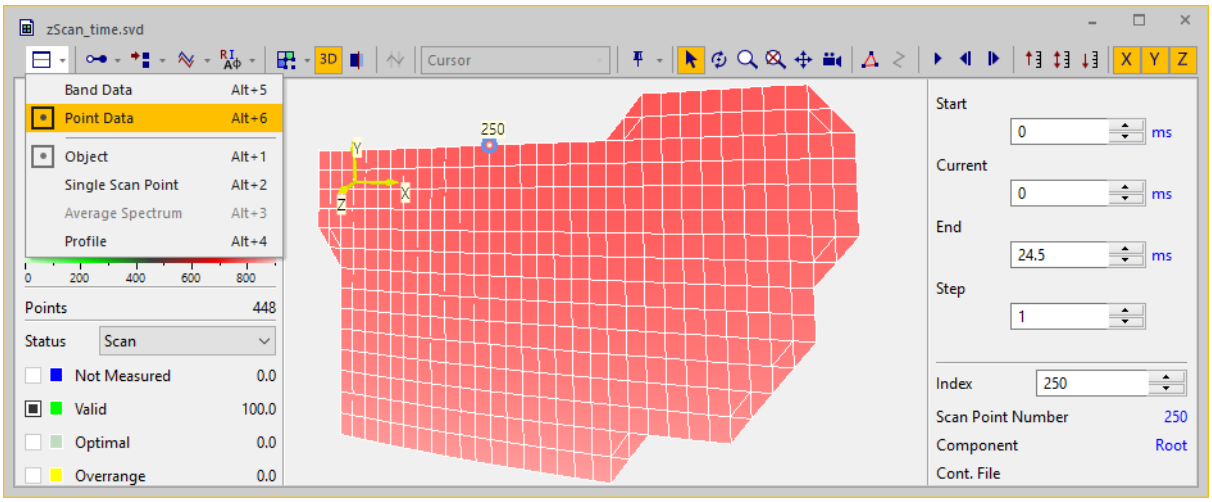


Figure 4.17: Selecting point data and Cursor entry in the frequency band list (Time)

Depending on the size of the file, the following message may appear:

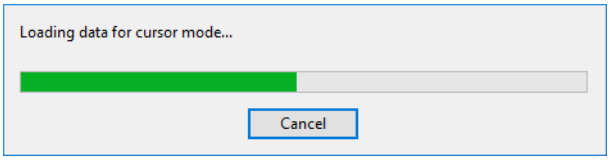


Figure 4.18: Loading data for cursor mode

ⓘ If you click Cancel during loading, the software shows the band data again.

2. If you do not have enough memory available, you will get the message that you need to reduce the amount of data.

   We recommend using the suggested value or entering a higher one. Thus you reduce the amount of data which is loaded. This means that only every n-th sample or respectively only the given number of frames is used. Below in the dialog you will get information on the memory.

3. Click OK.

   This means that the time data is reloaded and the z-axis scaled correspondingly.

ⓘ You can significantly reduce the necessary computing time for rescaling the z-axis if you deactivate filtering the data. See SECTION 4.1.2 on this.

ⓘ If you click Cancel, the software shows the band data again.

4. Display the 3D view by clicking  .

ⓘ You can animate in all view styles but not in the Status Points view style.

5. Select the section within the sample time in which the data is to be animated.

Set start and end

You have two options to set the start and end time:

- Enter the start and end time (Start and End) directly in the readout on the right in the presentation window. If the readout is not visible, select Presentation > Readout in the menu bar to display it.
- Display the analyzer by clicking  and selecting Single Scan Point in the pull-down menu. Click  and with a further mouse click, set the starting point directly in the diagram. Hold the mouse button pressed and drag it to the right. A second black line will appear which marks the end point. If the line is in the right place, release the mouse button.

  As the starting point is now identical to the current point in time, it is marked by a red line. Drag the line to the left and the starting point is shown with a black line.

Start animation

You have the following options to start an animation:

- You start the animation of an active presentation window by selecting Animation > Start in the menu bar (toolbar of the presentation window: ).

  You can see the defined time frame and the current status of the animation and also the step size in the readout. If the readout is not visible on the right of the object, select Presentation > Readout in the menu bar to display it.
- You start animations simultaneously in all open presentation windows by selecting Animation > Start All in the menu bar.

ⓘ You can increase the speed of the animation using the + key or decrease it respectively using the – key.

Stop animation

You have the following options to stop an animation:

- You stop the animation of an active presentation window by selecting Animation > Stop in the menu bar (toolbar of the presentation window: ).
- You stop all running animations by selecting Animation > Stop All in the menu bar.

Animation forwards or backwards

You can make an animation run forwards or backwards image by image. To do so, you have the following options:

- Step Forward: Select Animation > Step Forward in the menu bar (toolbar of the presentation window: ). You will see the next animation frame.
- Step Backward: Select Animation > Step Backward in the menu bar (toolbar of the presentation window: ). You will see the previous animation frame.

Animation faster or slower

You can make an animation run faster or slower. To do so, you have the following options:

- Faster: Select Animation > Increase Speed in the menu bar (+ key). The animation runs faster forwards or backwards, depending on your selection.
- Slower: Select Animation > Decrease Speed in the menu bar (– key). The animation runs slower forwards or backwards, depending on your selection.

ⓘ Every time the point data is displayed, the Step Size in the readout is updated with each change.

Scale the deflection

To change the degree of deflection in animation of non-scalar-data, you have the following options:

- You display the deflection intensified by selecting Presentation > Scale Up on the menu bar (toolbar: ).
- You display the deflection with default settings by selecting Presentation > Auto Scale in the menu bar (toolbar: ).
- You display the deflection attenuated by selecting Presentation > Scale Down in the menu bar (toolbar: ).

ⓘ Hold the Shift key pressed during scaling and the scaling is immediately changed by a factor of 20.

If you link windows, the scaling factor will be also transferred with the View setting.

Save animation

You can save animations in the AVI (*.avi) multimedia format or as an animated graphics in the GIF (*.gif) format. See SECTION 5.2.5 on this.

4.5 Editing Scan Points

You can also retrospectively edit scan points in the presentation mode. To do so, you have the following options:

- You can select scan point in different ways. See SECTION 4.5.1 on this.
- Outliers are measurement values that do not fit in an expected measurement series or generally do not meet the expectations. You can select outliers. See SECTION 4.5.2 on this.
- You can change the indices of selected scan points. See SECTION 4.5.3 on this.
- You can have connections between scan points defined automatically by the software. See SECTION 4.5.4 on this.
- You can define geometry components and assign scan points to the individual components. See SECTION 4.5.5 on this.

4.5.1 Selecting Scan Points

Select with the mouse

To select scan points with the mouse, you have the following options:

- To select several scan points adjacent to each other, hold the Shift key and the left mouse button pressed and draw one or more rectangles across the required scan points.
- To select several scan points that are not adjacent to each other, hold the Control key pressed and click the required scan points.

Select All

To select all scan points, you have the following options:

- Press the key combination Ctrl + A.
- Select Edit > Select All in the menu bar.
- Click the object with the right mouse button and select Select Points in the context menu. The Select Scan Points dialog appears. Select All and click OK.

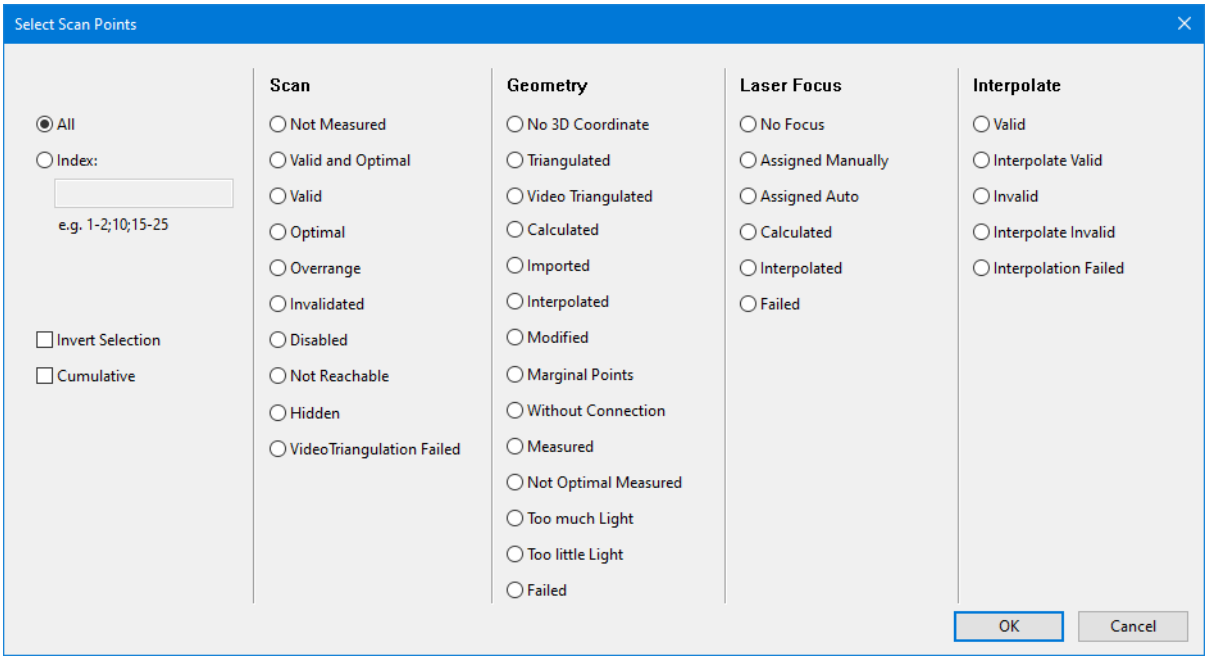


Figure 4.19: Select Scan Points dialog

Select with certain properties

You can select scan points which display certain properties. To do so, proceed as follows:

1. Select Edit > Select Points in the menu bar or click the video image with the right mouse button and select Select Points in the context menu.

   The Select Scan Points dialog appears (see above).

ⓘ You can select the scan points according to general criteria (left) or according to the corresponding status (Scan status, Geometry status, Laser Focus status or Interpolate status).

2. Select the criteria according to which you want to select points and click OK.

ⓘ If you want to select the scan points on the basis of their index, you can display the indices. See SECTION 3.1.2 on this.

3. If you want to select points with different properties, then also tick the Cumulative check box and repeat steps 1 and 2 as often as necessary.

ⓘ To invert the selection, also tick the Invert Selection box.

4. To clear the selection, click  .

Select via legend for status

You can display the scan points of a certain status in the video window. To do so, tick the corresponding check box of the point status. The other way round, you can select scan points in the video image using the mouse, and in the legend the statuses of the selected scan points are displayed.

Select outliers

Outliers are measurement values that do not fit in an expected measurement series or generally do not meet the expectations. You have several options to select these outliers. See SECTION 4.5.2 on this.

4.5.2 Selecting Outliers

Outliers are measurement values that do not fit in an expected measurement series or generally do not meet the expectations.

To select outliers, you have the following options:

- Select Edit > Select Outliers in the menu bar.
- Click the object with the right mouse button and select Select Outliers in the context menu.

The Select Outliers dialog appears.

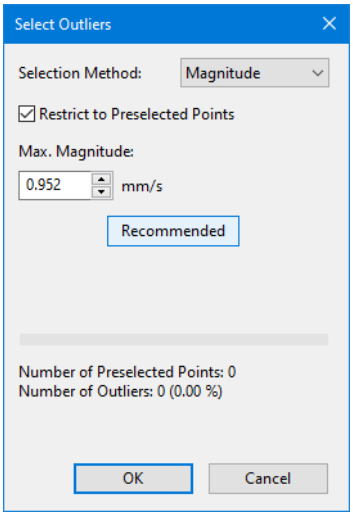


Figure 4.20: Select Outliers dialog for the Magnitude selection method

Selection Method: Select the required method from the list.

You have the following options:

- Magnitude
- 2D Filter
- 3D Filter

ⓘ The 3D Filter method is only available for scan points with 3D coordinates.

Restrict to Preselected Points: If you want to apply the filter only to the currently selected scan points, tick the check box

Otherwise, all scan points that fulfill one of the following properties are filtered:

- Valid or Optimal scan status. For an explanation of the scan status, see SECTION 4.1.1.
- Successfully interpolated if the Interpolate Points function is active. For an explanation of the Interpolate Points function, see SECTION 4.1.3.

Magnitude

With the Magnitude selection method, the maximum magnitude is used as a criterion for the selection of the outliers (figure see above).

Max. Magnitude: Enter the maximum magnitude value. The unit corresponds to that of the currently selected signal type for the scan points. Those scan points are marked as outliers, whose signal value in the amount exceeds the specified maximum magnitude.

Recommended: Click here to display the selection of outliers automatically calculated by the software.

For the calculation, the scan points according to the description of the Restrict to Preselected Points option are used. If the amount of scan points determined in this way is empty, a value of 0 is output as the recommended value. Otherwise the average of their magnitudes is calculated for these scan points, i.e. the average of the absolute amount of their amplitude values. If the magnitude of at least one scan point from the amount of the considered scan points is greater than twice the average magnitude, twice the average magnitude is output as the recommended value. Otherwise, the average magnitude is multiplied by a factor of 1.75 and output.

2D Filter

With the 2D Filter selection method, comparing the actual with the filtered signal values is used for calculating the outliers. For calculating the filtered data, a 2D approach is followed in which the neighboring points of each scan point are determined based on the length of the connecting lines between the scan points.

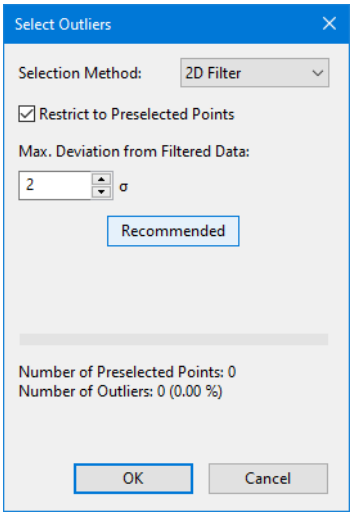


Figure 4.21: Select Outliers dialog for the 2D Filter selection method

Max. Deviation from Filtered Data: For all considered scan points, the standard deviation between actual and filtered data is calculated. Enter a threshold for the deviation in relation to the standard deviation.

Recommended: Click here to display the selection of outliers automatically calculated by the software.

For the calculation, the scan points according to the description of the Restrict to Preselected Points option are used. For Max. Deviation from Filtered Data, the value 2 is statically set as the recommended value.

3D Filter

With the 3D Filter selection method, comparing the actual with the filtered signal values is used for calculating the outliers. For calculating the filtered data, the neighboring points of each scan point are determined based on the distances of the scan points in 3D space.

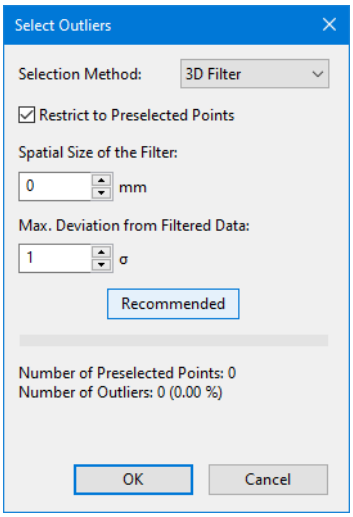


Figure 4.22: Select Outliers dialog for the 3D Filter selection method

Spatial Size of the Filter: Enter the spatial size of the filter. This parameter specifies the radius within which the neighboring points of the scan point are to be included in the filtering.

Max. Deviation from Filtered Data: For all considered scan points, the standard deviation between actual and filtered data is calculated. Enter a threshold for the deviation in relation to the standard deviation.

Recommended: Click here to display the selection of outliers automatically calculated by the software.

For the calculation, the scan points according to the description of the Restrict to Preselected Points option are used. For Spatial Size of the Filter the value 0.1 and for Max. Deviation from Filtered Data the value 2 are statically set as recommended values.

4.5.3 Changing Indices

Display indices

You set up the display of the scan point indices in the video image in the Display Properties dialog in 3D view. To open the dialog, display the 3D view by clicking 3D in the graphics toolbar.

In acquisition mode you open the dialog as folllows:

- Double-click the video image and display the Data page.
- Click the video image with the right mouse button and select Data Properties in the context menu.
- Activate the video window (to do so, click it) and select Scan > Properties in the menu bar. Display the Data page.

In presentation mode you open the dialog as follows:

- Double-click the object in the presentation window and display the Data page.
- Click the presentation window with the right mouse button and select Data Properties in the context menu.
- Activate the presentation window (to do so, click it) and select Presentation > Properties in the menu bar. Display the Data page.

You will also find a detailed description of the dialog in SECTION 3.1.

Change indices

To change the indices of selected scan points, you have the following options:

- Click one or several scan points with the right mouse button ans select Change Indices in the context menu.
- Select one or several scan points and select Edit > Modify Selected Points > Change Indices in the menu bar.

The Change Indices dialog appears.

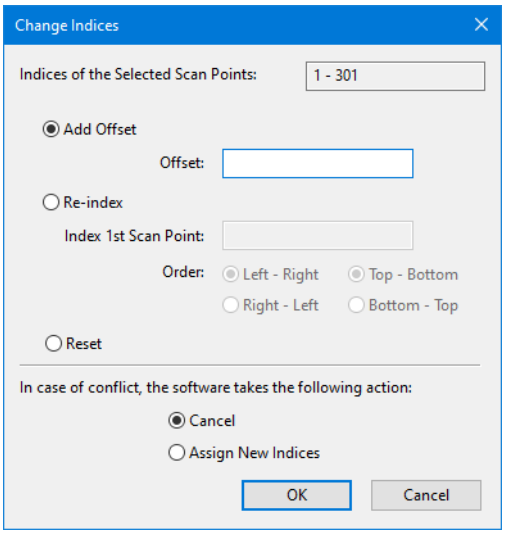


Figure 4.23: Change Indices dialog

Indices of the Selected Scan Points: The indices of the selected scan points are displayed.

Add Offset: Click here if you want to add a certain amount to the indices of the selected scan points or want to subtract it from them. Enter the required value in the Offset field. For a negative offset you have to enter an additional minus sign (–).

Example:

The selected scan points have the indices 1 - 21.

Offset: 10

The new indices will be 11 - 31.

Re-index: Click here if you want to renumber the selected scan points. In the Index 1st Scan Point field, enter the value at which the new numbering is to start. In addition to that, select the direction of the numbering, starting with the first scan point.

Example:

The selected scan points have the indices 27 - 52.

Index 1st Scan Point: 30

Order: Left - Right, Top - Bottom

The new indices are 30, 31, 32, etc. They are assigned to the selected scan points from left to right and from top to bottom.

Reset: Click here if you want to undo the changes made to the indices. The indices are then reset to the scan point number. The scan point numbers are assigned by the software in ascending order and without gaps starting with one. After the measurement, the scan point number and the index are displayed in the readout of the presentation window.

In case of conflict, the software takes the following action: Here you select how the software is to react if there is a conflict after changing the indices, i.e. numbers were assigned twice.

You have the following options:

- Cancel: The entire action is canceled, i.e. no changes are made.
- Assign New Indices: The selected scan points are assigned the required indices. Indices for other scan points which already exist are counted up, starting with the highest index +1.

Example:

You have defined 2500 scan points and want to newly assign the indices 1 - 200 to the scan points 200 - 400.

Index 1st Scan Point: 1

On conflict with these changes, take the following action: Assign New Indices

The selected scan points are given the indices 1 - 200. The scan points which originally had the indices 1 - 200, are given the indices 2501 - 2701.

ⓘ Independently of whether you abort assignment of new indices or carry it out, you will be notified whether a conflict has occurred.

4.5.4 Defining Connections between Scan Points

Connect scan points automatically

To automatically connect scan points, proceed as follows:

1. Change to 3D view by clicking  .
2. Click  .

   The Connect Points dialog appears.

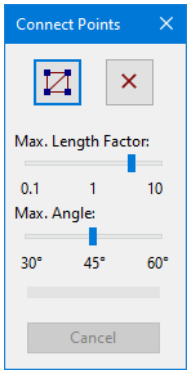


Figure 4.24: Connect Points dialog

3. Select the maximum length factor (Max. Length Factor). By moving this slider with the mouse you can set the length of connection still allowed (minimum 1/10 to maximum factor 10). The factor refers to the median of the length of the visible connecting lines that already exist.
4. Select the maximum permissible angle of the viewer (Max. Angle).

   The software only generate areas the normal vector of which includes a maximum of this angle with the line of sight of the viewer (minimum 30° to maximum 60°).
5. If applicable, select the scan points you want to connect as described in SECTION 4.5.1.

ⓘ If connections exist between the selected scan points, you have to delete these. Then you can connect the scan points automatically.

6. Click  in the Connect Points dialog.

   The software will connect all scan points or the selected ones respectively, creating triangles. Existing connections are retained here.
7. If necessary, rotate the object further and repeat steps 5 and 6 until you have made all connections.

To delete automatically created connections, proceed as follows:

8. Select the scan points as described in SECTION 4.5.1.
9. Click  in the Connect Points dialog.

4.5.5 Defining Geometry Components

In the video window of acquisition mode and in the presentation window, you can define geometry components and assign scan points to the individual components. The components are displayed in a list on the left in the video window or presentation window respectively. By default, only one component initially exists, the root component.

ⓘ In acquisition and presentation mode, you can show (hide) the scan points allocated to a component. This does not affect the average spectrum of the points.

If you have selected scan points in the video window, then the name of the associated geometry component is shown in the scanning head control.

If you have selected scan points in the presentation window, then the name of the associated geometry component is shown in the readout.

Edit component

To edit the root component or any other component, click the component with the right mouse button and select Edit in the context menu. The Edit Component dialog appears.

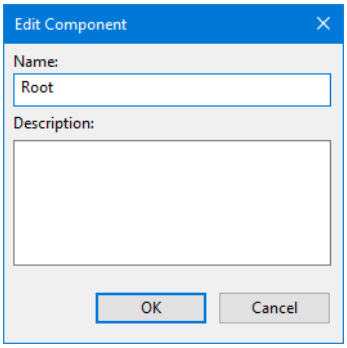


Figure 4.25: Edit Component dialog

Name: Enter the desired name for the component.

Description: Add a text to describe the component in greater detail.

ⓘ To change the name of the component, click the component with the left mouse button and press the F2 key. Enter the new name.

Add component

To add a component, you have the following options:

- Click the component below which you want to add a new component with the right mouse button and select Add in the context menu.
- Click the component wit the left mouse button and press Insert.

The software generates a New Component at the required position. You can edit this component as described above.

ⓘ You can create any structure for the components.

Remove component

To remove a component, you have the following options:

- Click the required component with the right mouse button and select Remove in the context menu.
- Click the component with the left mouse button and press Delete.
- Click the component with the left mouse button and press the Shift + Delete key combination. The selected component will be deleted without any queries.

ⓘ Scan points that were assigned to deleted components are automatically assigned to the root component.

Assign scan points

To assign scan points to the individual geometry components, proceed as follows:

1. Select the scan points you want to assign to a geometry component as described in SECTION 4.5.1.
2. Select the component which the scan points should belong to. To do so, click the component with the right mouse button and select Add Selected Points in the context menu.
3. Repeat these two steps until all scan points have been assigned to the required components.

ⓘ Scan points which you have not assigned to any other component are always automatically assigned to the root component.

Show and hide scan points

To show or hide the scan points of individual components, activate or deactivate the corresponding check box in front of the component's name.

Show number of scan points of component

To show the number of scan points in a component, click the name of the respective component and leave the mouse pointer there. The number of scan points is displayed and how many of these scan points have the Disabled scan status.

Select scan points of a component

To select the scan points of a component, you have the following options:

- Click the component with the right mouse button and select Select Points in the context menu.
- Double-click the component.
- To select the scan points of several components, double-click the required components while holding the Control key pressed.
- To select all scan points of a component, i.e. the scan points of the secondary components as well, click the component with the right mouse button and select Select All Points in the context menu.

5 Managing Data

The following functions are available for managing your measurement data:

You will find information on opening, loading and importing files, settings, and data in SECTION 5.1.

You will find information on saving and exporting files, settings, and data in SECTION 5.2.

5.1 Opening, Loading, Importing

You can display your data with the software using the following actions:

- The software can open, load, or import different file formats. You will find a summary about this as well as a description of the File Information in SECTION 5.1.1.
- You can open scans, single point measurements, and signal processor documents in Presentation mode. See SECTION 5.1.2 on this.
- You can load frequency band definitions from different formats. See SECTION 5.1.3 on this.
- In Presentation mode, you can import data (Point Signals) or frequency bands (Band Signals) from the different formats. See SECTION 5.1.4 on this.

5.1.1 Opening or Loading Files

You can evaluate sections of the measured spectrum. To do so, you have the following options:

Open files

The software can open the following files:

- Scans in proprietary binary format Scan Data (*.svd) - See SECTION 5.1.2 on this.
- Single point measurements in proprietary binary format Single Point Data (*.pvd) - See SECTION 5.1.2 on this.
- Signal processor files in proprietary binary format Signal Processor Files (*.spd) - See SECTION 5.1.2 on this.

Load files

The software can load or import the following files:

- Frequency band definition from different formats - See SECTION 5.1.3 on this.
- Settings in proprietary binary format Settings Files (*.set) - See the acquisition mode on this.
- Measurement data in Universal Files (*.unv, *.uff, *.asc) format - See SECTION 5.1.4 on this.
- Measurement data in ME'Scope5 Project Files (*.vtprj) or ME'Scope Structure Files (*.str) format - See SECTION 5.1.4 on this.
- Measurement data in ASAM ODS ATFX Files (*.atfx) format - See SECTION 5.1.4 on this.
- Geometries - See the acquisition mode on this.

File information

Once you have opened a file, you can display additional information which you have saved with the file.

To display it, select File > Info in the menu bar (toolbar:). The File Information dialog appears.

ⓘ You can also use the Info command in the project browser. See SECTION 1.5.4 on this as well.

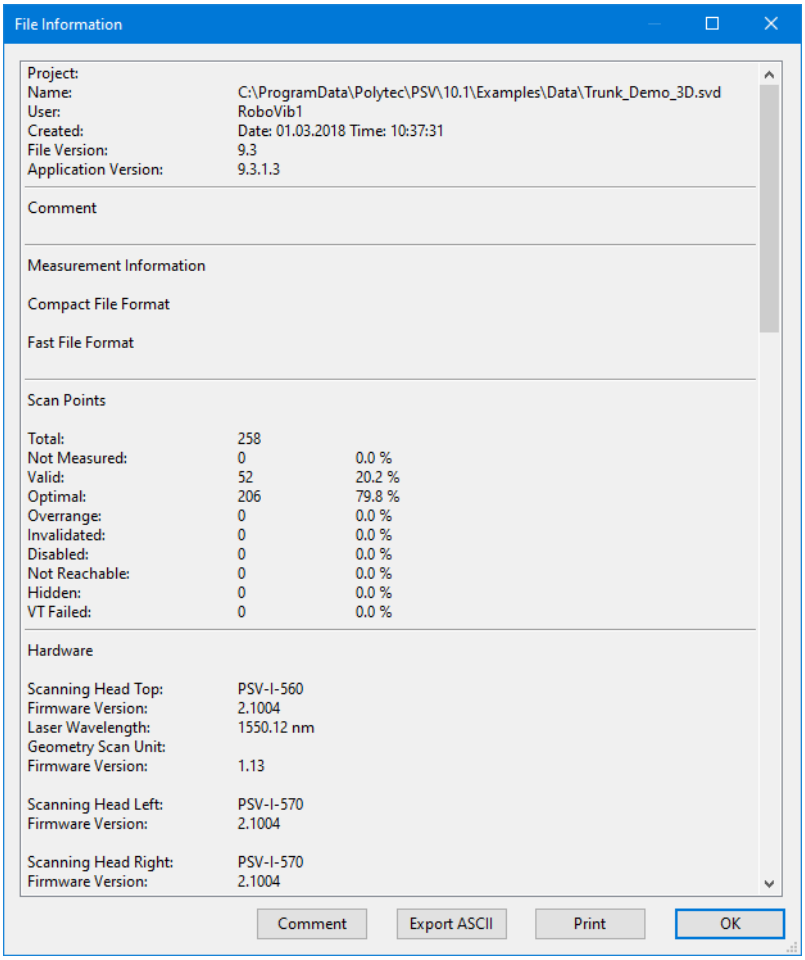


Figure 5.1: File Information dialog

ⓘ If errors occur during the scan like e.g. VideoTriangulation is failed or the status of the system has changed like e.g. status of the CoherenceOptimizer shows a too high temperature in the scanning head then these messages are displayed in the Comment.

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

Dialog description

Project: The file name with the complete file path, the user, the creation date, and the software version used to make the measurement are all shown.

Comment: Comments on the measurement are shown. You can add to comments by clicking Edit Comment at the bottom of the dialog. The Comment dialog appears. You can now add the text to the measurement and save it by clicking OK.

Measurement Information: Information on file formats or error messages during the measurement are displayed.

Hardware: The system equipment is shown (sensor head, junction box, data acquisition board, etc.).

General, Frequency, Sampling, Trigger, Channels, SE etc.: The measurement mode and other parameters for data acquisition are displayed.

Vibrometer 1: The front-end or controller equipment is shown respectively. The Bragg cell frequency is additionally displayed for the UHF.

Function Generator 1: If you used an internal function generator, the type and settings of the function generator are displayed.

Export ASCII: You can export the file information in ASCII Files (*.txt) format by clicking Exort ASCII at the bottom of the dialog. The Save As dialog appears. Navigate to the saving location, enter the file name and click Save.

Print: You can print out the file information by clicking Print at the bottom of the dialog. The Print dialog appears. Select the printer and set further print properties. To start the printing process, click OK.

Features

Scan Points: A summary of the scan point status of the scan points after saving the measurement is displayed.

5.1.2 Opening Scans, Single Point Measurements and Signal Processor Documents

In Presentation mode, you can open the following files:

- Scans in proprietary binary format Scan Data (*.svd)
- Single point measurements in proprietary binary format Single Point Data (*.pvd)
- Signal processor files in proprietary binary format Signal Processor Files (*.spd)

To do so, you have the following options:

- Open dialog
- List of most recent files in the File menu
- Project browser

ⓘ You can also open the last scan measured if you switch to Presentation mode. To do this, follow the instructions in the software.

If you get a message during opening, you have not defined a distance to the object. The distance must not be 0. Enter a correct distance and click OK.

Open file via dialog

In order to open a scan, a single point measurement or a signal processor document, proceed as follows:

1. Select File > Open in the menu bar (toolbar:).

   The Open dialog appears.

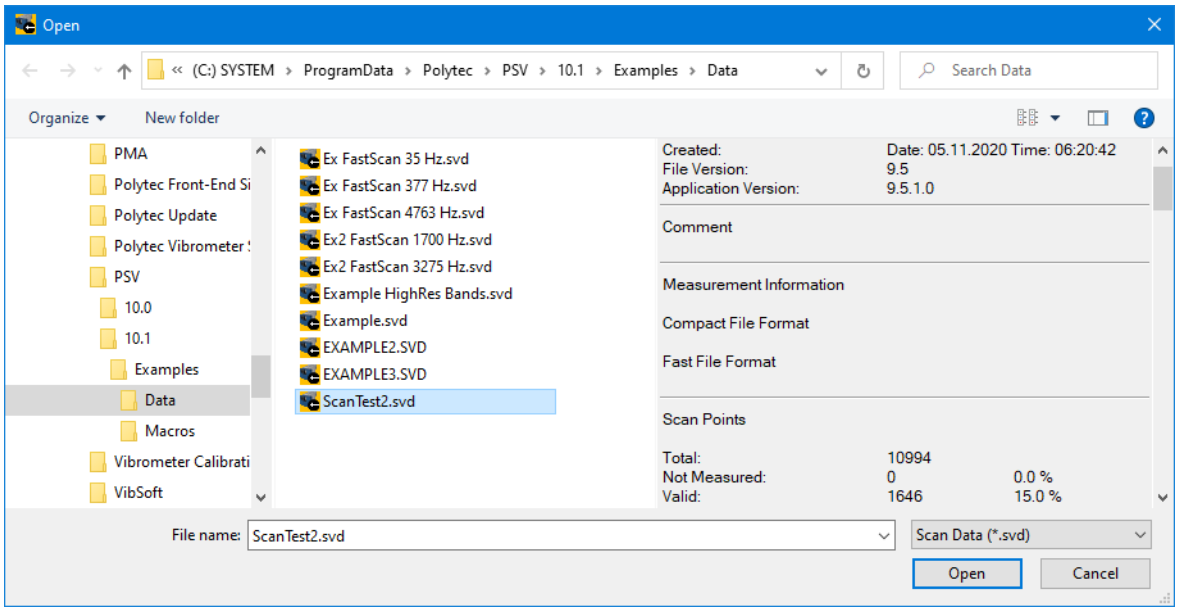


Figure 5.2: Open dialog for a scan

2. Navigate to the saving location and select Files of type and File name.
3. To identify the files more easily, show the preview by clicking .

   You will receive information on the project, on scan points, the hardware, and the setting used to measure the data.
4. Click Open.

   If you open a scan, a presentation window appears. If you open a single point measurement, an analyzer appears and if you open a signal processor document, a signal processor appears.

Open file from file list

At the bottom of the File menu, there is a list of up to nine files which you opened most recently. You can open one of these files by clicking it here in the list.

Open file via project browser

You can use the project browser to open the files. See SECTION 1.5.4 under Open on this as well.

5.1.3 Loading Frequency Band Definition

In presentation mode, you can load frequency band definitions from the following formats:

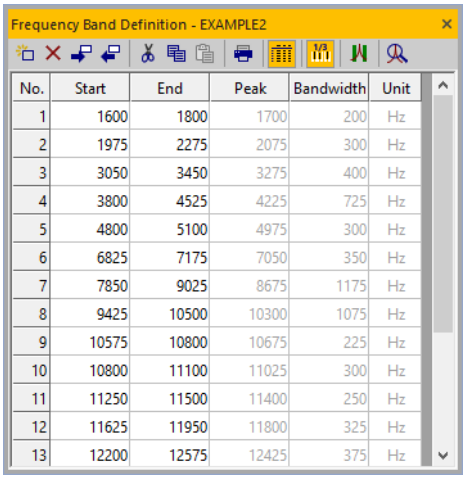
- Frequency Band Description (*.fbd)
- PSV Files (*.svd)
- ASCII Files (*.txt)

ⓘ When you load a frequency band definition, the existing one is overwritten.

To load a frequency band definition, proceed as follows:

1. Open the measurement which you want to load the frequency band definition for. See SECTION 5.1.2 on this.
2. To open the frequency band definition, select Setup > Frequency Bands in the menu bar (toolbar of the presentation window:).

   The Frequency Band Definition dialog appears.



No.	Start	End	Peak	Bandwidth	Unit
1	1600	1800	1700	200	Hz
2	1975	2275	2075	300	Hz
3	3050	3450	3275	400	Hz
4	3800	4525	4225	725	Hz
5	4800	5100	4975	300	Hz
6	6825	7175	7050	350	Hz
7	7850	9025	8675	1175	Hz
8	9425	10500	10300	1075	Hz
9	10575	10800	10675	225	Hz
10	10800	11100	11025	300	Hz
11	11250	11500	11400	250	Hz
12	11625	11950	11800	325	Hz
13	12200	12575	12425	375	Hz

Figure 5.3: Frequency Band Definition dialog

3. Click in the Frequency Band Definition dialog.

   The Open dialog appears.
4. Navigate to the saving location and select Files of type and File name.
5. Click Open.

   The frequency bands will be loaded.
6. To close the frequency band definition, select Setup > Frequency Bands in the menu bar again (toolbar of the presentation window:).

5.1.4 Importing Data in Universal File, ME'Scope or ASAM ODS ATFX Format

In presentation mode you can import data (Point Signals) or frequency bands (Band Signals) from the following formats:

- Universal Files (*.unv, *.uff, *.asc)
- ME'Scope5 Project Files (*.vtprj)
- ME'Scope Block Files (*.blk)
- ASAM ODS ATFX Files (*.atfx)

This data is saved as user defined data sets or bands and can then be analyzed. This means for example you can import modal adaptations to the spectra from ME'Scope and make comparisons with the original data in the software.

ⓘ If you import data from ME'Scope5 (new file extension *.vtprj), then ME'Scope5 is automatically started, the appropriate file opened, and then the data is imported. So make sure that ME'Scope5 is installed on your PC.

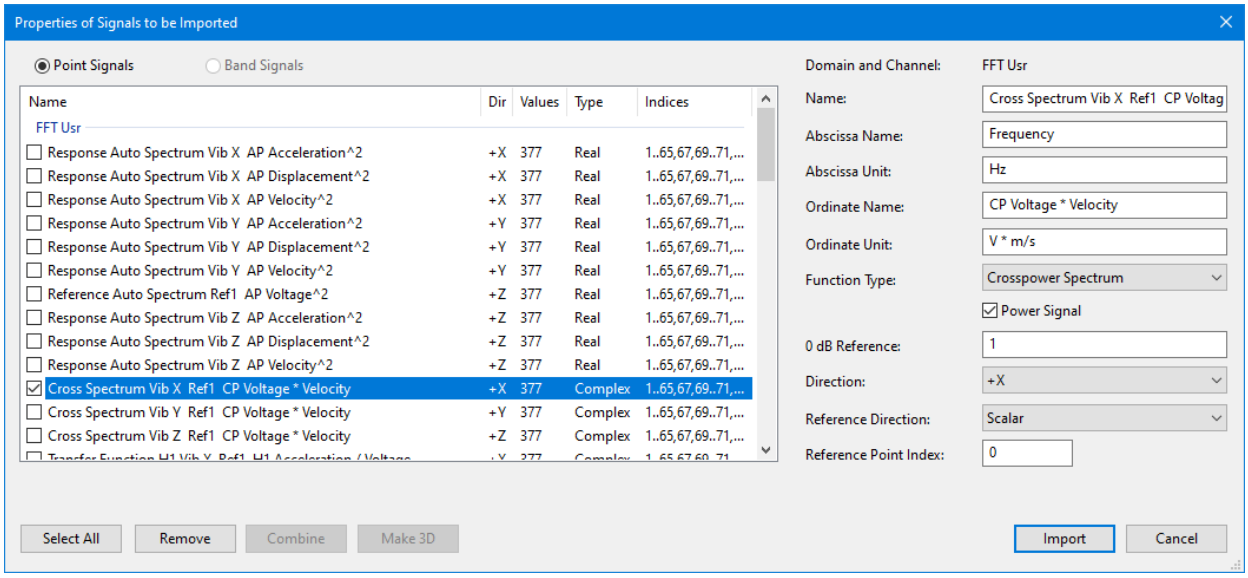


Figure 5.4: Properties of Signals to be Imported dialog (Point Signals)

Signals: On the left, select the signals which you want to import (Point Signals or Band Signals). Activate the corresponding Name check box. If you want to import all signals, click Select All. If you want to remove superfluous signals to be able to keep an overview, mark them and click Remove.

Domain and Channel: On the right, the domain, the channel as well as the signal name and other properties are shown. For every signal, you can adjust these properties for import. If you want to import bands, you will see the Band Definitions right at the bottom.

You can set the properties of several user defined signals to the same value. To do so, proceed as follows:

1. Mark the respective signals in the list. To do so, hold the Control key pressed and click the signals.
2. Enter the required common property on the right.
3. As soon as you now change the signal selection or click OK, the changes are applied to the signals.

Import

To import data in Universal File, ME'Scope or ASAM ODS ATFX format, proceed as follows:

1. Open the measurements which you want to import data into. See SECTION 5.1.2 on this.
2. Select File > Import in the menu bar.

   The Open dialog appears.

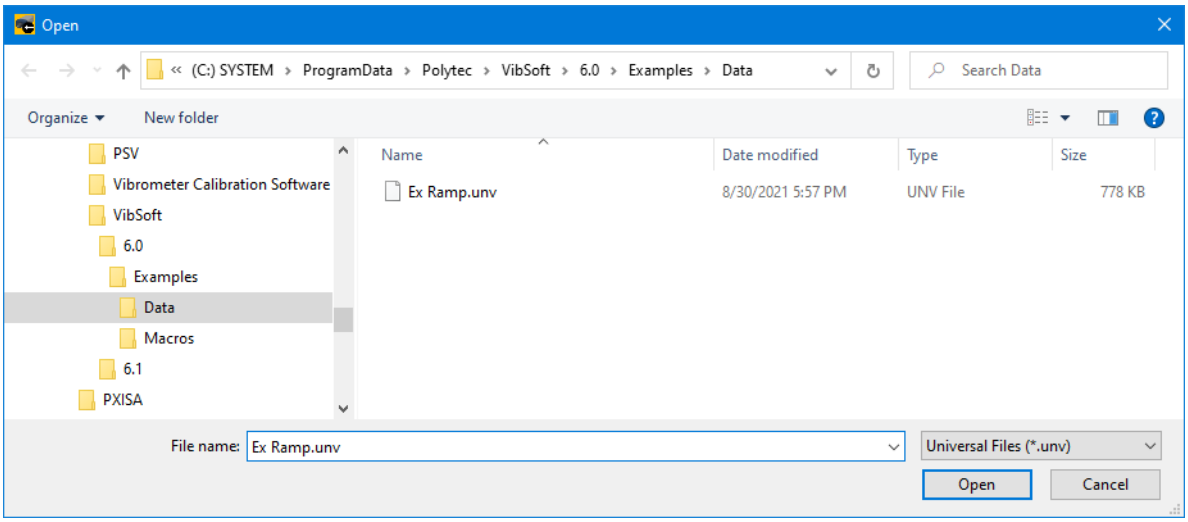


Figure 5.5: Open dialog for importing data (Universal File)

3. Navigate to the saving location and select Files of type and File name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

4. To identify the files more easily, show the preview by clicking  .

   You will receive information on the project, the hardware, and the setting used to measure the data.
5. Click Open.

ⓘ If you import an ME'Scope project file with several structure elements or an ASAM file with several base elements, first of all a query appears. Select the element required and click OK.

The data is imported and the Properties of Signals to be Imported dialog appears. See the figure above on this as well.

6. Mark the signals which you want to import and if necessary, set the parameters for import. You will find information on this in the following.
7. Click Import.

Combine signals

You can group signals with the same properties from different scan points (different indices). To do so, proceed as follows:

1. Mark the corresponding signals by either holding the Control key pressed while clicking the individual signals, or holding the Shift key pressed while clicking the first and last of the wanted signals. See the following figure on this.
2. Click Combine.

   The signals are combined and selected for import.

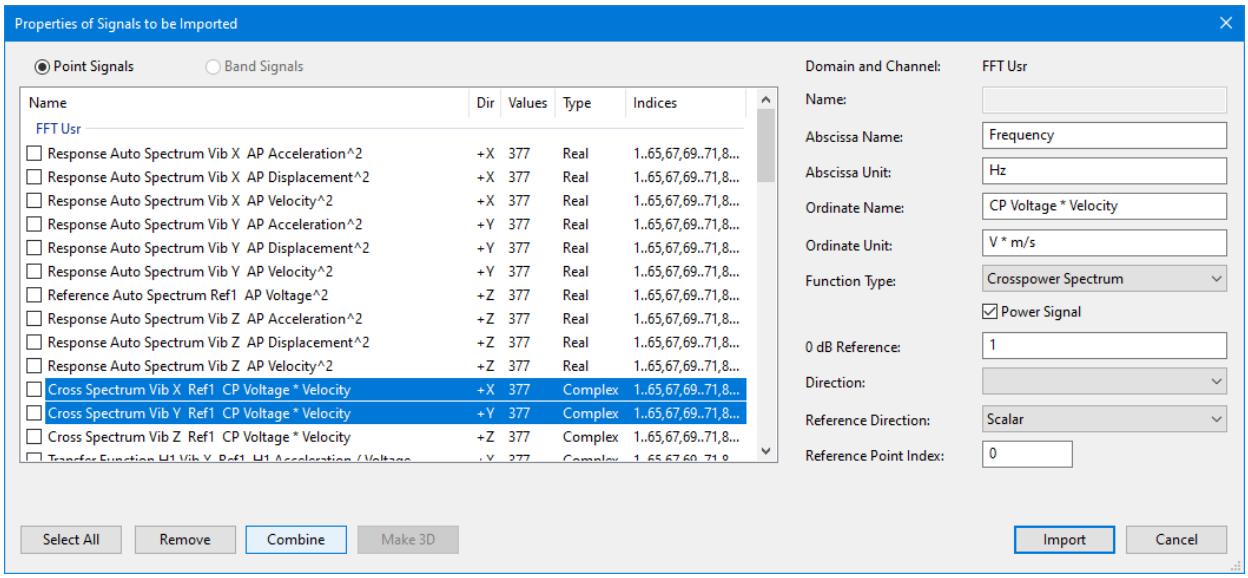


Figure 5.6: Marking signals to combine them

ⓘ If a combination is not possible you will receive a message which tells you which conditions need to be fulfilled. See the following figure on this as well.

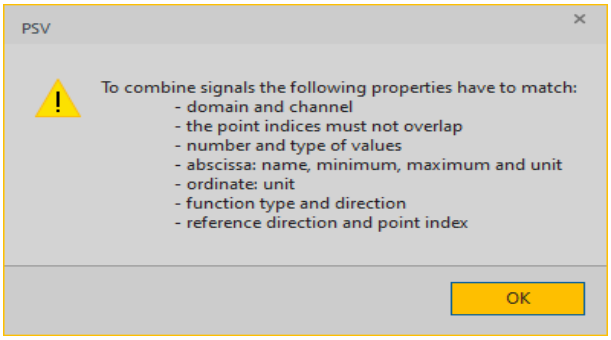


Figure 5.7: Conditions for combining signals

ⓘ You can also combine frequency bands. To do so, the signals need to have the same properties but different frequencies. These can then be combined to frequency bands of a single, user defined signal. The frequency band is then selected in the presentation window using the frequency band list.

Generate 3D signals

You can combine signals with the same properties and indices, but different directions, into one 3D signal. To do so, proceed as follows:

1. Mark the corresponding signals by either holding the Control key pressed while clicking the individual signals or also holding the Shift key pressed while clicking the first and last of the wanted signals. See the following figures on this.
2. Click Make 3D.

   The signals are grouped together to a 3D signal, are positioned under a separate headline and are marked for import. See the following figures on this.

ⓘ In case no 3D signal is possible, you will get a message which states which conditions must be fulfilled. See the following figures on this.

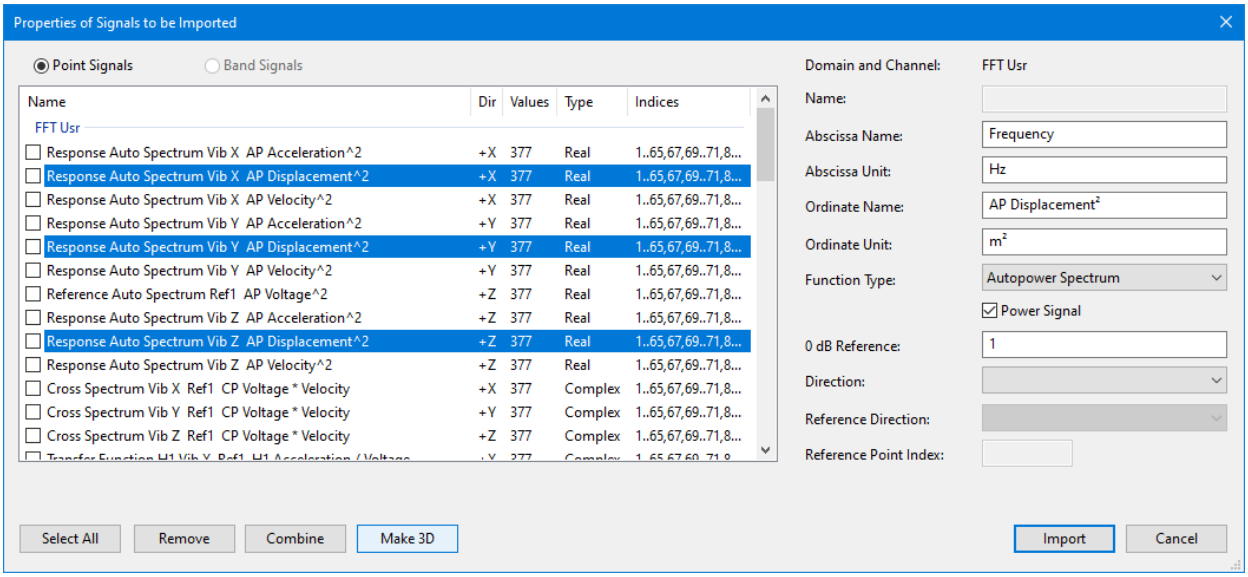


Figure 5.8: Marking signals to generate a 3D signal

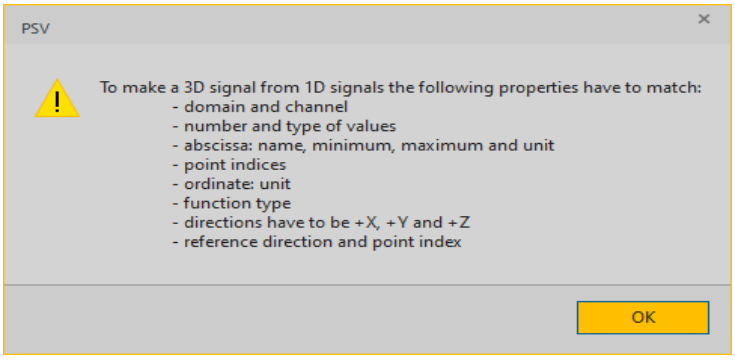


Figure 5.9: Conditions for generating 3D signals

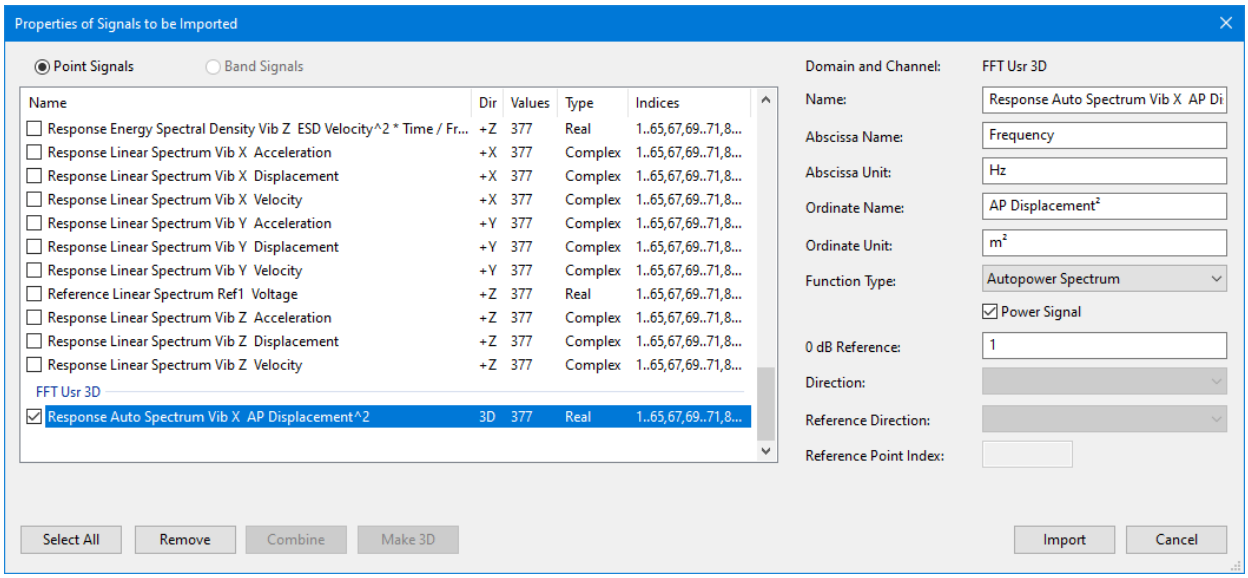


Figure 5.10: 3D signal generated

5.2 Saving and Exporting

The software offers the following options for saving and exchanging data:

- Saving scans in proprietary binary format Scan Data (*.svd) - See SECTION 5.2.1 on this.
- Saving scans with frequency band data, but without point data in proprietary binary format Scan Data (*.svd) - See SECTION 5.2.1 on this.
- Saving scans with user defined data sets, but without frequency band data and original point data in proprietary binary format Scan Data (*.svd) - See SECTION 5.2.1 on this.
- Saving single point measurements and average spectrum in proprietary binary format Single Point Data (*.pvd) - See SECTION 5.2.2 on this.
- Saving frequency band definitions in the ASCII Files (*.txt) format or in the Frequency Band Description (*.fbd) format - See SECTION 5.2.3 on this.
- Saving windows in various graphics formats - See SECTION 5.2.4 on this.
- Saving animations in different multimedia formats - See SECTION 5.2.5 on this.
- Saving recalculated scans as a signal processor document in Signal Processor Data (*.spd) format - See SECTION 3.7.5 on this.
- Copying measurement data in ASCII format, signal processor data, graphics, and windows settings onto the clipboard - See SECTION 5.2.6 on this.
- Exporting measurement data in ASCII Files (*.txt) format - See SECTION 5.2.7 on this.
- Exporting measurements in Universal Files (*.unv, *.uff, *.asc) format - See SECTION 5.2.8 on this.
- Exporting measurement data in Sound Files (*.wav) Wave Sound format - See SECTION 5.2.9 on this.
- With ExpME option: Exporting measurements in ME'Scope Files (*.str, *.blk, *.shp) format - See SECTION 5.2.10 on this.
- With ExpASAM option: Exporting measurements in ASAM ODS ATFX Files (*.atfx) format - See SECTION 5.2.11 on this.
- With GeoWorks option: Exporting geometries in STL Files (*.stl) stereo lithography data format - See SECTION 5.2.12 on this.

5.2.1 Saving a Scan

Complete scan

When you start a scan, the software automatically sets up a file for the scan and asks you to save.

During scanning, the software saves the following data in proprietary binary format Scan Data (*.svd):

- Settings
- Spectra of all scan points which have been measured completely (single point spectra)
- Average spectrum of all scan points
- RMS signal of amplitudes over the entire bandwidth

Even if you cancel a scan, the software has saved all scan points already measured.

In Presentation mode, you can define frequency bands for a scan and draw profile lines. The software automatically saves the frequency band definitions and profile lines in the scan.

You define frequency bands as described in SECTION 4.2. You draw profile lines as described in SECTION 4.3.

Frequency bands

In Presentation mode, often only certain sections of the frequency range measured are of interest. For these sections you usually define frequency bands and then evaluate only those frequency bands. You can save this scan without the single point spectra, as the frequency bands contain all relevant information. This means you can save a lot of disk space.

The scan is then subject to certain limitations:

- You cannot define any other frequency bands.
- The Single Scan Point view is not available.

The saved average spectrum corresponds to the spectrum which is created by removing invalid single points. See SECTION 4.1 on this as well. In the file generated, the average spectrum does not change if you make points Valid or Invalidated as the single spectra necessary for this are not available.

For saving a scan without single spectra, proceed as follows:

1. Open the scan you want to save. See SECTION 5.1.2 on this.
2. Define the frequency bands which you want to evaluate. See SECTION 4.2 on this.
3. Select File > Save Bands in the menu bar.

   The Save As dialog appears.
4. Navigate to the saving location and enter the file name. The software suggests the file name Scan name_b. You can choose a different file name if you prefer.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

5. Click Save.

5.2.2 Saving Single Point Measurement and Average Spectrum

In acquisition mode, you can save the measurement in the active analyzer in proprietary binary format Single Point Data (*.pvd). In Presentation mode, you can save the single scan point or the average spectrum displayed from a scan.

ⓘ The average spectrum saved corresponds to the original data and not to the spectrum created by removing invalid single points. See SECTION 4.1 on this as well.

At the same time, the software saves the following data:

- Time signals
- Spectra
- Parameters for data acquisition and evaluation

To save single point measurements or spectra, proceed as follows:

1. Select File > Save Single Point Data As in the menu bar.

   The Save As dialog appears.

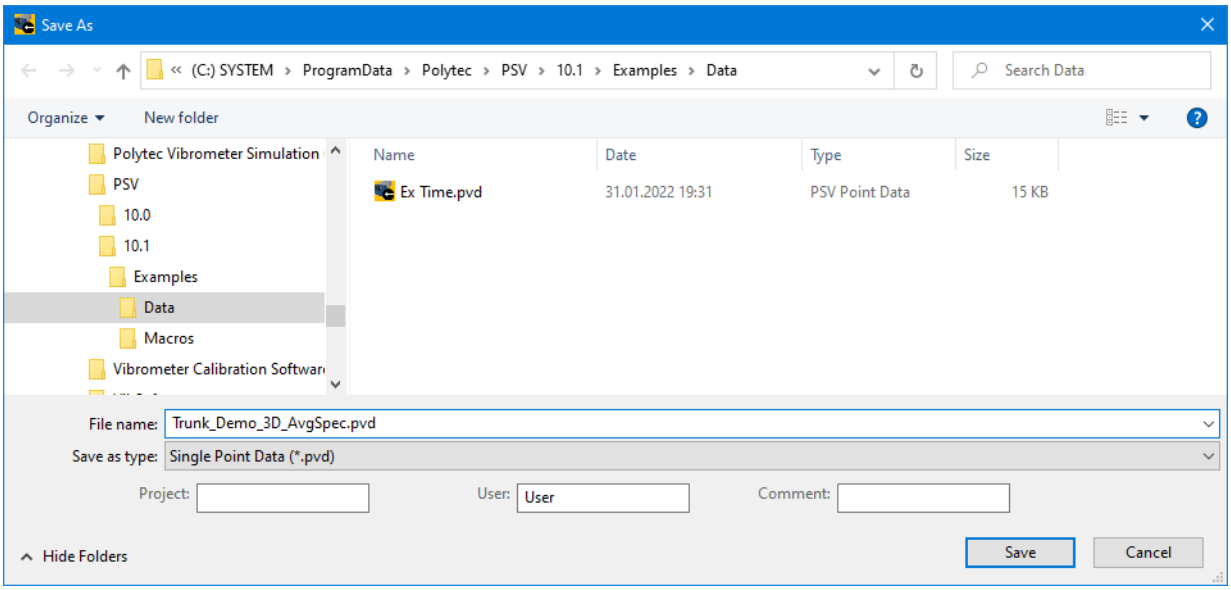


Figure 5.11: Save As dialog for saving single point data

2. Navigate to the saving location and enter the file name. Depending on the measurement mode or view, the software suggests a file name. You can choose a different file name if you prefer.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

3. At the bottom in the dialog you can enter additional information to characterize the file more accurately.

ⓘ This information is part of the file properties which you can also view and edit in the Explorer. It is displayed when you open a file in presentation mode and also in the File Information dialog. See SECTION 5.1.1 on this as well.

4. Click Save.

5.2.3 Saving Frequency Band Definition

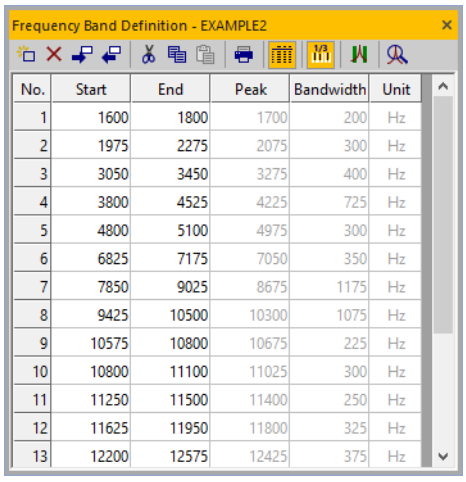
In presentation mode, you can save the current frequency band definition in the following formats:

- Frequency Band Description (*.fbd)
- PSV Files (*.svd)
- ASCII Files (*.txt)

To save frequency bands, proceed as follows:

1. Open the measurement for which you want to save the frequency band definition. See SECTION 5.1.2 on this.
2. To open the frequency band definition, select Setup > Frequency Bands in the menu bar (toolbar of the presentation window: ).

   The Frequency Band Definition dialog appears.



No.	Start	End	Peak	Bandwidth	Unit
1	1600	1800	1700	200	Hz
2	1975	2275	2075	300	Hz
3	3050	3450	3275	400	Hz
4	3800	4525	4225	725	Hz
5	4800	5100	4975	300	Hz
6	6825	7175	7050	350	Hz
7	7850	9025	8675	1175	Hz
8	9425	10500	10300	1075	Hz
9	10575	10800	10675	225	Hz
10	10800	11100	11025	300	Hz
11	11250	11500	11400	250	Hz
12	11625	11950	11800	325	Hz
13	12200	12575	12425	375	Hz

Figure 5.12: Frequency Band Definition dialog

3. Click  in the Frequency Band Definition dialog.

   The Save As dialog appears.

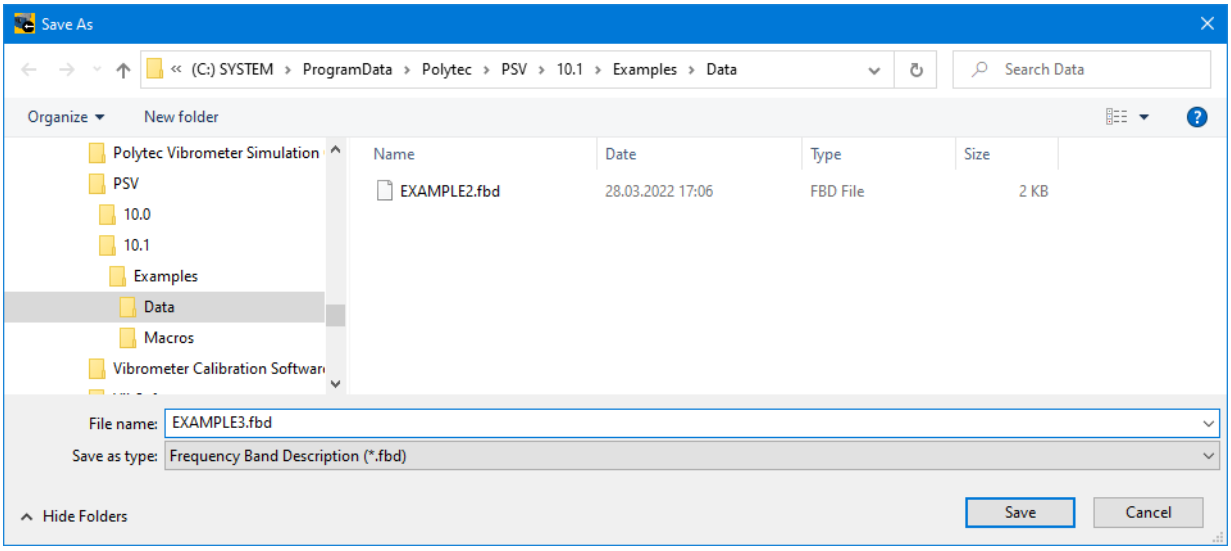


Figure 5.13: Save As dialog for frequency bands

4. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

5. Click Save.

   The displayed frequency band definition will be saved.

6. To close the frequency band definition, select Setup > Frequency Bands in the menu bar (toolbar of the presentation window:).

5.2.4 Saving Graphics

You can save the active window in different graphics formats. The title bar and the frame are not saved.

The following graphics formats are available:

- Windows Bitmap Files (*.bmp)
- Portable Network Graphics Files (*.png)
- Jpeg File Interchange Format Files (*.jpg)
- Tagged Image File Format Files (*.tif)
- Graphics Interchange Format Files (*.gif)

ⓘ If you want to save an animation, read SECTION 5.2.5.

To save a graphic, proceed as follows:

1. Open the measurement you want to save a graphic for. See SECTION 5.1.2 on this.
2. Select File > Save Graphics in the menu bar.

   The Save As dialog appears.

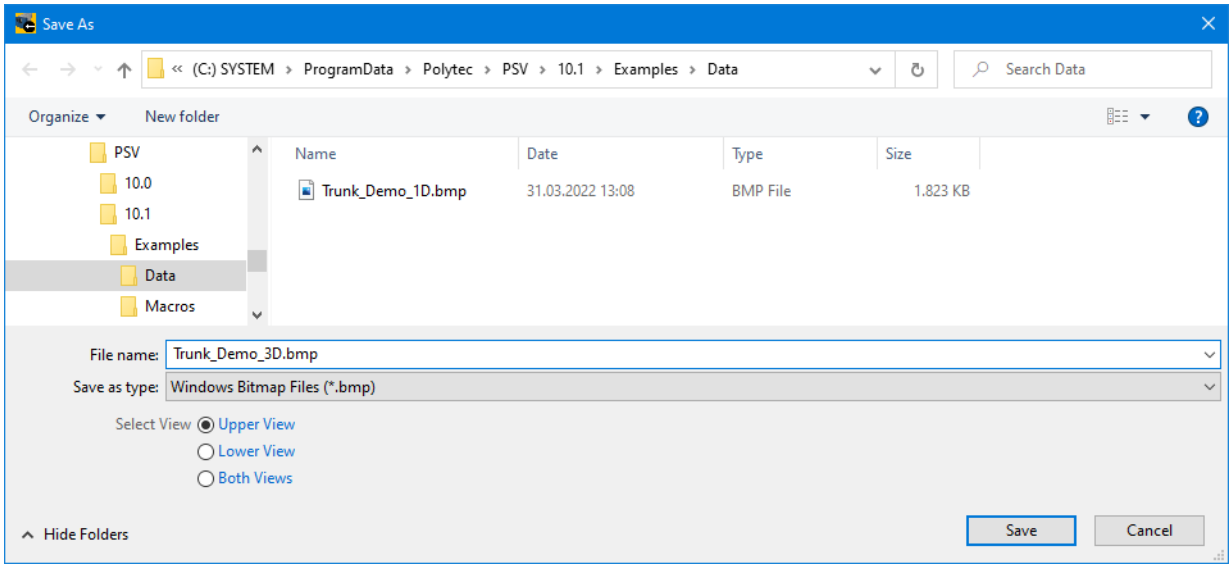


Figure 5.14: Save As dialog for graphics

ⓘ If the presentation window includes an analyzer, you can choose if you want to save an image for the Upper View, the Lower View or for Both Views.

3. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

4. Click Save.

5.2.5 Saving Animation

In Presentation mode, you can save animations in different formats. You will find information on animations in SECTION 4.4.

The following formats are available:

- Animated GIF (*.gif)
- MP4 Video Files (*.mp4)
- Windows Media Files (*.wmv)

ⓘ If you would like to save the animation in the Animated GIF (*.gif) format, we recommend hiding the video image and setting the transparency of the data to 0%. See SECTION 3.1.3 on this as well. This leads to a higher quality of the saved animation and at the same time to a lower file size.

To save animations, proceed as follows:

1. Open the measurement which you want to save an animation for. See SECTION 5.1.2 on this.
2. Set up your animation as described in SECTION 4.4.1 or SECTION 4.4.2.
3. Select File > Save Animation in the menu bar.

   The Save As dialog appears.

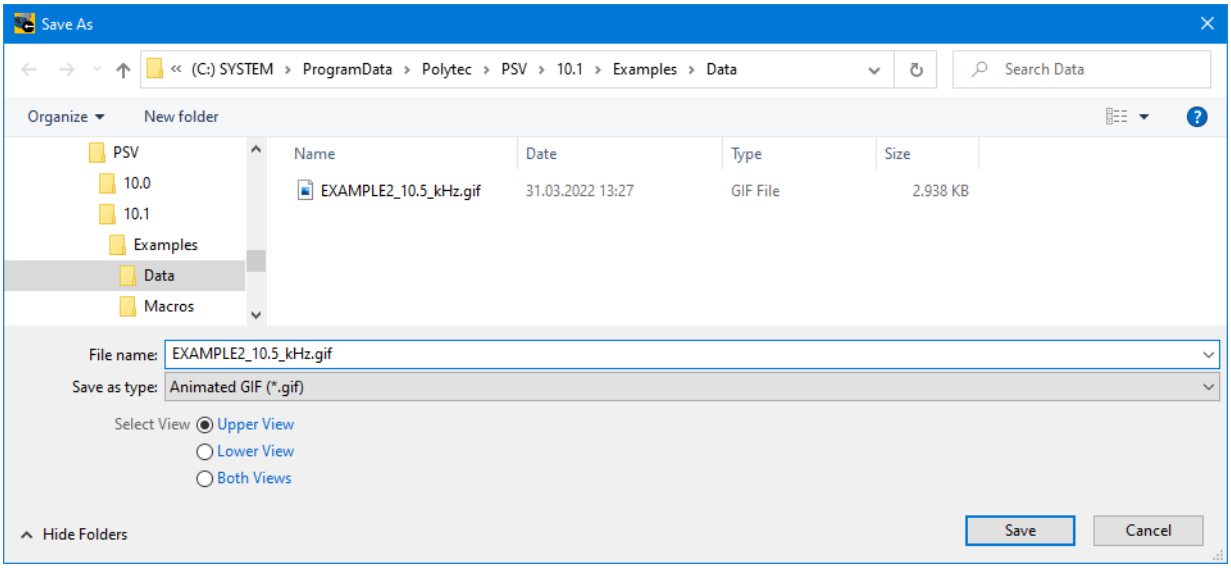


Figure 5.15: Save As dialog for animations

ⓘ If the presentation window includes an analyzer, you can choose if you want to save an image for the Upper View, the Lower View or for Both Views.

4. Navigate to the saving location, select Save as type and enter the file name. If you save the animation from the surface view, the software suggests the Scan name_Frequency file name. You can choose a different file name if you prefer.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

5. Click Save.

5.2.6 Copying Data onto Clipboard

You can copy data onto the clipboard from the active window as graphics, as signal processor data or in ASCII format. You can also copy the window settings onto the clipboard. If you select the graphics, then the title bar and the frame are not copied. The live video image is copied as graphics without scan points.

ⓘ Depending on the window selection or the view, you are offered a more or less comprehensive dialog.

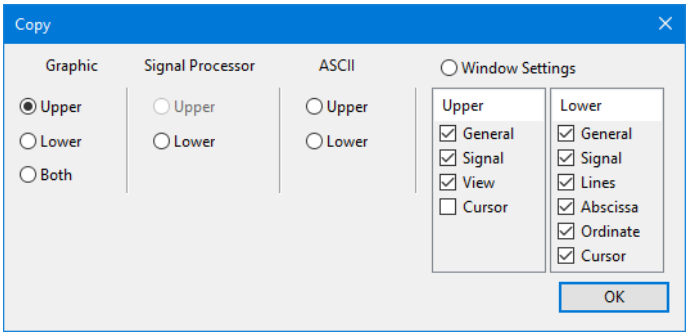


Figure 5.16: Copy dialog for clipboard

ⓘ When copying in profile view, pay attention that you do not have selected any profile lines. Otherwise these profile lines will be copied onto the clipboard.

Copy onto clipboard

To copy onto the clipboard, proceed as follows:

1. Activate the required window or the required view. To do so, click it.
2. In the menu bar, select Edit > Copy (toolbar: ).

   The Copy dialog appears (refer to figure above).
3. Select the data you want to copy onto the clipboard.
4. Click OK.

   You can now paste the data from the clipboard into other applications.

ⓘ You can also press the Ctrl + C key combination. The Copy dialog appears as long as you hold the Control key pressed. Select which data you want to copy and release the Control key. The data is copied.

Copy view onto clipboard

You can also selectively copy a certain part of the window onto the clipboard. To do so, proceed as follows:

1. Click the view with the right mouse button and select Copy in the context menu.

   The Copy dialog appears.

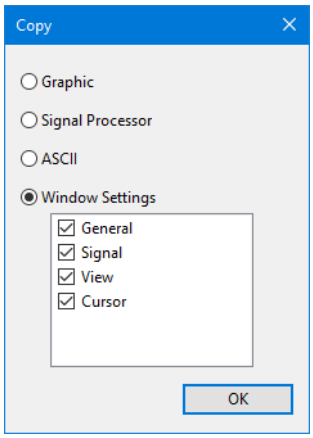


Figure 5.17: Copy dialog in Object view

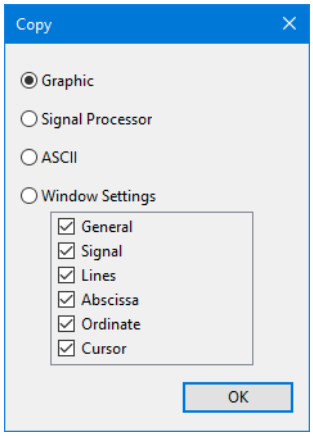


Figure 5.18: Copy dialog in the analyzer

2. Select the data you want to copy onto the clipboard.
3. Click OK.

   You can now paste the data from the clipboard into other applications.

5.2.7 Exporting Measurement Data in ASCII Format

You can export the data in the active analyzer or presentation window as an ASCII file. In doing so, the coordinates shown in the readout are exported.

To calculate the angles of the scanner mirrors from the coordinates, you will need the information from the 2D or 3D alignment respectively. You will get this from Polytec File Access.

ⓘ In principle, when exporting in ASCII format only the data which has been displayed in the presentation window is exported. If you have hidden components, this data will not be exported. If you have evaluated data (Filter Data or Interpolate Data), the evaluated data is exported.

Export ASCII

To export the measurement data in ASCII format, proceed as follows:

1. Open the measurement which you want to export as an ASCII file. See SECTION 5.1.2 on this.
2. Select File > Export > ASCII in the menu bar.

   The Save As dialog appears.

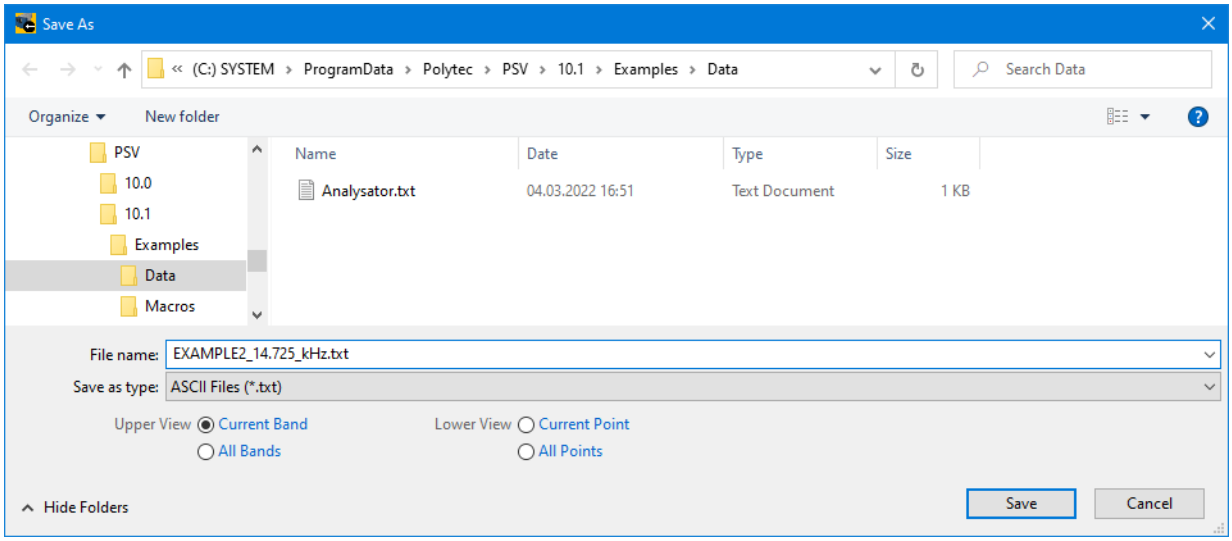


Figure 5.19: Save As dialog for ASCII export

3. Navigate to the saving location, select Save as type and enter the file name. If you export in presentation mode, the software suggests the File name_Frequency file name. You can choose a different file name if you prefer.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

4. If you export in presentation mode, in the dialog at the bottom you will find a selection suitable for your selected view. Select which data you want to export from the upper or the lower graphic. You can export different data consecutively several times.

ⓘ If you are exporting All Bands or All Points, then a file is generated for every frequency band or every scan point.

5. Click Save.

5.2.8 Exporting Universal File

Data sets

In both acquisition and presentation mode, you can export data in Universal File format to use it in a modal analysis software.

The following data sets are exported:

Data set	Description	Format
Header	General information, e.g. file name, date, comment	151
Units	Units of the signals and coordinates	164 Double Precision
Nodes (Geometry)	Coordinates of the scan points	2411 Double Precision
Elements (Trace Lines)	Graphic representation of the scan points and their connections	2412 + 82 Integer
Frequency bands	A data set for every signal and every frequency band	55 Single Precision
Scan Points	A data set for every signal and every scan point	58 Single Precision
Scan Points in binary format	A data set for every signal and every scan point	58b Single Precision

Features

You need to take the following points into consideration when exporting:

- Scan points with the Not Measured status are not exported.
- Data evaluation with Presentation > Filter Data and Presentation > Interpolate Data does not have any effect on export, the measured data is exported.
- Signals which have been gained from the original signals through integration and differentiation can be exported.
- Only MSA-500 with VDD option: Measurements in I/Q measurement mode cannot be exported as Universal File.
- Only in Presentation mode: Open the scan you want to export. See SECTION 5.1.2 on this.
- Only PSV with MultiFrame option: If you are exporting a MultiFrame measurement, the software will generate a file for all frames. The frame numbers are written in the head of the file or in the signal description respectively. You will find information on MultiFrame measurements in the acquisition mode.

Export UFF data To export data in Universal Files (*.unv, *.uff, *.asc) format, proceed as follows:

1. In the menu bar, select File > Export > Universal File.

   The Save As dialog appears.

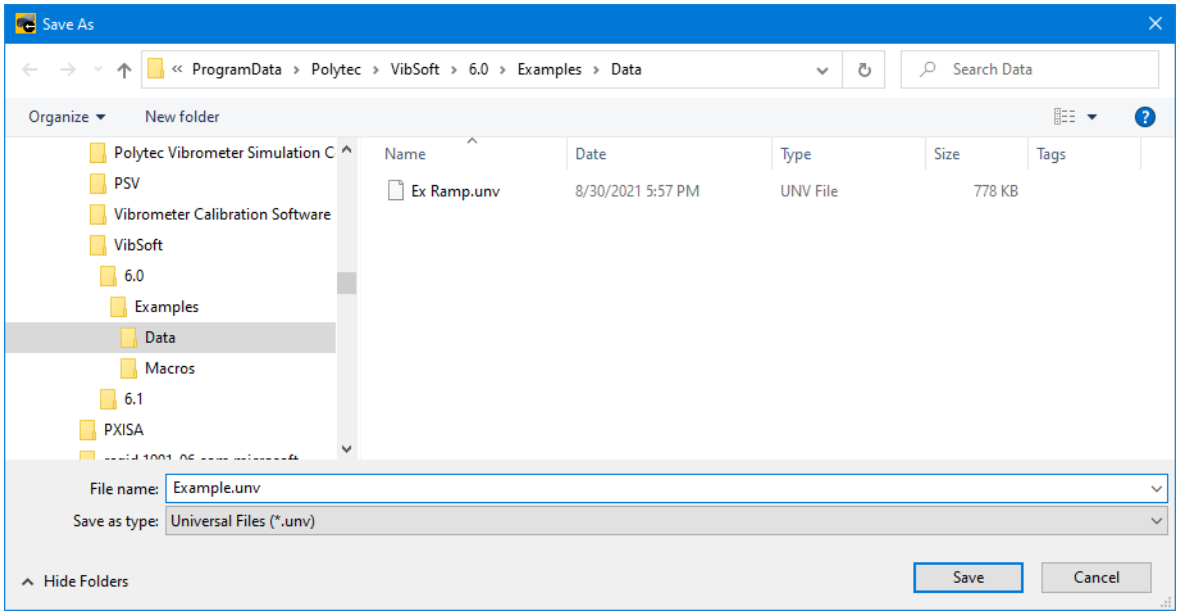


Figure 5.20: Save As dialog for Universal File export

2. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

3. Click Save.

   The Universal File Export dialog appears. See the figure below on this as well.

4. Mark the signals which you want to export and set the parameters for the export.

5. Click OK.

ⓘ You can correct the entries from a measurement for the current export again and can also enter new settings for files which already exist. However the changes have no effect on the saved measurement.

Dialog

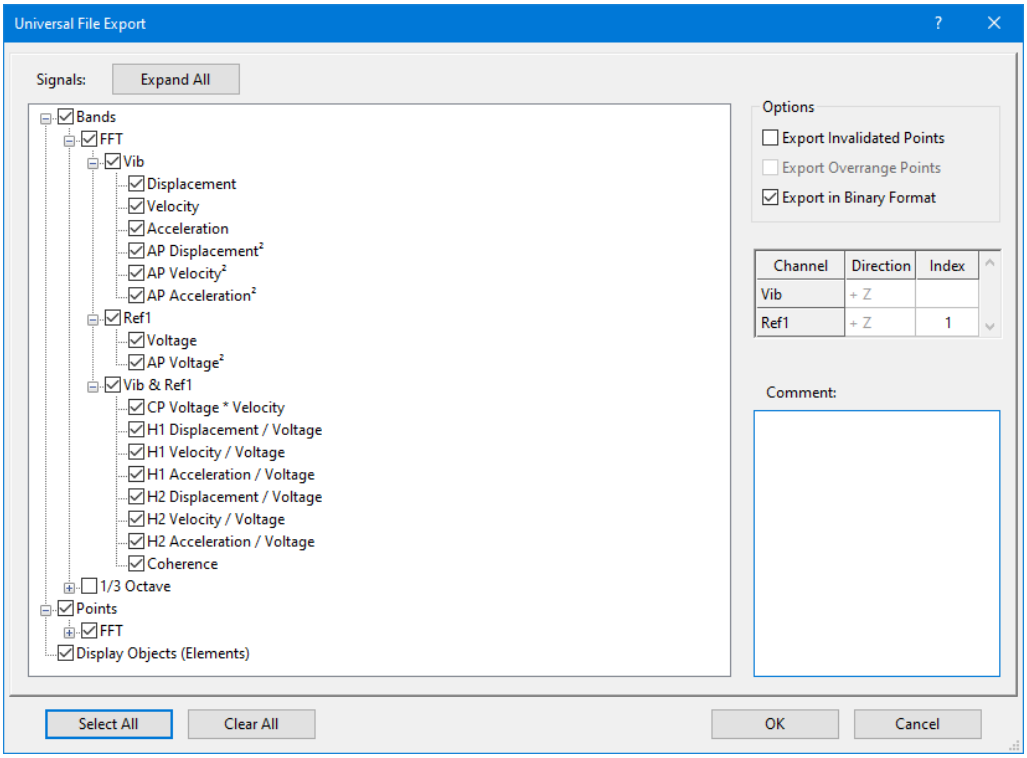


Figure 5.21: Universal File Export dialog

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

Dialog description

On the left you mark the signals and on the right you set the parameters for the export.

ⓘ If you have defined geometry components as described in SECTION 4.5.5, then only the component names for the data sets 82 and 58 will be taken into account for export.

Signals: Mark the signals which you want to export. The selection depends on which signals you have measured and whether frequency bands have been defined. If Display Objects (Elements) is ticked, then a graphic representation of the scan points and their connections is exported.

ⓘ You can export user defined data sets but not average spectra.

Expand All: Click here to display the entire tree in detail.

Select All: Click here to select all signals.

Clear All: Click here to clear the signal selection.

Export Invalidated Points: If you want to export signals from points which you have marked as Invalidated, tick the check box.

Export Overrange Points: If you want to export signals from points which have been assigned the Overrange scan status, tick the check box.

Export in Binary Format: For a faster export, tick the check box to export the data in Binary Universal File Format. If point data (format 58b) is exported, the file size is reduced to approx. 30%.

Channel: The channels you are exporting are displayed.

Direction: The directions of the vibration and the reference excitation are displayed. You can only change the direction of the vibration in the readout of the presentation window.

ⓘ It is not possible to specify the vibrational direction for the Vibrometer 3D channel.

Index: If you have selected a reference channel and have measured the reference signal at one of the scan points, enter the index of this reference point.

Enter an index for a response channel so that the software exports only one signal for that channel. This avoids, for example, that a Driving-Point function is exported for each measured scan point.

ⓘ The settings for the vibrational direction and the reference point have no effect on exporting user defined signals (Usr or Usr X, Usr Y, Usr Z channel respectively). We recommend that you check the properties of the user defined signals before exporting and if necessary, change them as described in SECTION 3.7.5.

Comment: You can enter a comment up to 256 characters long.

5.2.9 Exporting Measurement Data in Wave Sound Format

You can export data (time signals and spectra) in the active analyzer or presentation window as a file in Sound Files (*.wav) Wave Sound format. When exporting spectra, an inverse FFT is calculated and a "Resampling" matching the audio frequency is performed. For this purpose you will need a valid phase and need Rectangle set as the window function, to ensure you get the results you expect.

ⓘ For this reason, we only recommend exporting spectra in Wave Sound format if you have a complex signal for the measurement, and if you have used a trigger and the Rectangle window function.

Export Wave Sound format

To export data in Wave Sound format, proceed as follows:

1. Open the file which you want to export in Wave Sound format. See SECTION 5.1.2 on this.
2. Select File > Export > Wave Sound Format in the menu bar.

   The Save As dialog appears.

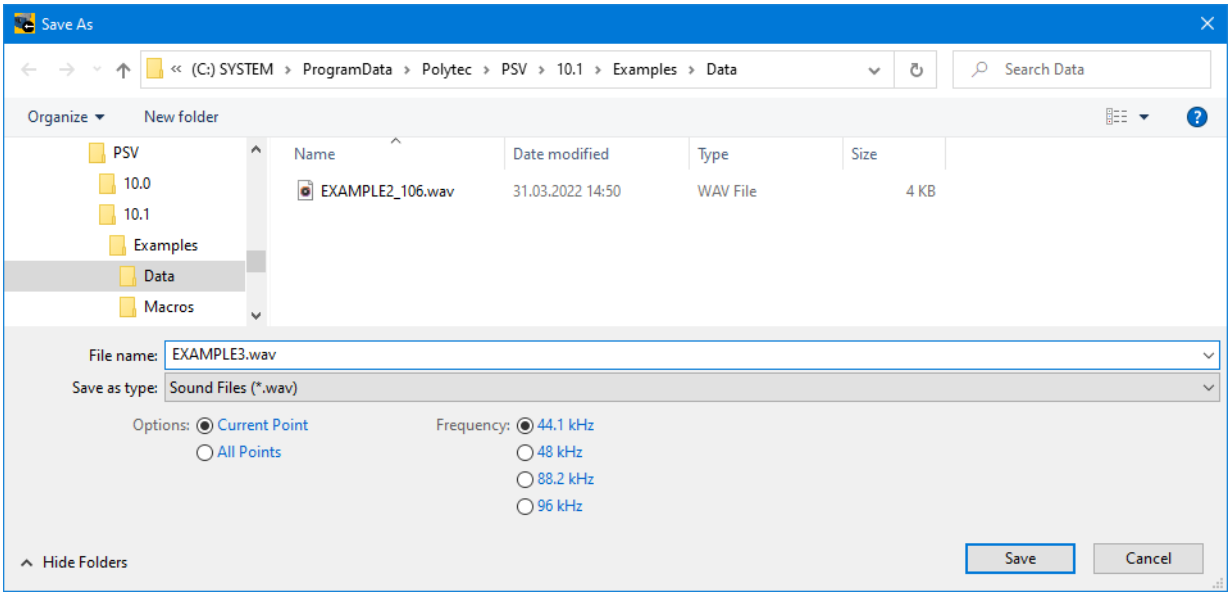


Figure 5.22: Save As dialog for Wave Sound export

3. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

4. At the bottom of the dialog, select which Frequency is used for the "Resampling". In Single Scan Point view, you can select the export options Current Point or All Points.
5. Click Save.

5.2.10 Exporting ME'Scope

Data sets

You can export files in ME'Scope format. For this, you need the ExpME option.

Export will provide up to three files with the following data:

- Single points in the ME'Scope Block Files (*.blk) format
- Frequency bands in the ME'Scope Shape Files (*.shp) format
- Geometry in the ME'Scope Structure Files (*.str) format

Features

You need to take the following points into consideration when exporting:

- Scan points with the Not Measured status are not exported.
- Data evaluation with Presentation > Filter Data and Presentation > Interpolate Data does not have any effect on export, the measured data is exported.
- Only MSA-500 with VDD option: Measurements in I/Q measurement mode cannot be exported as ME'Scope.
- Only in Presentation mode: Open the scan you want to export. See SECTION 5.1.2 on this.
- Only PSV with MultiFrame option: If you are exporting a MultiFrame measurement, the software will generate a file for all frames. The frame numbers are written in the head of the file or in the signal description respectively. You will find information on MultiFrame measurements in the Acquisition software manual.

Export ME'Scope data

To export data in ME'Scope Files (*.str, *.blk, *.shp) format, proceed as follows:

1. In the menu bar, select File > Export > ME'Scope.

   The Save As dialog appears.

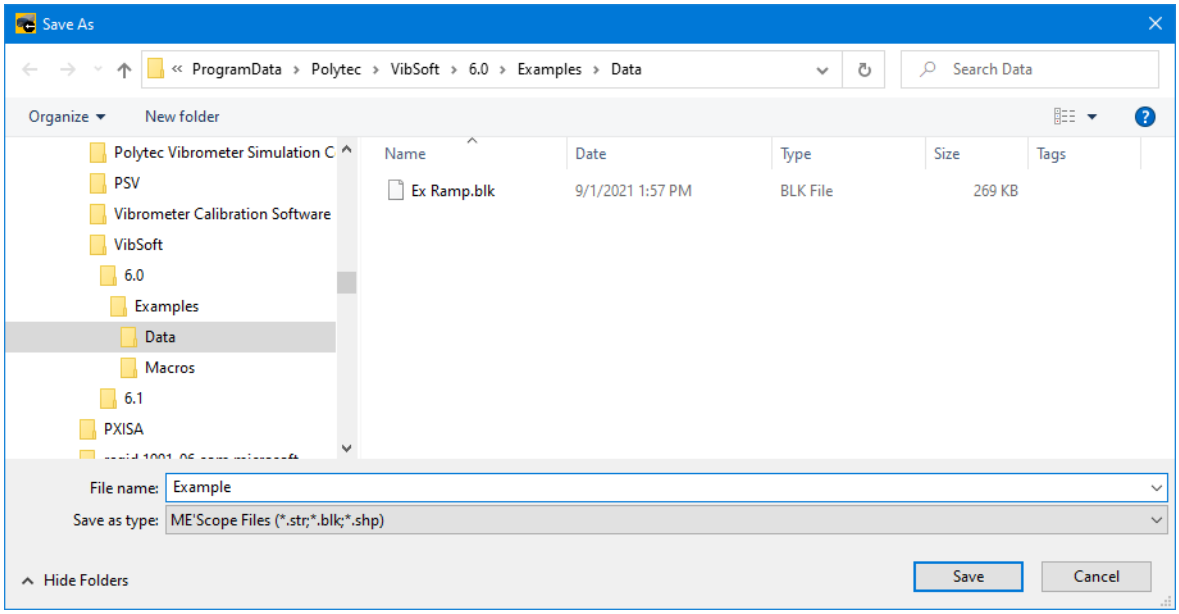


Figure 5.23: Save As dialog for ME'Scope export

2. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

3. Click Save.

   The ME'Scope Export dialog appears. See the following figure on this as well.

4. Mark the signals which you want to export and set the parameters for the export.

ⓘ Signals with different x-axes, e.g. Time and FFT domain signals, can not be exported together. Export these signals one after the other respectively.

5. Click OK.

ⓘ You can correct the entries from a measurement for the current export again and can also enter new settings for files which already exist. However the changes have no effect on the saved measurement.

Dialog

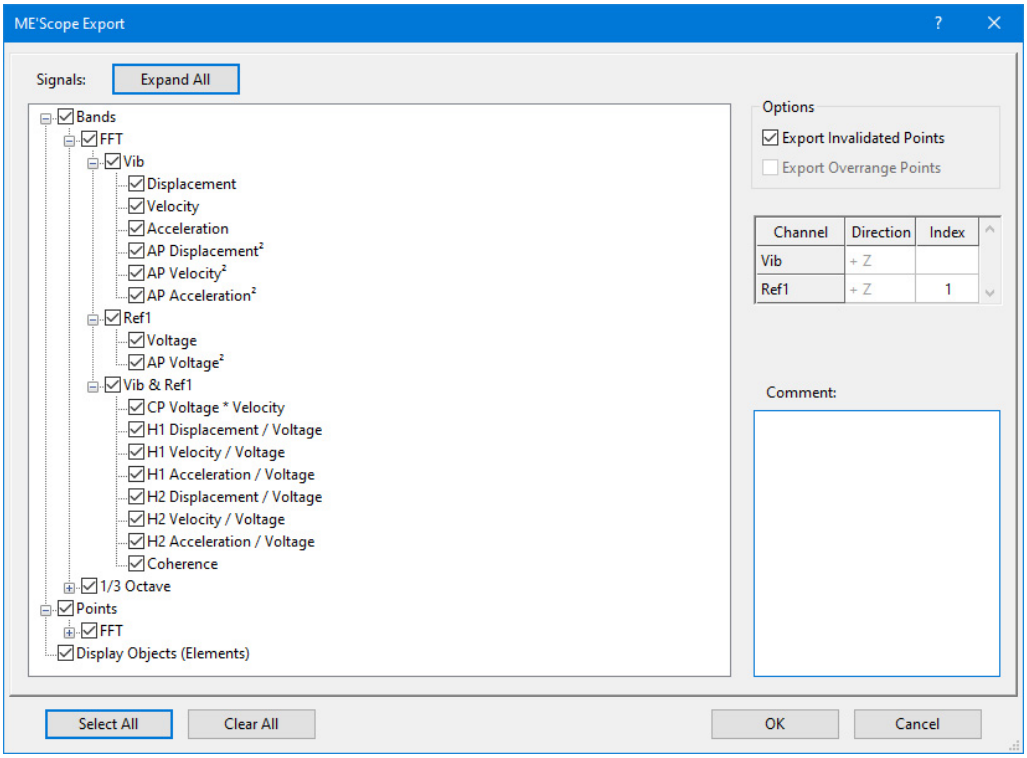


Figure 5.24: ME'Scope Export dialog

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

Dialog description

On the left you mark the signals and on the right you set the parameters for the export.

Signals: Mark the signals which you want to export. The selection depends on which signals you have measured and whether frequency bands have been defined. If Display Objects (Elements) is ticked, then a graphic representation of the scan points and their connections is exported.

ⓘ You can export user defined data sets but not average spectra.

Expand All: Click here to display the entire tree in detail.

Select All: Click here to select all signals.

Clear All: Click here to clear the signal selection.

Export Invalidated Points: Tick the check box if you want to export signals of points which you have marked as Invalidated.

Export Overrange Points: Tick the check box if you want to export signals of points which have been assigned the Overrange scan status.

Channel: The channels you are exporting are displayed.

Direction: The directions of the vibration and the reference excitation are displayed. You can only change the direction of the vibration in the readout of the presentation window.

ⓘ It is not possible to specify the vibrational direction for the Vibrometer 3D channel.

Index: If you have selected a reference channel and have measured the reference signal at one of the scan points, enter the index of this reference point.

Enter an index for a response channel so that the software exports only one signal for that channel. This avoids, for example, that a Driving-Point function is exported for each measured scan point.

ⓘ The settings for the vibrational direction and the reference point have no effect on exporting user defined signals (Usr or Usr X, Usr Y, Usr Z channel respectively). We recommend that you check the properties of the user defined signals before exporting and if necessary, change them as described in SECTION 3.7.5.

Comment: You can enter a comment up to 256 characters long.

5.2.11 Exporting ASAM ODS ATFX

Data sets

You can export measurement data in ASAM format (default ASAM ODS 5.3.0). For this, you need the ExpASAM option.

Export will provide two files with the following data:

- XML file in ASAM ODS ATFX Files (*.atfx) format
- Binary file with the included data in (*.btf) format

ⓘ You always have to save or move both files together. During import, only the ASAM file is selected. But all data is needed for an import.

Features

You need to take the following points into consideration when exporting:

- Scan points with the Not Measured status are not exported.
- Data evaluation with Presentation > Filter Data and Presentation > Interpolate Data does not have any effect on export, the measured data is exported.
- Only MSA-500 with VDD option: Measurements in I/Q measurement mode cannot be exported as ASAM ODS ATFX.
- Only in Presentation mode: Open the scan you want to export. See SECTION 5.1.2 on this.
- Only PSV with MultiFrame option: If you are exporting a MultiFrame measurement, the software will generate a file for all frames. The frame numbers are written into the signal description. You will find information on MultiFrame measurements in the Acquisition software manual.

Export ASAM data

To export the data in the ASAM ODS ATFX Files (*.atfx) format, proceed as follows:

1. In the menu bar, select File > Export > ASAM ODS ATFX.

   The Save As dialog appears.

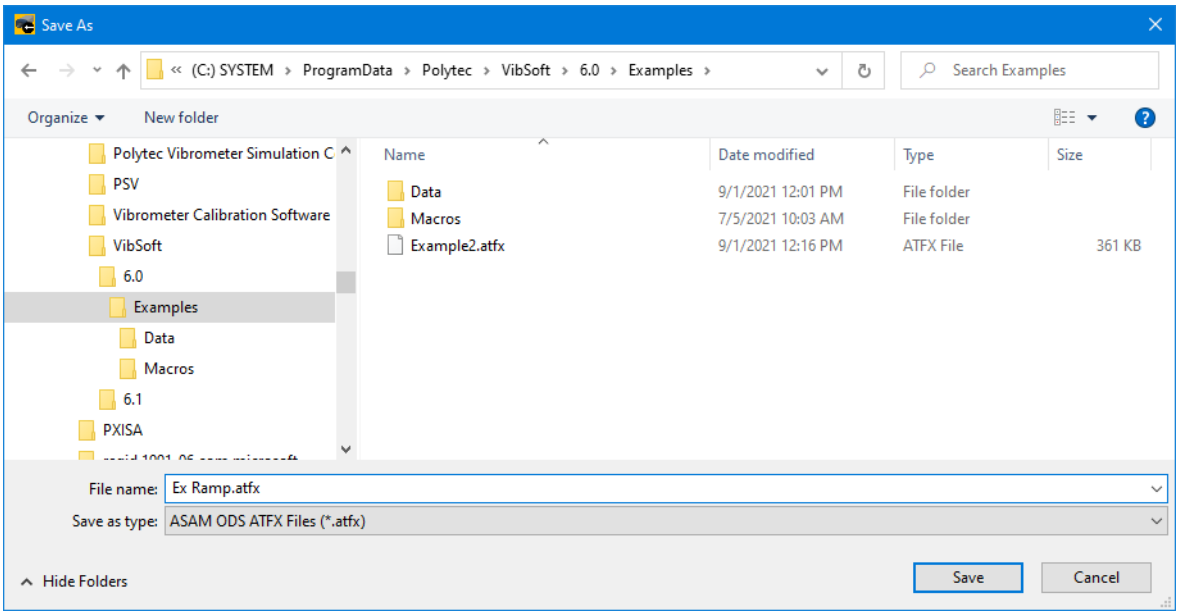


Figure 5.25: Save As dialog for ASAM ODS ATFX export

2. Navigate to the saving location and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

3. Click Save.

   The ASAM ODS ATFX Export dialog appears. See the following figure on this as well.

4. Mark the signals which you want to export and set the parameters for the export.

5. Click OK.

ⓘ You can correct the entries from a measurement for the current export again and can also enter new settings for files which already exist. However the changes have no effect on the saved measurement.

Dialog

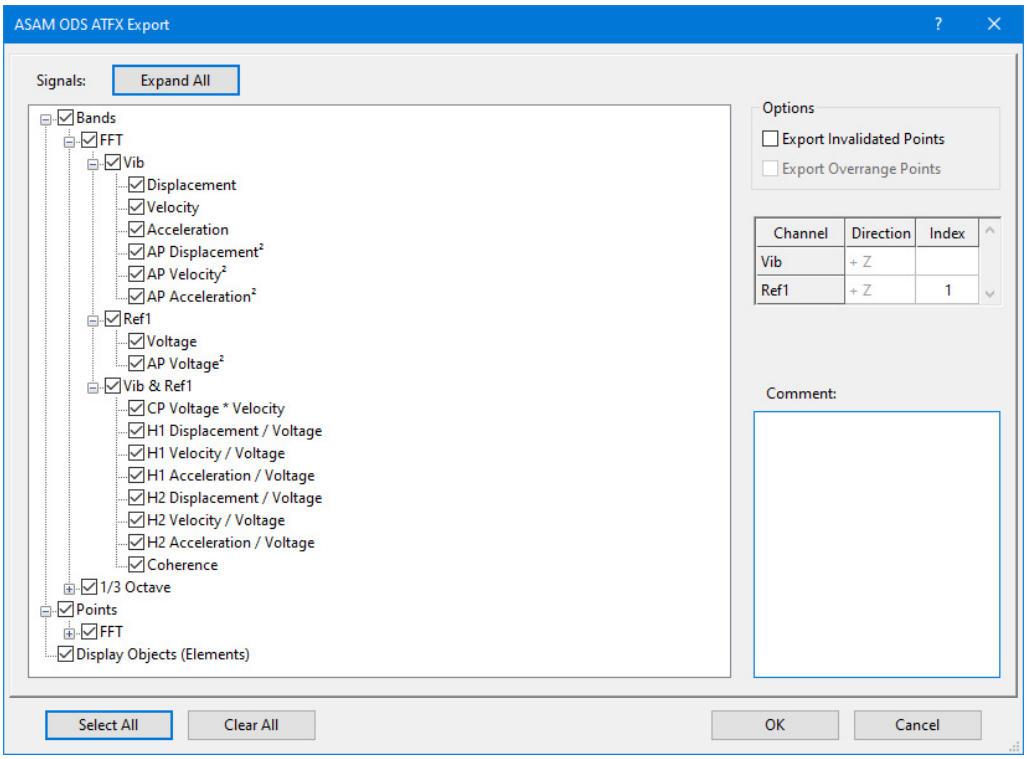


Figure 5.26: ASAM ODS ATFX Export dialog

ⓘ You can freely adjust the size of the dialog (width/height) to be able to read all the information.

Dialog description

On the left you mark the signals and on the right you set the parameters for the export.

Signals: Mark the signals which you want to export. The selection depends on which signals you have measured and whether frequency bands have been defined. If Display Objects (Elements) is ticked, then a graphic representation of the scan points and their connections is exported.

ⓘ You can export user defined data sets but not average spectra.

Expand All: Click here to display the entire tree in detail.

Select All: Click here to select all signals.

Clear All: Click here to clear the signal selection.

Export Invalidated Points: Tick the check box if you want to export signals of points which you have marked as Invalidated.

Export Overrange Points: Tick the check box if you want to export signals of points which have been assigned the Overrange scan status.

Channel: The channels you are exporting are displayed.

Direction: The directions of the vibration and the reference excitation are displayed. You can only change the direction of the vibration in the readout of the presentation window.

ⓘ It is not possible to specify the vibrational direction for the Vibrometer 3D channel.

Index: If you have selected a reference channel and have measured the reference signal at one of the scan points, enter the index of this reference point.

Enter an index for a response channel so that the software exports only one signal for that channel. This avoids, for example, that a Driving-Point function is exported for each measured scan point.

ⓘ The settings for the vibrational direction and the reference point have no effect on exporting user defined signals (Usr or Usr X, Usr Y, Usr Z channel respectively). We recommend that you check the properties of the user defined signals before exporting and if necessary, change them as described in SECTION 3.7.5.

Comment: You can enter a comment up to 256 characters long.

5.2.12 Exporting Geometries in STL Format

You can export the geometries as an STL file in stereo lithography data format STL ASCII File (*.stl) or STL Binary File (*.stl).

Export STL

To export the geometries in STL format, proceed as follows:

1. Select File > Export > STL in the menu bar.

   The Save As dialog appears.

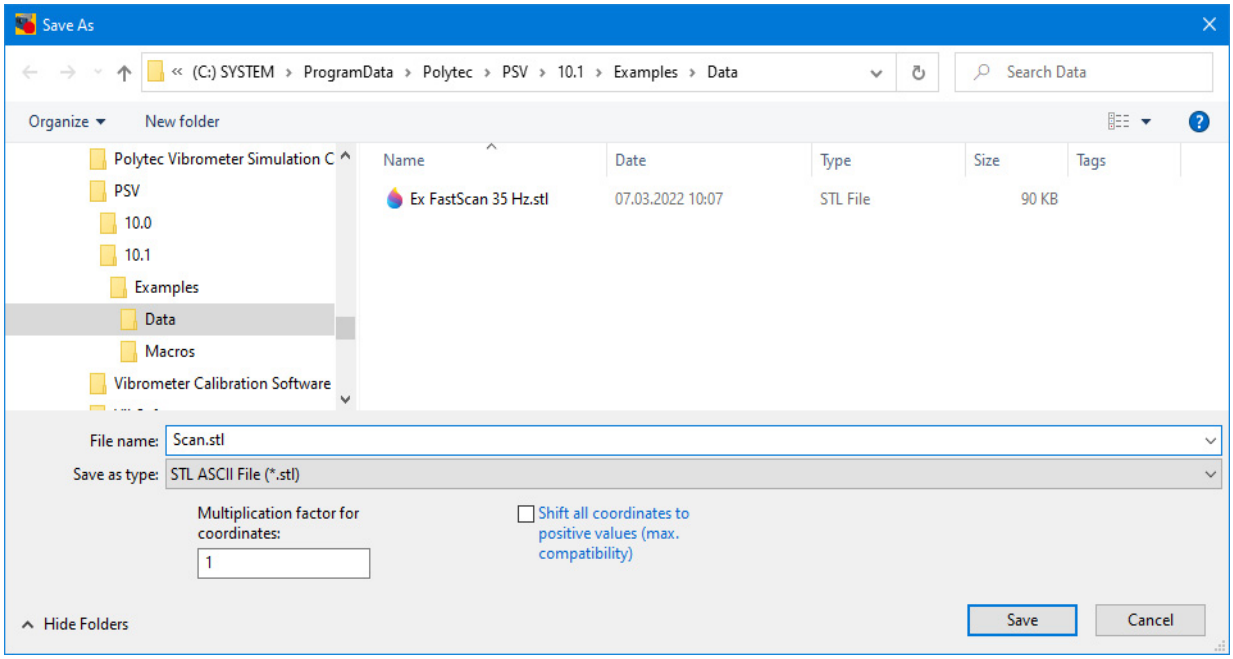


Figure 5.27: Save As dialog for STL export

2. Navigate to the saving location, select Save as type and enter the file name.

ⓘ Once you have displayed the project browser, you will then automatically be shown the directory marked there as saving location. Refer also to SECTION 1.5.4.

3. At the bottom of the dialog, enter which multiplication factor should be used for the coordinates.
4. If you want to reach maximum compatibility, tick the Shift all coordinates to positive values check box.
5. Click Save.

   The data is saved in the selected format.

Appendix A: Key Combinations in the Software

General

Key combination	Definition
Ctrl + O	Open Document
Ctrl + Q	Start acquisition mode

In the Presentation Window

Key combination	Definition
Ctrl + I	Display file information
Ctrl + S	Save document
Ctrl + G	Save the presentation window as graphic
Ctrl + P	Print the presentation window
Ctrl + Z	Undo the last action
Ctrl + Y	Redo the previously undone action
Ctrl + C	Copy selection onto the clipboard
Ctrl + V	Insert selection from the clipboard
Ctrl + A	Select all
Ctrl + F	Open/close Frequency Band Definition dialog
Alt + 1	Display Object view
Alt + 2	Display Single Scan Point view
Alt + 3	Display Average Spectrum view
Alt + 4	Display Profile view
Alt + 5	Display Band Data view
Alt + 6	Display Point Data view
Shift + Y	Show the sight of the video camera
Alt + Enter	Change display properties
Ctrl + Space	Set the scan status for the selected scan point to Invalidated
Shift + Space	Set the scan status for the selected scan point to Valid
Ctrl + R	Interpolate selected scan point
Ctrl + T	Delete the interpolation of the selected scan point
Delete	3D view: Delete marked objects
T	3D view and Connect Points dialog open: Generate triangles for selected scan points
Space bar	Start/stop animation
+	During running animation: Increase the animation speed
-	During running animation: Decrease the animation speed

In the Analyzer

Key combination	Definition
Ctrl + K	Scale ordinate to min/max values of current data
Ctrl + Delete	Delete all frequency bands

In the Project Browser

Key combination	Definition
Shift + Delete	Irrevocably delete selected file without security popup
Delete	Irrevocably delete selected file with security popup
F2	Rename selected file or folder
Enter	Execute default action in the context menu

In the Signal Processor

Key combination	Definition
Ctrl + C / Ctrl + Insert	Copy selected cell
Ctrl + X / Shift + Delete	Cut selected cell
Ctrl + V / Shift + Insert	Insert selected cell
Ctrl + Z / Alt + Backspace	Undo last action
Ctrl + Y	Redo previously undone action
Ctrl + #	For all cells switch display to show the formulae

Index

Numerics

1/3 octave bands, frequency band definition 4-17
2D filter, selection method for outliers 4-33
3D filter, selection method for outliers 4-34
3D geometry
  display video image (3D view) 3-8
  line of sight 3-7
  presentation 3-26
  projection (3D view) 3-8
  scan range (3D view) 3-9
3D view
  display properties, presentation window 3-6
  view style of data 3-26
3D view, clipping planes 3-9

A

acquisition
  mode in the software 1-5
analyzer
  description 1-13
  icons 1-14
  in the signal processor 1-20
  key combinations A-2
  open 1-13
  print 1-22
  properties 3-10
animation
  FFT/FastScan/MultiFrame measurement mode 4-23
  of the vibration 4-23
  save 5-17
  scale deflection 4-25, 4-29
  set up frame rate 2-4
  time measurement mode 4-27
application window, description 1-8
arrows, view style 3-25
ASAM ODS ATFX
  export 5-29
  import 5-6
ASCII
  export 5-20
  load frequency band definition 5-5
Assigned Auto, laser focus status 4-5
Assigned Man., laser focus status 4-5
autoscale, in analyzer 3-18
average spectrum, view 3-24

B

background color
  in presentation window 3-3
  in the analyzer 3-21
band cursor 3-13
band data, evaluate 4-12
Basic Engine manual 1-24

C

Calculated from 2D, geometry status 4-4
Calculated, laser focus status 4-5
calculation table, in the signal processor 1-20
cascade, arrange windows/analyzers 1-20
channels
  display data set 3-29
clipboard, copy data 5-18
clipping planes, define 3-33
close all, windows/analyzers 1-20
color palette 3-28
color-coding 3-28
colors of the data, in the readout (analyzer properties) 3-21
colors, select in presentation window 3-3
combined file
  display 3-41
  general information 3-35
  generate 3-37
command description, help 1-23
component, readout 3-27
components, define geometry component 4-38
connections between scan points 4-37
connections between scan points (APS)
  define automatically 4-37
context-sensitive help 1-23
cursor mode
  FFT measurement mode 4-19
  time measurement mode 4-27
cursor values, copy 3-16
cursor, move in the analyzer 3-16
cursor, set in presentation window 3-26
cursor, set in the analyzer 3-11

D

data
  analyzer properties 3-20
  display properties, presentation window 3-4
  manage 5-1
  process/evaluation 4-1
data set, display 3-28
data, read in presentation window 3-26
data, read in the analyzer 3-11
dB reference
  define 0 (presentation) 3-32
  setup 2-2
devices, initialize on start 1-5

diagram, in the analyzer 1-14
differential cursor 3-12
direct help 1-23
DirectX, technology for graphics 2-4
Disabled, scan status 4-2
display properties
  3D view 3-6
  data 3-4
  general, presentation window 3-3
  set up display of object 3-2
display type, show data set 3-30
display, setup 2-3
domain, display data set 3-29
Dongle 1-2

E

evaluation
  sections of a spectrum 4-12
export 5-11
  ASAM ODS ATFX 5-29
  ASCII 5-20
  ME'Scope 5-26
  STL 5-32
  Universal File 5-21
  Wave Sound format 5-25

F

Failed
  geometry status 4-4
  laser focus status 4-5
file information, display 5-1
file list 5-4
filter, smooth data 4-6
fitted curve 3-14
folder, add to project browser 1-16
folder, remove in project browser 1-16
formulae
  find and replace in signal processor 3-65
  save in signal processor 3-64
formulae editor, in the signal processor 1-19
frame rate, animation 2-4
frequency band definition 4-12
  automatically 4-15
  load 5-5
  manually 4-13
  save 5-14
  select/edit frequency bands 4-17
frequency band, display in surface view 3-33
frequency bands
  calculate 1/3 octave bands 4-17
  cut/copy 4-18
  define 4-13
  delete 4-17
  determine peak frequencies again 4-17
  select 4-17
function wizard
  activate in the signal processor 3-48
  example 3-78
functions
  signal processor 3-49

G

general
  analyzer properties, set up diagram 3-21
  display properties 3-3
geometry component, define 4-38
geometry status 4-4
graphic representation, technology 2-4
grid, display in the analyzer 3-21

H

Hardlock 1-2
harmonic cursor 3-15
help texts 1-23
Hidden, scan status 4-2
horizontally, arrange windows/analyzers 1-20

I

icons for commands
  in the analyzer 1-13
  in the application window 1-8
image section, pan 1-22
import, ASAM ODS ATFX/ME'Scope/Universal File 5-6
Imported, geometry status 4-4
index, readout 3-27
indices
  change 4-35
  display 4-35
individual measurements
  apply frequency bands 3-43
  merge (stitching) 3-37
  sort (combined file) 3-43
Info
  about PSV 1-24
  command in project browser 1-17
initial operation
  start Polytec DMS 1-2
interpolate status 4-10
interpolate, points 4-6
Interpolated
  geometry status 4-4
  laser focus status 4-5
Invalidated
  assign status 4-7
  scan status 4-2
isolines, view style 3-25

K

key combinations
- analyzer A-2
- general A-1
- presentation window A-1
- project browser A-2
- signal processor A-2

L

laser focus status 4-5
legend for color-coding 1-12
license file 1-2
list of components 1-12
list of open windows/analyzers 1-20
log file for messages 2-6
log on 1-4

M

magnitude, selection method for outliers 4-33
ME'Scope
- export 5-26
- import 5-6

Measured, geometry status 4-4
measurement data, evaluate 4-1
measurement data, set up view style 3-4
measurement mode
- FFT/FastScan/MultiFrame, animate vibration 4-23
- time, animate vibration 4-27

messages
- log file 2-6
- set up display 2-5

modes, in the software 1-5
Modified, geometry status 4-4

N

new, presentation window 1-20
No 3D Coor., geometry status 4-4
No Focus, laser focus status 4-5
Not Measured, scan status 4-2
Not Reachable, scan status 4-2

O

object, setup presentation 3-3
object, view 3-24
online help 1-23
OpenGL, technology for graphics 2-4
Optimal Meas., geometry status 4-4
Optimal, scan status 4-2
options, display software options 1-24
outliers, select scan points 4-32
Overrange, scan status 4-2

P

page view, print preview 1-23
page, setup 1-22
pan
- zoomed diagram section 3-17
- zoomed image section 1-22

password 1-4
peak frequencies, frequency band definition 4-17
peak-fit 3-14
point data, evaluate 4-19
presentation
- measurement data/scan 4-1
- mode in the software 1-6

presentation window
- description 1-10
- icons 1-11
- key combinations A-1
- open 1-10
- print 1-22
- set up view 3-22

presentation, setup
- dB reference 2-2
- display 2-3
- messages 2-5

print preview 1-23
print, active window 1-23
printer, setup 1-22
profile line, draw/edit 4-21
profile, view 3-24
profiles, display 4-21
project browser
- add project folder 1-16
- commands 1-17
- copy and move objects 1-16
- description 1-15
- key combinations A-2
- open/position 1-15
- remove project folder 1-16

project folder, add 1-16
project folder, remove 1-16
projection, 3D geometry (3D view) 3-8
properties, display of the object 3-2
PSV, about 1-24

R

readout
- in the presentation window 1-12

readout, in the analyzer 1-14
redo 4-22
restore window layout 2-3

S

samples, time domain 2-4
save graphics 5-16
scale, axes ranges in analyzer 3-18
scan data
  open 5-3
  save 5-11
scan file, create new 3-79
scan point
  show number of a component 4-39
scan point number, readout 3-27
scan points
  assign, to components 4-39
  automatically connect 4-37
  change indices 4-35
  display indices 4-35
  edit 4-30
  select (not APS) 4-30
  select outliers 4-32
  select, of a component 4-39
  select, point status 4-2
  show/hide, of a component 4-39
  status display 4-2
  view style 3-25
scan points (point object)
  connect automatically 4-37
scan range, 3D geometry (3D view) 3-9
scan status 4-2
scan, evaluate 4-1
scroll bar, in the analyzer 1-14, 3-17
select file, signal processor 3-65
setting files, storage 1-4
sight, 3D geometry 3-7
signal description, in the signal processor 3-60
signal processor
  application examples 3-67
  calculation examples 3-58
  cell links 3-57
  conventions 3-57
  copy measurement data 3-46
  copy/calculate measurement data 3-45
  description 1-19
  display result 3-63
  display/calculate measurement data 3-44
  find and replace in formulae 3-65
  fix view style 3-63
  function wizard 3-78
  functions 3-49
  key combinations A-2
  open documents 5-3
  paste/edit measurement data 3-47
  print 1-22
  save formulae 3-64
  save user defined data sets 3-61
  select file 3-65
  signal description 3-60
  use icons/function wizard 3-48
signal, display data set 3-29
single cursor 3-11
single point data
  open 5-3
  save 5-13
single spectrum, view 3-24
smooth, presentation of measurements/scans 4-6
software
  display options/version 1-24
  install 1-3
  modes 1-5
  start 1-4
spectrum, evaluate 4-12
start DMS 1-2
status bar 1-9
status display
  geometry status 4-4
  laser focus status 4-5
  readout 3-27
  scan status 4-2
status points, view style 3-26
stitching
  display combined file 3-41
  general information 3-35
  merge individual measurements 3-37
STL format, export 5-32
surface, view style 3-25

T

technology, graphic representation 2-4
theory manual 1-24
time domain, samples 2-4
title bar
  in the analyzer 1-13
  in the application window 1-8
title, show
  in presentation window 3-3
  in the analyzer 3-21
Too much/little Light, geometry status 4-4
toolbar
  general icons 1-7
  icons in the analyzer 1-14
  icons in the presentation window 1-11
  in the analyzer 1-13
  in the application window 1-8
  in the presentation window 1-10
  in the signal processor 1-19
  software-specific icons 1-9
Triangulated, geometry status 4-4

U

undo 4-22
Universal File
  export 5-21
  import 5-6
user defined data sets
  create new scan file 3-79
  properties 3-61
  save 3-61

V

Valid
  assign status 4-7
  scan status 4-2
version, display software version 1-24
vertically, arrange windows/analyzers 1-20
vibration, animate 4-23
vibrational direction x, y, z
  display for animation 4-26
  display in the analyzer 3-19
vibrational direction, change 4-11
video image
  display (3D view) 3-8
Video Triangulated, geometry status 4-4
view style, data 3-25
view, presentation window 3-23
VT Failed, scan status 4-2

W

Wave Sound format, export 5-25
window list 1-20
window settings 1-21
windows
  arrange 1-20
  link 1-21, 3-16
Windows settings, adapt 1-3
wireframe, view style 3-25

Z

zoom
  end 1-22
  in the analyzer 3-17
  in windows 1-22
  pan diagram section 3-17
  pan image section 1-22
  undo 1-22
zoom functions
  in the analyzer 1-14
  in the presentation window 1-12

Polytec Europe

Germany (DE)
Polytec GmbH
Headquarters
Polytec-Platz 1–7
76337 Waldbronn

Tel.: +49 7243 604-0
Fax: +49 7243 69944
E-mail: info@polytec.de
Internet: https://www.polytec.de

Germany (DE)
Polytec GmbH
Sales and Consulting Office Berlin
Schwarzschildstraße 1
12489 Berlin

Tel.: +49 7243 604-4540
Fax: +49 7243 604-484540
E-mail: polytecberlin@polytec.de
Internet: https://www.polytec.de

France (FR)
Polytec France S.A.S.
TECHNOSUD II, Bâtiment A
99, Rue Pierre Semard
92320 Châtillon

Tel.: +33 1 49 65 69-00
Fax: +33 1 57 19 59 60
E-mail: info@polytec.fr
Internet: https://www.polytec.com/fr

Great Britain (GB)
Polytec Ltd.
Unit 8, The Cobalt Centre
Siskin Parkway East
Middlemarch Business Park
Coventry CV3 4PE

Tel.: +44 2475 267 020
E-mail: info@polytec-ltd.co.uk
Internet: https://www.polytec.com/uk

Polytec Worldwide

ASEAN Countries
Polytec South-East Asia Pte. Ltd.
Blk 4010, Ang Mo Kio Ave 10
#06-06 TechPlace I
Singapore 569626

Tel.: +65 64510886
Fax: +65 64510822
E-mail: info@polytec-sea.com
Internet: https://www.polytec-sea.com

China (CN)
Polytec China Ltd.
Room 402, Tower B Minmetals Plaza,
No. 5 Chaoyang North Avenue,
Dongcheng District, Beijing 100010

Tel.: +86 10 6568 2591
Fax: +86 10 6568 8291
E-mail: info-cn@polytec.com
Internet: https://www.polytec.com/cn

Japan (JP)
Polytec Japan
13th floor, Arena Tower,
3-1-9 Shinyokohama, Kohoku-ku
Yokohama-shi, 222-0033 Kanagawa

Tel.: +81 45 478-6980
Fax: +81 45 478-6981
E-mail: info.jp@polytec.com
Internet: https://www.polytec.com/jp

USA West
Polytec, Inc.
North American Headquarters
16400 Bake Parkway, Suites 150 & 200
Irvine, CA 92618

Tel.: +1 949 943-3033
Fax: +1 949 679-0463
E-mail: info@polytec.com
Internet: https://www.polytec.com/us

USA Central
Polytec, Inc.
Central Office
47909 Halyard Dr.
Plymouth, MI 48170

Tel.: +1 734 253-9428
Fax: +1 508 281-4725
E-mail: info@polytec.com
Internet: https://www.polytec.com/us

USA East
Polytec, Inc.
East Coast Office
1 Cabot Road, Suites 101 & 102
Hudson, MA 01749

Tel.: +1 508 417-1040
Fax: +1 508 281-4725
E-mail: info@polytec.com
Internet: https://www.polytec.com/us